The Social Determinants of Bank Runs

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Abstract

Motivated by new evidence from the 2023 U.S. regional bank crisis, I develop a theory of how depositor social structure influences bank runs. Rumors about bank financial health spread via interpersonal communication, triggering withdrawals that may lead to collapse. I show that the occurrence, speed, and scale of runs depend on depositor connectedness—the likelihood that agents know and communicate with each other. Higher connectedness accelerates information spread, making runs faster and more severe. In contrast, dispersed depositor bases slow down runs, and can even prevent them entirely.

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1 Introduction

Explaining the run on Silicon Valley Bank in March 2023, Federal Reserve historian Jonathan Rose (2023) writes: *"The most significant departure from historical comparisons is that depositors were unusually connected or similar to each other. As a result, they withdrew in a coordinated or similar way."* Rose's phrasing is precise: *coordinated* appears adverbially, describing *how* agents withdrew rather than *whether* they withdrew—suggesting that coordination is not merely a binary switch between equilibria but a process that unfolds through a network. If withdrawal behavior propagates through social connections, then the structure of those connections ought to matter for bank-level fragility.

Most theories of runs, however, do not model the process through which depositors coordinate, making it difficult to formalize such insights. My paper addresses this limitation by developing a model in which information diffuses through interpersonal communication.

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Agents learn about potential runs from others and decide when to withdraw based on their beliefs. Each period, they assess bank fragility based on fundamentals and coordination pressure: how many others they expect to have learned. The interplay between information diffusion and belief updating generates testable bank-level dynamics: a more dispersed depositor base can slow runs and even prevent them. The key mechanism is that recovery dynamics—agents rationally returning their deposits as survival evidence accumulates—feed back into the run phase: when reentry is fast enough, simultaneous withdrawals never breach solvency, eliminating the very risk that motivated the run.

The model I develop builds on Abreu and Brunnermeier (2003) and He and Manela (2016)'s frameworks of sequential awareness in bubbles and runs. It features a continuum of depositors and a single bank. Depositors derive utility from deposits in the form of a convenience flow but face the risk of losing their funds in a run. The run is prompted by a solvency shock that arrives at a Poisson rate. This shock can leave the bank in one of two states: fragile or healthy. A fragile bank can sustain only a fraction κ of withdrawals before collapsing; a healthy bank can withstand any level of withdrawals—the shock was a false alarm.

Agents learning about the shock may fear that others withdraw, and withdraw themselves, prompting a self-fulfilling run. The run is not instantaneous however, because agents do not learn about the shock simultaneously. Instead, information spreads gradually through social contacts. I model this diffusion process explicitly using a tool from mathematical epidemiology: the Susceptible–Infected (SI) process, widely used to model disease transmission. Agents meet each other at rate β; when an uninformed agent meets an informed one, they learn about the shock. This generates logistic diffusion dynamics: information initially accelerates as more agents become aware, then decelerates as the informed population saturates. Agents receiving information learn that a shock has occurred but cannot observe whether the bank is fragile or healthy. Cases in which the shock leaves the bank healthy can thus be interpreted as false rumors—information spreads about a solvency shock that poses no actual threat.

The meeting rate β captures the intensity of interpersonal communication and can arise from various sources: geographical deposit concentration (the measure I use for the motivational evidence in Section 2), social media exposure (Cookson et al. 2023), or any channel affecting the speed of information diffusion. A higher β implies faster information transmission and thus lower coordination frictions, which I show makes banks more vulnerable to runs in equilibrium.

Coordination frictions matter not only because they determine how quickly information spreads, but also because they discipline how rational depositors update their beliefs and time their actions. When agents learn about a potential solvency shock, they must decide *when* to withdraw—or when to return if they come to believe the rumor was false. Their decision depends on beliefs about both the bank's true state and the anticipated withdrawal behavior of others. Solving this problem in full generality is intractable, so I focus on a natural subset of equilibria—*stationary equilibria*. In such equilibria, agents' strategies depend only on the time elapsed since becoming informed, not on calendar time.

At each instant the depositor faces a dynamic trade-off: holding deposits for longer provides utility, but continued exposure increases the risk of being caught in a crash. The key statistic governing this decision is the perceived instantaneous crash probability, or *hazard* *rate*. The hazard rate can be decomposed into two components: the conditional probability that a fragile bank collapses, which rises with withdrawal pressure, and the posterior belief that the bank is indeed fragile. These two components evolve in opposite directions. An agent anticipating the collapse understands that more depositors will learn about the shock and withdraw, raising the conditional crash probability. The bank's continued survival, on the other hand, provides Bayesian evidence that the shock was likely a false rumor: beliefs about fragility decline over time. In the period immediately following agents' learning, the first force dominates, prompting agents to withdraw when hazard exceeds utility benefits. But as survival evidence accumulates and the logistic learning curve saturates—decelerating information flow—the second force takes over, and hazard starts decreasing. Agents who withdrew earlier now rationally reenter, confident the rumor was false. Agents' gradual exit and reentry generate hump-shaped aggregate dynamics—withdrawals rise as panic spreads, then fall as confidence rebuilds.

The model's central result is an existence threshold for runs. Because aggregate withdrawals are hump-shaped, they necessarily peak. The height of this peak depends on depositor connectedness β. When depositors are dispersed, information spreads slowly and withdrawals are staggered over time, lowering the peak. Conversely, rapid diffusion compresses withdrawals in time, raising the peak. There exists a critical threshold: when connectedness is sufficiently low, aggregate withdrawals always remain within the bank's solvency capacity. In that case, runs become impossible—even at the height of panic, withdrawals never breach solvency, eliminating any incentive to run.

Deposit utility shapes recovery dynamics similarly. When deposits are attractive, agents reenter quickly to enjoy their benefits despite remaining crash risk, reducing the number of agents withdrawn at any given moment and lowering the peak. Enhancing deposit attractiveness can serve as a viable run-prevention policy, an insight consistent with the findings of Cipriani et al. (2024) that many banks raised rates to moderate outflows during the 2023 crisis. However, as I show in Section 5, the effectiveness of such measures depends on the underlying social structure of depositors, β.

The model's tractability enables structural estimation. As a methodological exercise, I illustrate this by fitting the model's closed-form withdrawal path to intraday stock returns during the March 2023 crisis. This allows me to recover bank-specific estimates of depositor connectedness β and other model parameters directly from observed price movements.

Three extensions, solved numerically, enrich the baseline framework: heterogeneous learning speeds across depositor groups (which generate inequality in run exposure and asymmetric recovery dynamics), interest-bearing deposits (which create technical difficulties but no new insights beyond higher deposit utility), and a social-learning environment in which agents learn by observing others' withdrawals rather than through word-of-mouth.

Literature review. Understanding how depositors coordinate on bank runs has been a central question in banking theory since Diamond and Dybvig (1983). My paper is part of an established branch of this literature that focuses on the *dynamics* of coordination. Dynamic aspects were first brought into the Diamond-Dybvig model through sequential service constraints (Green and Lin (2003), Peck and Shell (2003), Gu (2011)). More recent efforts model run dynamics in tractable frameworks using alternative models: He and Xiong

(2012), for example, introduced a dynamic debt run model based on firm debt rollover.

Within that last strand, I am particularly indebted to He and Manela (2016), which itself builds on Abreu and Brunnermeier (2003). The latter introduced "sequential awareness" in the context of rational bubbles: agents become aware of the bubble sequentially but are uncertain about their position in the sequence, gambling that they are early enough to delay exit and thereby prevent standard backward induction.[1] He and Manela (2016) adapted this framework to bank runs, introducing uncertainty about whether the bank is fragile or healthy and allowing agents to acquire additional noisy information. My paper retains the fragile-versus-healthy uncertainty but replaces information acquisition with explicit modeling of information diffusion using Susceptible-Infected (SI) epidemiological dynamics. In contrast to both Abreu and Brunnermeier (2003) and He and Manela (2016), this learning process generates natural saturation, allowing survival evidence to dominate and trigger recovery. The description of recoveries itself is new to my paper. The real contribution, however, is the observation that recovery dynamics feed back into the run phase itself—if coordination frictions or deposit utility push agents to reenter early enough, runs can be eliminated entirely even at fragile banks, a fundamentally new insight.

Empirical evidence supports the importance of social structure in coordinating depositor behavior. Kelly and Ó Grada (2000) show that during historical New York bank runs, withdrawal decisions traveled along immigrants' social networks. Studying a run in India, Iyer and Puri (2012) find that an individual's withdrawal probability rises when neighbors and acquaintances withdraw. Kiss et al. (2014) provide experimental evidence of similar mechanisms. Schmidt et al. (2016) discuss the informational advantage of institutional investors during the 2008 runs on Money Market Funds, and Liu et al. (2023) highlight the role of online exposure during the Terra Luna run. Finally, Cookson et al. (2023) and Benmelech et al. (2024) provide similar evidence during the 2023 run.[2] My paper complements this empirical work by developing a theoretical framework that shows how micro-level social interactions aggregate to determine bank-level run dynamics.

The information diffusion process I study builds on Jackson and Lopez-Pintado (2011) and López-Pintado (2006, 2008, 2012), who applied mean-field epidemiological approximations developed by Pastor-Satorras and Vespignani (2001) to model product adoption as disease transmission. The critical distinction is that in those papers, agents are mostly concerned with a coordination problem: they face strategic complementarities in adoption decisions. The setting in my paper is more complex, mixing coordination and competition incentives (agents want to withdraw before others).

Stepping back, my focus on information diffusion connects to a broader theoretical literature on how information structure shapes coordination in financial crises. The global games tradition, pioneered by Morris and Shin (2001) for currency attacks and adapted by Peck and Shell (2003) to bank runs, demonstrates that coordination outcomes depend critically on the precision of public versus private information. Angeletos and Pavan (2004) show that greater transparency in public information can either facilitate or hinder coordination depending on the degree of strategic complementarity. Morris and Shin (2002) establish that enhanced public information dissemination can reduce welfare when agents have socially

[1]This type of game is sometimes referred to as a "clock game" (Brunnermeier and Morgan (2010)).

[2]See Appendix B for a detailed discussion on the empirical literature on the 2023 regional banking crisis.

valuable private information. More recently, Parlatore (2024) demonstrates that more precise information increases an economy's vulnerability to bank runs as a whole. Llambias and Ordonez (2024) find a non-monotone effect in a dynamic setting: faster access to information initially facilitates coordination on runs, but if sufficiently fast, makes runs less frequent because agents are less afraid of being left holding the bag. My setting embeds information precision in a dynamic diffusion process: agents must infer not only fundamentals but also how widely information has spread, introducing a backward-looking assessment of run pressure that shapes whether withdrawals coalesce into an effective run—or dissipate, even when the bank is fragile.

The paper proceeds as follows. Section 2 presents empirical evidence on the complementarity between fundamental and coordination risk. Section 3 develops the theoretical framework. Section 4 characterizes stationary equilibria and derives the hazard rate decomposition. Section 5 analyzes how coordination frictions and deposit attractiveness determine run dynamics, establishing the main result. Section 6 explores how the model can be used for structural estimation using crisis-period stock returns. Section 7 explores extensions with heterogeneity, interest rates, and social learning.

2 Empirical Evidence: The Role of Depositor Concentration

While principally theoretical, my paper is motivated by Rose (2023)'s conjecture that depositor connectedness accelerates bank runs. In this section, I illustrate that this conjecture appears grounded empirically. I establish that banks with a concentrated depositor base experienced bigger runs in March 2023 and that this effect interacted with traditional sources of fundamental risk.

2.1 Measure of fundamental risk and coordination frictions

I construct measures that capture both standard fundamental balance-sheet risks faced by banks and the coordination risk dimension generated by depositor concentration.

Fundamental risk My measure of fundamental risk is drawn directly from the work of Drechsler et al. (2025).[3] They show that a bank's total economic value can be decomposed as the sum of the marked-to-market value of assets net of the book value of total deposits and the franchise value of deposits, providing a quantitative framework to estimate each component from the Call Reports. After the 2023 rate hikes, SVB (and other banks affected by the crisis) experienced a shift in the composition of their values: the value of their assets dropped, while the value of their deposit franchise increased. The new balance-sheet structure was fragile because the deposit franchise is runnable. Indeed, if depositors run, the franchise value plunges and can make the bank insolvent, justifying the run in the first

[3] I am grateful to Itamar Drechsler, Alexi Savov, Philipp Schnabl, and Olivier Wang for generously sharing their replication code.

place.[4] From the bank value decomposition, I define the solvency measure κ as the critical fraction of the total deposit franchise that must be lost for the bank to reach a critical solvency threshold $\underline{v}$:[5]

$$\underbrace{A - D}_{\text{MTM Equity Value}} \quad + \quad (1-\kappa) \times \qquad \underbrace{DF}_{\text{Deposit Franchise Value}} \qquad = \underline{v}$$

Banks with $\kappa < 1$ face genuine fundamental risk: a sufficiently large run can render them insolvent. Banks with $\kappa > 1$ remain solvent even after losing their entire deposit franchise, making runs unlikely in the first place. Appendix B.3 discusses the construction of this measure in detail.

Coordination frictions. To measure coordination frictions, I use a simple, transparent proxy: the geographic concentration of deposits across bank branches. I measure concentration using the Herfindahl-Hirschman Index (HHI) of branch-level deposits. Formally, the HHI is defined as: $\sum_{i \in \text{Branches}} \left(\frac{\text{Deposits in } i}{\sum_i \text{Deposits in } i} \right)^2$. HHI has a natural interpretation as the probability that two randomly selected deposit dollars originate from the same branch. When deposits are geographically concentrated, depositors are more likely to share local networks and encounter each other, allowing information about bank health to spread more rapidly. Depositors can better assess how many others have learned about potential issues and are likely withdrawing, reducing coordination frictions.

While other measures could be proposed, HHI offers a parsimonious measure of the network structure relevant for information transmission. It is computed from publicly available regulatory data (FDIC's Summary of Deposits) and can be readily constructed for other crisis episodes. Appendix B provides additional discussion of HHI's distribution across banks, its time-series evolution, and its correlation with other bank characteristics.

2.2 Empirical Strategy and Results

To test the independent role of coordination risk during the crisis, I examine cumulative stock returns during the week of March 6–13, 2023—around SVB's collapse.[6] Absent high-frequency data on deposit outflows, stock returns provide the best available proxy for run severity during this period and have been standard in the 2023 banking crisis literature.[7]

Figure 1 presents the relationship between stock returns during the crisis week, fundamental risk (κ), and coordination risk (HHI), with color intensity indicating return magnitude. The figure reveals that banks experiencing large stock price declines cluster in the bottom-right region: low κ combined with high HHI. SVB, First Republic, Signature Bank, Western

[4]A more thorough account of the deposit franchise run risk, its role in the 2023 crisis and a review of other sources of risk accounted by the literature is available in Appendix B.

[5]Following Drechsler et al. 2025, I set $\underline{v}$ to 3% of assets in my empirical analysis. 3% corresponds to the Basel III recommendation for banks in good standing.

[6]Stock returns are sourced from CRSP; see Appendix B.3 for construction details.

[7]Cookson et al. (2023) and Benmelech et al. (2024), for example, directly use stock returns as the testable outcome of their bank vulnerability measures. Importantly, Cipriani et al. (2024) validate this approach using confidential Federal Reserve data, showing that stock returns exhibit a strong correlation with actual deposit outflows during the run phase.

Alliance, and PacWest are all in that category. In contrast, banks with $\kappa \geq 1$ generally show small price drops, consistent with the interpretation that such banks remain solvent even after complete deposit franchise destruction. Wells Fargo, JPMorgan Chase, and Citigroup fall in this category. Among fragile banks ($\kappa < 1$), those with low HHI experienced substantially smaller stock price declines despite their fundamental weakness—US Bank and Bank of America exemplify this pattern. The theoretical model I develop in subsequent sections focuses particularly on explaining this protective effect: why dispersed depositor bases can prevent runs even at fundamentally fragile institutions.

Table 1 formalizes these patterns in OLS regressions. Columns (1)–(4) confirm the scatter plot's message: both depositor concentration and fundamental fragility independently predict worse stock performance, with largely orthogonal effects. The key result appears in Column (5): among safe banks ($\kappa > 1$), HHI has a small, barely significant effect; among fragile banks ($\kappa < 1$), the effect is substantially larger and highly significant. Notably, fragility alone (at zero HHI) has no effect—both ingredients appear necessary to generate severe runs. Additional robustness checks are in Appendix B.4.2.

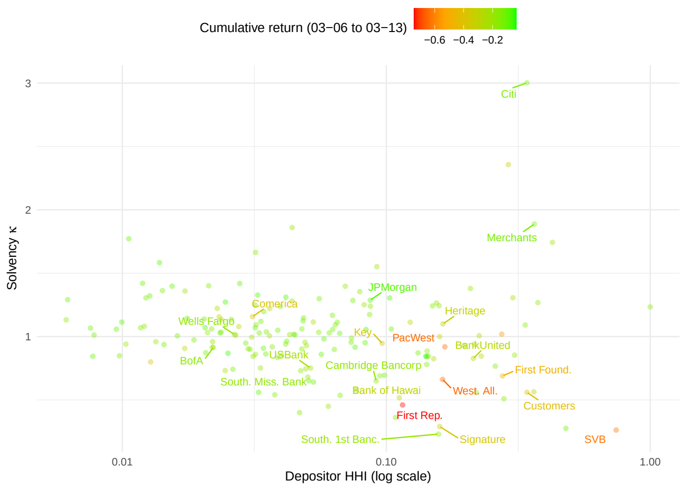


Figure 1: Stock returns, fundamental risk, and coordination risk.

3 Model: A Fragile Bank and a Rumor Spreading

Empirically, both fundamental fragility and depositor concentration seem necessary to foster runs: even fragile banks seem protected when their depositor base is dispersed. This section develops a model formalizing a role for depositor dispersion that explains why. Information about bank health spreads through social contacts. Dispersion slows diffusion among depositors, making coordinated withdrawals harder to achieve.

3.1 Many Agents, One Risky Bank

Time is continuous and extends forever. There is a continuum of agents of measure one, each endowed with one unit of wealth. Stagnant wealth (cash) yields no utility, but at time $t = 0$,

Dependent Variable:	Cumulative Stock Return (Mar 6-13, 2023)				
	(1)	(2)	(3)	(4)	(5)
Solvency κ	0.067***		0.065***	0.033	
	(0.023)		(0.022)	(0.034)	
Depositor HHI		-0.284***	-0.280***	-0.440***	-0.132*
		(0.061)	(0.059)	(0.142)	(0.076)
Depositor HHI $\times\kappa$				0.155	
				(0.126)	
Indicator ($\kappa < 1$)					0.000
					(0.019)
$\kappa < 1\times$ Depositor HHI					-0.354***
					(0.120)
Constant	-0.250***	-0.154***	-0.221***	-0.188***	-0.152***
	(0.025)	(0.010)	(0.024)	(0.036)	(0.012)
Observations	176	176	176	176	176
R^2	0.046	0.112	0.155	0.163	0.179

*Note: *** p<0.01, ** p<0.05, * p<0.1.*

Table 1: Deposit concentration, franchise value and bank returns.

a bank is established and offers a utility flow of u per unit of deposits.[8] I remain agnostic on specific interpretations for u, treating it as a reduced-form representation of non-pecuniary benefits such as convenience.

Agents decide whether and *when* to deposit to maximize their expected utility flow. Depositing yields utility flow but incurs a risk in the form of a run triggered by a solvency shock. A single such shock arrives at a random time t_0, drawn from a Poisson process with rate λ, and is one of two types. With probability p, the shock is *strong*, leaving the bank *fragile*: it can no longer withstand withdrawal of its entire deposit base. With probability $(1-p)$, the shock is *weak*, leaving the bank *healthy* and fully solvent—a false alarm. A healthy bank remains solvent regardless of the level of withdrawals. The parameter $\kappa < 1$ captures shock severity: a fragile bank can withstand withdrawal of at most a fraction κ of its deposit base before collapsing.

Crucially, the shock's *type* is never revealed to depositors, and its *arrival* is initially observed only by a small fraction G_0 of agents at t_0. The remaining agents learn that a shock has occurred gradually, through interpersonal communication, as described in Section 3.2 below. Throughout the information diffusion process, agents remain uncertain about the type of the shock. Weak shocks propagate exactly as strong ones do—they can be interpreted as false rumors.

Bank fragility depends on both the shock probability p and the capacity κ; the empirical measure "solvency κ" in Section 2 serves as a proxy for their combination. Banks with $\kappa > 1$ remain solvent in expectation; banks with $\kappa < 1$ face genuine fragility risk.

I assume that to be viable the bank needs to gather a positive depositor base at $t = 0$. As discussed in Section 5, there are environments where run risk deters participation entirely, precluding bank formation.

[8]Deposits do not accrue interest; this reduced-form specification is more tractable than explicitly modeling interest. Section 7 extends the model to interest-bearing deposits, which doesn't change the core mechanics.

3.2 The Spread of a Panic

Runs stem from depositors learning about and reacting to potential fragility. A simple backward induction argument shows that if the shock were perfectly observed by all agents, collapse would be immediate. Instead, I assume a form of *sequential awareness* (Abreu and Brunnermeier 2003): a small fraction G_0 of agents initially observes the shock and gradually spreads the information to others via interpersonal communication.

Specifically, I model information diffusion as the following process.[9] The mass of informed agents evolves from an exogenous initial condition G_0 according to the ordinary differential equation $dG = (1 - G)\, G\, \beta\, dt$. A mass $1 - G$ of uninformed agents meets other agents at rate β, and each meeting has probability G of being with an informed agent who passes along information about the shock. Throughout, I parameterize G by time elapsed since the shock: $G(0) = G_0$ is the exogenous initial seed at t_0, and $G(s)$ is the informed fraction at calendar time $t_0 + s$.

The meeting rate β is a key parameter capturing coordination frictions: higher β implies faster information transmission, which I will show increases "coordination risk" from the bank's point of view. In the empirical analysis of Section 2, I proxy coordination frictions using deposit concentration (HHI). More broadly, β can be interpreted as reflecting exposure to social media, shared investment networks, or any channel determining how rapidly information spreads between depositors. Figure 2 displays the S-shaped learning dynamics for $\beta \in \{0.5, 1, 2\}$: diffusion starts slowly (few informed agents), accelerates as informed agents multiply, then saturates as the rumor runs out of room to spread.

A natural concern is that information diffusion could depend on withdrawal behavior—agents who see others withdrawing may update faster. Section 7 extends the model to this endogenous learning channel, where agents learn by observing others' actions rather than through word-of-mouth. This "social learning" mechanism introduces additional complexity that requires numerical methods, but the core insights of the model remain unchanged.

Following He and Manela (2016), I assume the shock is eventually revealed publicly.

Assumption 1 (Awareness Window)**.** *The shock is revealed publicly once a fraction $1 - \varepsilon$ of agents have learned. Let η denote the time of revelation, defined by $G(\eta) = 1 - \varepsilon$. Assume ε is such that:*

- *$G(\eta) - G(0) \geq \kappa$ (sufficient learning for potential collapse),*
- *$G(\eta) > 1 - \kappa$ (learning population exceeds crash casualties),*
- *$\eta \leq 1/\lambda$ (learning timescale shorter than shock arrival).*

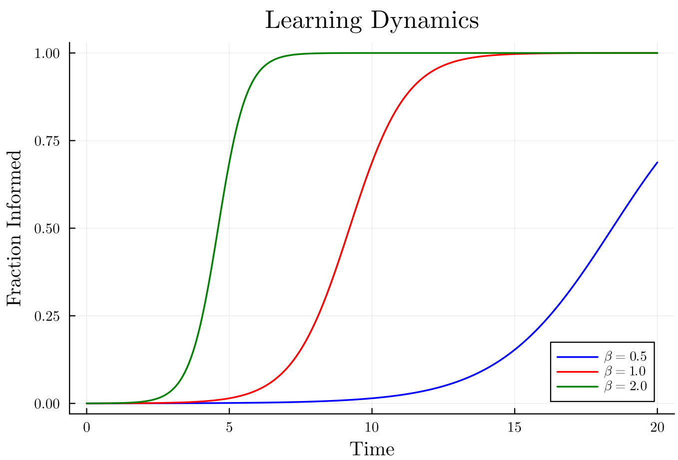
This assumption is primarily a technical device keeping agents' belief supports bounded and time-invariant. The first two conditions are sanity checks (the problem is well-defined and interesting). The third, $\eta \leq 1/\lambda$, requires that information spreads to near-saturation

[9] The structural results of the paper—equilibrium characterization and existence of threshold regimes (Propositions 1 and 2)—extend to any learning process whose implied density g is strictly log-concave (including normal and gamma densities). I focus on the logistic specification because it yields a single rate parameter β with a transparent economic interpretation, which is the basis for the comparative statics in Corollary 3.

faster than new shocks arrive—natural in the banking context. It plays a role in disciplining comparative statics in β.

Finally, note that agents only learn once: they ignore any information from subsequent meetings. Relaxing this assumption becomes intractable quickly, as it requires tracking beliefs conditional on the timing of each meeting—a rich state-space problem that I leave for future research.

Figure 2: Dynamics of informed agents for different communication speeds $\beta \in \{0.5, 1, 2\}$.



3.3 Depositor Optimization

Given the information diffusion process, agents face a dynamic withdrawal decision. Let $V(t)$ denote the expected value of \$1 in wealth at time t. Since deposits do not accrue interest, wealth remains constant absent a crash. The value function satisfies the Hamilton-Jacobi-Bellman equation:

$$0 = \dot{V}(t) + \max_{\alpha \in [0,1]} \{u(1-\alpha) + h(t)[\alpha - V(t)]\},$$

where α is the fraction of wealth held in cash and $h(t)$ is the hazard rate—the instantaneous probability of a crash conditional on survival—*perceived* by the depositor. The first term captures utility flow from deposits; the second captures expected value from a crash, equal to the probability $h(t)$ times the net gain $\alpha - V(t)$ (the agent retrieves the cash portion α but loses the continuation value $V(t)$).

The perceived hazard rate $h(t)$ is formed via Bayesian updating and depends on the agent's information set: whether and when they learned about the shock—denoted t_i, with the convention $t_i = \varnothing$ if the agent has not yet learned—and the absence of a crash up to time t. I write $h(t; t_i)$ to make the dependence on both sources of information explicit. Because agents learn at different times, the hazard they estimate at any given calendar time differs—this heterogeneity is what generates the gradual, aggregate run dynamics analyzed in Section 4.

Linearity in α implies bang-bang solutions: if $h(t; t_i) > u$, the agent withdraws everything ($\alpha^* = 1$); if $h(t; t_i) < u$, the agent holds everything in deposits ($\alpha^* = 0$). At the margin, moving one unit of wealth to cash costs u in utility flow but provides expected crash protection $h(t;t_i)$. The following lemma formalizes this result.

Lemma 1. *The optimal holding $\alpha^*(t;t_i)$ of an agent with information t_i at time t is given by:*

$$\begin{cases} \alpha^*(t;t_i) = 1 & \text{if } h(t;t_i) > u \\ \alpha^*(t;t_i) = 0 & \text{if } h(t;t_i) < u \\ \alpha^*(t;t_i) \in [0,1] & \text{if } h(t;t_i) = u \end{cases}$$

Note that u can be seen as a risk-tolerance threshold: agents maintain deposits when crash risk stays below u and withdraw when it exceeds it. Determining $h(t;t_i)$ requires solving for equilibrium, which I turn to next.

4 Analysis: Run Equilibria and Evolution of Beliefs

Understanding how individual withdrawals aggregate into runs requires establishing when each agent withdraws, which in turn requires characterizing how agents' beliefs about crash risk evolve. Because beliefs are endogenous to others' strategies, their evolution is inherently complex. I ensure tractability by focusing on *stationary* equilibria, where behavior depends only on time since learning rather than calendar time. This restriction yields closed-form expressions for hazard rates and a simple characterization of equilibrium as a system of three equations.

4.1 Stationary Equilibrium

Stationary equilibria are Perfect Bayesian Equilibria where strategies—when and how much to withdraw, or redeposit—are independent of calendar time, as formalized below.

Definition 1. *A* ***stationary equilibrium*** *is a Perfect Bayesian Equilibrium where each agent's holdings $\alpha(t;t_i)$:*

(i) depend only on elapsed time since learning, $\tau = t - t_i$, if the agent has learned ($t_i \neq \varnothing$);

(ii) is a constant α_{un} for $t \geq \eta$, if the agent has not yet learned ($t_i = \varnothing$).

Stationary equilibria with sequential awareness were first introduced by Abreu and Brunnermeier (2003) and adopted by He and Manela (2016). Stationary equilibria follow naturally from the memoryless property of the initial shock t_0; they prove particularly tractable because strategies depend simply on a single variable and are symmetric in the learning type t_i. Throughout the paper, I will restrict analysis to realizations of the shock $t_0 \geq \eta$, so that belief supports are time-invariant at the moment agents learn, ensuring stationarity is well-defined.

An immediate property of stationary equilibria is that (when the bank is fragile) the crash occurs at a fixed (endogenous) interval ξ after t_0. To see why, define (net) aggregate withdrawals at time t, given a shock at t_0, as $AW(t;t_0) = \int_{t_0}^{t} \alpha(t;t_i)\, dG(t_i - t_0)$. This expression integrates each depositor's holdings at t, $\alpha(t;t_i)$ of all agents, weighted by the density of agents who learned at each time $t_i \in [t_0, t]$. The collapse happens if and when $AW(t;t_0)$ reaches the threshold κ.

In a stationary equilibrium, agents' optimal holdings depend only on $\tau = t - t_i$. Substituting into the expression for aggregate withdrawals and letting $\xi = t - t_0$ denote time since the shock, we obtain $AW(\xi) = \int_0^\xi \alpha(\xi - s)\, dG(s)$. The crash time is therefore characterized by $\xi^* = \inf\{\xi : AW(\xi) = \kappa\}$, which guarantees that the crash happens after a fixed amount of time, at $t_0 + \xi^*$.

4.2 After Learning

Agents face two distinct decisions: the *running decision*—whether to withdraw after learning about a potential shock—and the *participation decision*—whether to hold deposits before learning. I begin with the running decision, taking participation as given. Focusing on stationary equilibria allows a sharp characterization of how hazard rates evolve after learning. Section 4.3 returns to participation, establishing when uninformed agents find it optimal to hold deposits in the first place.

Lemma 2. *Suppose agents' strategies and beliefs form a stationary equilibrium such that crash happens at $t_0 + \xi^*$. Consider $h(t_i + \tau; t_i)$, the hazard rate of collapse for an agent who learned at time t_i and evaluates it at τ periods after learning. The following properties hold:*

- ***Stationarity:*** *$h(t_i + \tau; t_i)$ is a function of τ only, i.e., it suffices to write $h(\tau) := h(t_i + \tau; t_i)$.*
- ***Decomposition:*** *$h(\tau) = \pi(\tau) h_f(\tau)$ where:*
  - *$h(\tau)$ is unimodal.*
  - *$\pi(\tau)$ is the posterior belief that the bank is fragile, and is decreasing in τ.*
  - *$h_f(\tau)$ is the posterior hazard rate of collapse of a fragile bank (which collapses with probability 1 if κ is reached), and is increasing in τ.*
  - *We have:* $\begin{cases} d\pi(\tau) = -(1-\pi(\tau))\pi(\tau) h_f(\tau) d\tau, \\ dh_f(\tau) = \left(h_f(\tau) - \left(\frac{G''(\xi^*-\tau)}{G'(\xi^*-\tau)} + \lambda\right)\right) h_f(\tau) d\tau. \end{cases}$

Proof. See Appendix A. □

Lemma 2 shows that in stationary equilibrium, hazard rates themselves depend only on the time since learning. Intuitively, if the crash occurs at $t_0 + \xi^*$, agents need only estimate t_0. Consider two agents who learn at different calendar times. The memoryless property of the exponential distribution ensures they update their estimate of t_0 identically. An agent's learning time reveals nothing about their position in the sequential awareness process. Consequently, all informed agents share the same initial hazard rate, regardless of when they learn. As time passes, the only new information—that no crash has yet occurred—is common knowledge, causing them to update beliefs identically and maintain synchronized hazard rates over time.

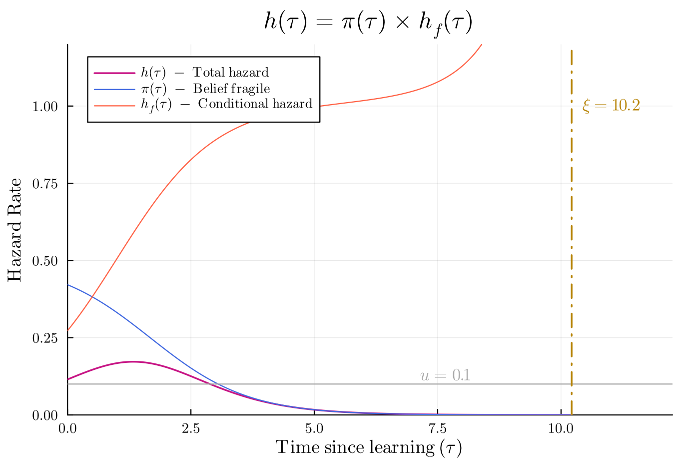


Figure 3: Hazard rate decomposition into posterior fragility $\pi(\tau)$ and conditional hazard $h_f(\tau)$.

Figure 3 illustrates the dynamics captured by Lemma 2.[10] The total hazard rate $h(\tau)$ (purple curve) exhibits a characteristic hump shape. The decomposition $h(\tau) = \pi(\tau)h_f(\tau)$ reveals two opposing forces driving this pattern. The posterior belief that the bank is fragile ($\pi(\tau)$, blue curve) decreases monotonically as agents update based on survival. Each period without a crash provides evidence against fragility, with the strength of this evidence proportional to how likely a crash would have been had the bank been fragile (captured by h_f). The differential equation $d\pi = -(1-\pi)\pi h_f d\tau$ has logistic structure, making updates fastest when uncertainty is highest (π near $1/2$) and slowest when beliefs approach certainty.

The dynamics of the conditional hazard h_f are best interpreted viewed backwards from the moment of collapse ξ (marked by the golden bar). In reversed time, h_f follows a logistic curve strictly *above* a time-varying capacity $\lambda + G''/G'$: it starts at infinity—reflecting imminent crash—and decays toward this floor as we recede from the deadline. The baseline component λ is the fundamental shock arrival rate; the correction G''/G' captures the curvature of information diffusion, raising the effective risk floor when diffusion accelerates and lowering it when diffusion decelerates.

The interaction of these forces produces the unimodal total hazard rate. Early on, rapid increases in h_f dominate gradual declines in π, causing h to rise. Later, as survival evidence mounts and learning saturates, declining π dominates, causing h to fall. Since the hazard rate crosses the threshold u at most twice—once rising, once falling—agents optimally exit and reenter deposits once, as formalized in the following corollary.

Corollary 1. *In a stationary equilibrium with crash at $t_0 + \xi$, agents optimally exit at time τ_O and optimally reenter at time τ_I, with $\tau_O, \tau_I \in [0, \xi]$. We have $\tau_O < \tau_I$, $\tau_O = \inf\{\tau : h(\tau) > u\}$ and $\tau_I = \sup\{\tau : h(\tau) > u\}$.*

If $h(\tau) < u$ for all $\tau \in [0, \xi]$, the agents never get out, and we write by convention $\tau_O = \tau_I = \xi$.

[10] All figures (excluding figures on empirical results in Section 2 and Section 6) correspond to equilibrium quantities computed numerically. See Appendix C for details on the computational implementation. Baseline parameter values and figure-specific parameters are documented in Tables 6 and 7.

4.3 Before Learning

With the run dynamics in hand, I now turn to the participation decision: under what conditions do uninformed agents willingly hold deposits? If runs are possible, agents may consider that depositing in the bank is too risky, even *before* they learn about any shock.

Lemma 3. *Given an equilibrium conjecture ξ^*, consider an uninformed agent at time $t \geq \eta$. The following properties hold:*

- ***Stationarity:*** *The uninformed hazard rate $h_{un}(\xi^*)$ depends only on the conjectured crash time, not on calendar time.*
- ***Participation condition:*** *The agent optimally holds deposits if and only if $u > h_{un}(\xi^*)$.*

Proof. See Appendix A. □

The parallel with Lemma 2 is exact: stationarity again follows from the memoryless property of the exponential prior for t_0. The participation condition mirrors the informed agent's bang-bang rule (Lemma 1)—since the uninformed hazard rate is constant, the decision reduces to a single threshold comparison.

The key difference from the informed case lies in the Bayesian updating. Uninformed agents condition on *not having learned*, so their posterior weights the survival probability $1-G(t-t_0)$ rather than the density $g(t-t_0)$ that appears for informed agents.[11] Note that the participation decision depends on the equilibrium that *would* obtain if agents participated, which is potentially off-path.

4.4 Characterization of Stationary Equilibria

The simplicity introduced by stationary equilibrium allows a clear characterization of equilibrium through a concise three-equation system, formalized in Proposition 1.

Proposition 1. *A stationary equilibrium comprises a learning distribution G, optimal exit time τ_O^*, reentry time τ_I^*, and crash time ξ^* satisfying:*

- ***Participation condition:*** $u > h_{un}(\xi^*)$
- ***Learning dynamics:*** $dG = (1-G)G\beta dt$
- ***Exit and reentry conditions:*** $\tau_O^* = \inf_{[0,\xi]}\{\tau : h(\tau) > u\}, \quad \tau_I^* = \sup_{[0,\xi]}\{\tau : h(\tau) > u\}$
- ***Crash condition:*** $\xi^* = \inf\{\xi : G(\xi - \tau_O^*) - G(\xi - \tau_I^*) = \kappa\}$

[11] The restriction $t \geq \eta$ ensures stationarity of the uninformed agent's beliefs. For $t < \eta$, the belief support is time-dependent—the agent can infer that any shock must be recent—and the hazard rate increases from zero toward h_{un}. When $u > h_{un}$, this transient is harmless: agents hold deposits throughout. When $u \leq h_{un}$, a backward unraveling argument shows that non-participation extends to $t = 0$ in equilibrium (see the proof of Proposition 2, Part 3(c)).

The exit and reentry conditions follow from Corollary 1. The crash condition exploits the single-exit, single-reentry structure: aggregate withdrawals equal the mass of agents who have exited but not yet reentered, which is simply $G(\xi - \tau_O^*) - G(\xi - \tau_I^*)$: the difference between the curves for exit and reentry, evaluated at crash time ξ^*.

The characterization exhibits a structure common to macroeconomic and mean-field game frameworks: (1) individual optimality conditions—the exit and reentry rules from Lemma 1 and the participation condition from Lemma 3; (2) forward dynamics governing aggregate state variables—the logistic learning process that determines how information spreads through the depositor network; and (3) an aggregate consistency requirement. This system of equations can be solved numerically to compute equilibrium quantities (see Appendix C).

The system admits two types of equilibria. A "no-run" equilibrium occurs when $\xi^* = \infty$: the hazard rate never exceeds u, so agents never withdraw, and the bank survives (justifying the flat hazard rate at 0). Such an equilibrium always exists (agents can always ignore the rumor). Conversely, a "run" equilibrium occurs when $\xi^* < \infty$: agents do withdraw, and the bank crashes at $t_0 + \xi^*$ if genuinely fragile. The paper is concerned with whether run equilibria exist. I show that this depends on whether deposit utility (u) is sufficiently low and communication speed (β) sufficiently high to make collective withdrawals self-fulfilling. I characterize these existence conditions precisely in Section 5.

A notable feature of run equilibria concerns the timing of withdrawals. Since the crash condition and the hazard rates[12] depend only on time differences ($\xi^* - \tau_O^*$ and $\xi^* - \tau_I^*$), the system is underdetermined in the level of τ_O^*. For generic parameter values, this forces a corner solution: agents withdraw immediately upon learning.

Corollary 2. *For generic parameter sets, any existing run equilibrium involves immediate withdrawal upon learning ($\tau_O^* = 0$).*

The intuition is a feedback loop. If $h(0) < u$, an agent prefers to wait $\tau > 0$ periods before withdrawing. By symmetry, all agents delay similarly, pushing the crash from ξ to $\xi + \tau$. Since shifting the crash shifts the entire hazard schedule by the same amount (Appendix A),[13] agents now want to wait 2τ, and so on—waiting breeds more waiting, and the system unravels to $\xi^* = \infty$. The resulting no-run equilibrium resembles a bubble: agents keep deposits even if they believe the bank is fragile, speculating that others will do the same.

The non-generic knife-edge case occurs if parameters make agents indifferent at $\tau = 0$. Then the hazard-shifting logic above produces renewed indifference at every horizon, generating a continuum of run equilibria at arbitrary delays. The *shape* of these runs—the time between exit and reentry—is identical to the generic case. Because this knife-edge requires a precise parameter alignment that vanishes under perturbations, I focus throughout on $h(0) > u$, forcing immediate withdrawals.

[12] See Appendix A.

[13] This is because agents care about how far away they estimate being from the crash.

5 Flattening the Curve

Runs exhibit a characteristic hump-shaped pattern of withdrawals—first rising, then peaking, and finally falling if the bank survives. As in epidemic control, spreading out withdrawals can "flatten" this curve, reducing peak stress and potentially preventing crashes altogether if the peak stays within the bank capacity.

5.1 The Anatomy of a Run

Figure 4 displays typical run dynamics. Consider $G(t)$, the cumulative mass of informed agents, represented in red with a dotted line. Since agents withdraw immediately upon learning (Corollary 2), this curve also tracks the mass of agents who have withdrawn. The blue curve is $G(t - \tau_I)$, the mass of agents who have reentered after initially withdrawing. Visually, it is simply $G(t)$ shifted by τ_I. The vertical distance between these curves equals aggregate withdrawals $AW(t) = G(t) - G(t - \tau_I)$ at each instant, in solid red (for $t \leq \tau_I$, no agents have yet reentered, so $AW(t) = G(t)$). The crash happens whenever AW crosses the horizontal line marking κ. For a fragile bank, the trajectory stops at crash time ξ^*; the continuation beyond this point shows the recovery dynamics that would occur for a healthy bank that survives the peak.

The hump arises because exits initially outpace reentries, then reverse as early exiters become reassured the rumor was false. Figure 5a and Figure 5b show that faster communication produces a sharper, taller peak, while lower deposit utility widens the gap between exit and reentry curves (agents must wait longer for the hazard to fall below u).

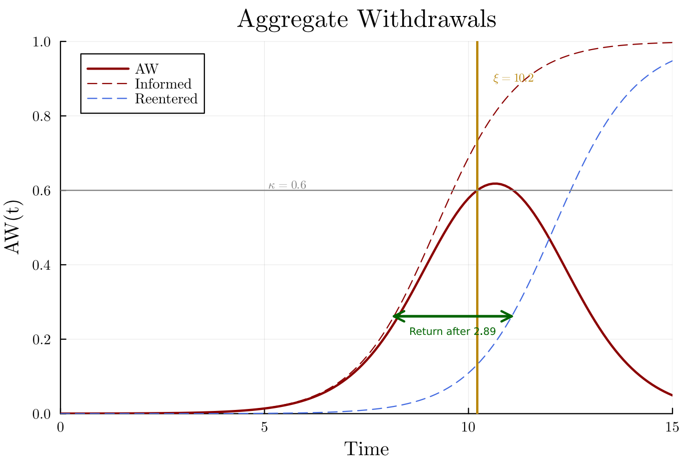
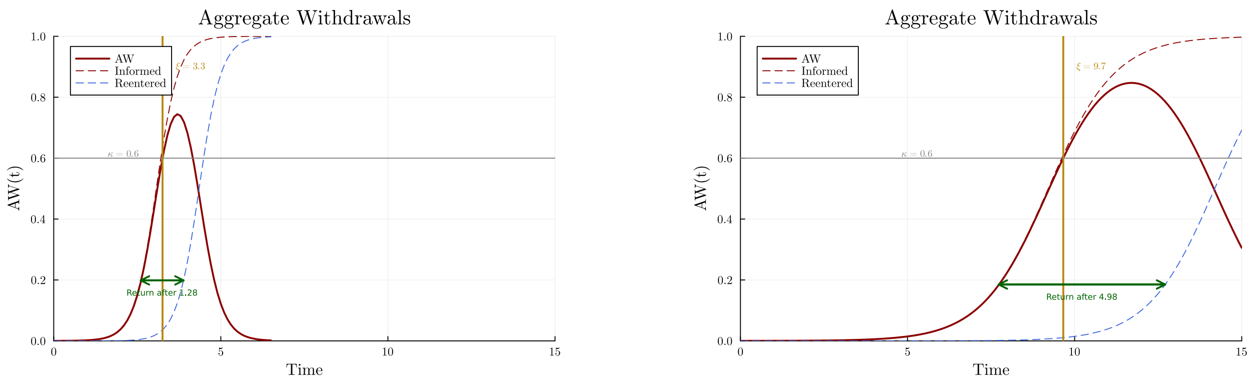


Figure 4: Dynamics of a run equilibrium.

5.2 Reentry Time

The dynamics of runs mirror those of an epidemic. Peak stress depends on two forces: how fast rumors spread (the "contact rate" β) and how long agents remain withdrawn (the



(a) Fast communication

(b) Low deposit utility

Figure 5: Dynamics under alternative contexts.

"sickness period" τ_I).[14] But unlike standard epidemiological models, the sickness period is endogenous: agents choose when to reenter based on their assessment of crash risk and the cost of foregoing deposit utility. The following lemma formalizes how parameter changes affect this duration.

Lemma 4 (Reentry Time Comparative Statics)**.** *Consider a stationary run equilibrium and assume it continues to exist in a neighborhood of the parameters. Then, reentry time τ_I^* is locally:*

- *non-increasing with respect to increases in u and η.*
- *non-decreasing with respect to increases in p.*

Furthermore, if $\tau_I^* < \xi^*$ then the above relationships hold strictly: τ_I^* is strictly decreasing in u, η and strictly increasing in p.

Proof. See Appendix A. The Appendix also discusses parameter restrictions ensuring $\tau_I^* < \xi^*$. □

Higher deposit utility or stronger fundamentals (lower p) shorten the withdrawal window τ_I, reducing the time during which agents are simultaneously withdrawn and thereby lowering peak stress.

5.3 How to Kill Runs

Can we reduce the peak sufficiently to prevent runs altogether? The answer is yes—there exists a threshold level of deposit utility above which run equilibria cease to exist. Recall that a no-run equilibrium always exists (agents can ignore the rumor). The proposition

[14] The analogy with epidemiology extends further: longer withdrawal durations stress bank balance sheets like extended patient stays stress hospital capacity. Standard epidemiology is also concerned with how long agents are *contagious*. In my model, agents are contagious forever, because informed agents always pass on the rumor, even after reentry. However, in the social learning extension (Section 7), agents are contagious only while withdrawn, creating richer dynamics.

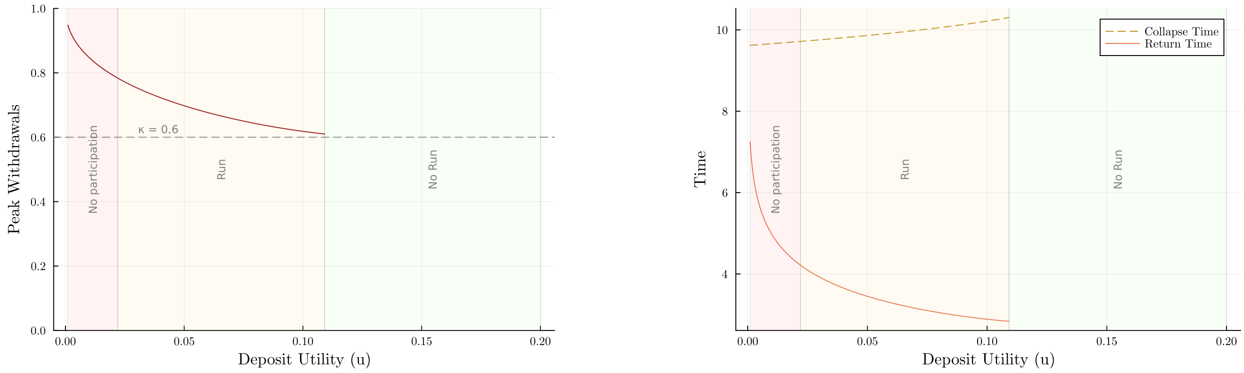


Figure 6: Effect of deposit utility on withdrawals and collapse.

below characterizes when a run equilibrium additionally exists and its consequences for participation.

Proposition 2. *Given parameters (β, κ, p, η), there exist thresholds $\underline{u} \leq \bar{u}$ such that:*

- ***No participation****: If $u \leq \underline{u}$, no run equilibrium with participation exists.*
- ***Participation with runs****: If $u \in (\underline{u}, \bar{u})$, agents participate and a unique run equilibrium exists.*
- ***No runs****: If $u \geq \bar{u}$, agents participate but no run equilibrium exists.*

Additionally, the peak withdrawal level $\bar{AW} = \max_t AW(t)$ is non-increasing in u for $u \in (\underline{u}, \bar{u})$.

Proof. See Appendix A. □

The run threshold $\bar{u}$ is the central object: as u increases, agents reenter sooner, lowering peak withdrawals until, at $\bar{u}$, the peak falls below κ and the run equilibrium unravels. The participation threshold $\underline{u}$ captures the converse: below it, the implied run is so severe that even uninformed agents find deposits too risky to hold.

Figure 6 illustrates these dynamics. Panel (a) shows that peak withdrawals decrease monotonically in u, approaching κ from above and vanishing at threshold $\bar{u}$. Panel (b) shows that crash time ξ^* increases with u as agents reenter sooner, slowing withdrawal accumulation. In both panels, the curves are also shown in the no-participation region ($u < \underline{u}$), showing the counterfactual run dynamics if agents were forced to participate.

From a normative perspective, Proposition 2 suggests that increasing deposit attractiveness—through higher interest rates[15] or improving deposit convenience—can prevent runs altogether. The next subsection establishes that the effectiveness of such policies depends critically on communication speed.

[15]Section 7.2 extends the model to interest-bearing deposits ($r > 0$). The interest rate r plays a similar role to the utility flow u: higher r induces earlier reentry and reduces peak withdrawals.

5.4 The Role of Depositor Connectedness

Depositor connectedness β affects run dynamics first through a direct channel—faster communication steepens the learning curve G, concentrating withdrawals in a shorter time window. However, faster communication also alters the hazard rate h that agents form, potentially changing their reentry decisions. Piecing together effects from h and the direct effect in the shape AW is not obvious. Fortunately, the model exhibits a useful scaling property that allows studying the effect of β analytically.

The key observation is that β rescales time in the model. To see this, note that all time derivatives in the equilibrium system are proportional to β. If we transform time by $t \to t/\beta$, then the differential equations remain unchanged except that $\beta \to 1$, $u \to u/\beta$, and $\lambda \to \lambda/\beta$. Since $\eta := G^{-1}(1-\varepsilon)$ represents the time to reach awareness saturation, it scales inversely with β. Specifically, in the non-scaled system there exists a $\bar{\eta}$ such that $\eta = \bar{\eta}/\beta$. Under the time transformation, $\eta \to \bar{\eta}$, which is independent of β, ensuring the model structure remains intact.

Increasing β is therefore equivalent to jointly reducing deposit utility and the prior distribution for t_0. Under Assumption 1, that is, when shocks arrive more rarely than information diffuses, the first effect dominates.

Corollary 3 (Fragility Regimes)**.** *Assume $\kappa\bar{\eta} + u/\lambda \geq 1$. There exist thresholds $\underline{\beta} \leq \bar{\beta}$ such that:*

- ***No runs**: If $\beta \leq \underline{\beta}$, agents participate but no run equilibrium exists.*
- ***Participation with runs**: If $\beta \in (\underline{\beta}, \bar{\beta})$, agents participate and a unique run equilibrium exists.*
- ***No participation**: If $\beta \geq \bar{\beta}$, no run equilibrium with participation exists.*

Additionally, peak withdrawals $\bar{AW} = \max_t AW(t)$ are non-decreasing in β for $\beta \in (\underline{\beta}, \bar{\beta})$.

Proof. See Appendix A. □

Corollary 3 establishes a regime structure governed by communication speed. Below $\underline{\beta}$, coordination frictions prevent runs. Above it, withdrawals coordinate into crashes with peak stress increasing in β. At the extreme ($\beta \geq \bar{\beta}$), crashes unfold so fast that uninformed agents have little hope of learning in time to withdraw, making deposits unattractive from the start—the run equilibrium destroys itself as in Proposition 2. This aligns with Section 2: banks with high HHI (a proxy for high β) experienced substantially larger stock price declines during the 2023 crisis.

The condition $\kappa\bar{\eta} + u/\lambda \geq 1$ is mild (it holds whenever the awareness window is long enough or the shock arrival rate is slow enough). It is needed only for the participation threshold $\bar{\beta}$; the run threshold $\underline{\beta}$ and monotonicity of peak withdrawals hold without it.[16]

The fact that deposit utility becomes u/β under rescaling has an interesting normative implication: the effectiveness of deposit-enhancing policies depends critically on coordination

[16]Together with $\eta \leq 1/\lambda$ (Assumption 1), the condition guarantees monotonicity of uninformed participation in β.

frictions. The threshold $\bar{u}$ required to prevent runs scales roughly linearly with β—exactly linear if we ignore the λ/β term in the rescaling: when coordination frictions are twice as low (higher β, faster communication), deposits must be made twice as attractive to achieve the same run-prevention effect.

Figure 7 illustrates this interaction through a heatmap of peak withdrawals over $(u, 1/\beta)$. The run-prevention frontier is approximately hyperbolic, validating the linear $\bar{u} \propto \beta$ scaling. The implication for bank risk management is direct: banks with concentrated, well-connected depositor bases must offer substantially higher yields to prevent runs, facing a sharper stability-profitability tradeoff than banks with dispersed depositors.

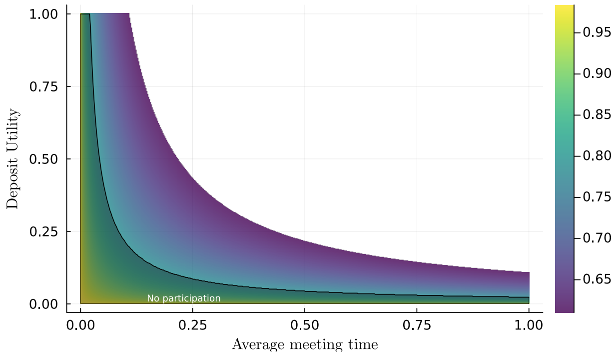


Figure 7: Interaction of u and $1/\beta$ on the peak of withdrawals.

6 Fitting the Model to Crisis Dynamics

In this section I illustrate the model predictions by fitting the closed-form expression for aggregate withdrawals to intraday stock returns during the crisis period, treating stock prices as a proxy for deposit dynamics. The exercise demonstrates the model's tractability: the closed-form solution makes it feasible to recover structural parameters from data.

6.1 Methodology

For each bank i, I compute high-frequency cumulative returns relative to the market open on March 6, 2023:

$$R_{it} = \frac{P_{it} - P_{i,0}}{P_{i,0}},$$

where P_{it} is the price of bank i stock at time t. I use 10-minute price data smoothed with a 1-hour rolling window.[17] The sample window extends through March 21, capturing both the crisis and early recovery.[18]

[17] See Appendix B.5 for details on data source and construction.

[18] Stock returns correlate strongly with deposit outflows during the run phase (Cipriani et al. 2024), though less so during recovery. Moreover, stock prices are publicly observable, whereas my model assumes agents cannot see aggregate withdrawals in real time. The exercise is therefore best interpreted as illustrating the model's quantitative tractability rather than providing a definitive empirical test.

Treating cumulative returns as a proxy for aggregate withdrawals, I fit the model-implied withdrawal path:

$$R_{it} = \underbrace{\frac{1}{1+e^{-\beta_i(t-t_{\text{mid},i})}}}_{\text{exits } G(t)} - \underbrace{\frac{1}{1+e^{-\beta_i(t-\tau_{I,i}-t_{\text{mid},i})}}}_{\text{reentries } G(t-\tau_I)}.$$

The right-hand side is the equilibrium aggregate withdrawals $G(t) - G(t-\tau_I)$, using the closed-form logistic for G. The midpoint $t_{\text{mid},i}$ captures when the shock reached half the population. I estimate three parameters per bank via nonlinear least squares: communication speed β_i, reentry time $\tau_{I,i}$, and timing $t_{\text{mid},i}$.

Note that the communication speed β is an exogenous structural parameter, determined by the depositor network's infrastructure. In contrast, the reentry time τ_I is an endogenous equilibrium outcome—a model-implied parameter that depends on exogenous primitives (β, p, u, etc.) through equilibrium actions. My estimation strategy treats both as parameters in the reduced-form equation, then examines how they vary with observable bank characteristics.

6.2 Fit

Figure 8 presents model fits for six representative banks spanning different crisis experiences. Each panel plots daily closing prices, smoothed intraday prices, and the fitted model curve. The model captures both rapid declines (SVB, PacWest) and gradual dynamics (Citi, US Bank) reasonably well, achieving a median $R^2 = 0.822$.[19] The heterogeneity arises naturally from variation in the estimated parameters: banks experiencing severe crises have higher estimated β (faster information spread) and higher τ_I (longer withdrawal periods).

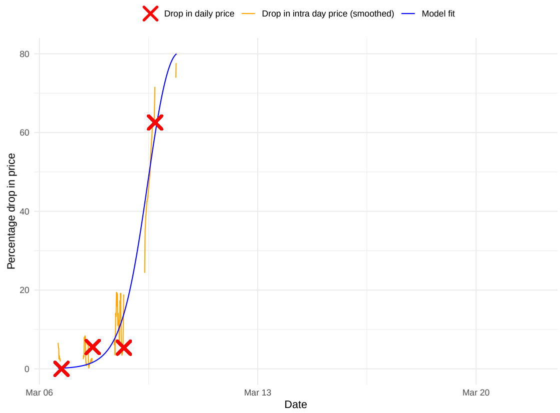
6.3 Analysis of Estimated Parameters

Using the estimated model parameters, I examine whether these model-implied measures correlate with the exogenous proxies for fundamental risk and coordination frictions introduced in Section 2.

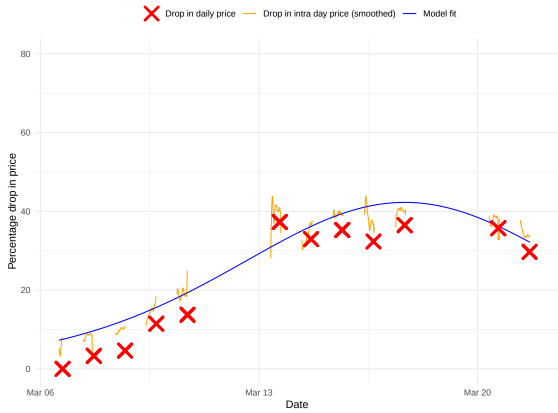
I first examine whether depositor concentration proxies for communication speed. If the HHI measure captures information diffusion rates as the model suggests, one should observe a positive correlation between HHI and estimated $\hat{\beta}$. Table 2 reports the results. Columns (2) to (4) show that HHI is indeed strongly positively correlated with $\hat{\beta}$, supporting the interpretation that depositor concentration facilitates information transmission.

Interestingly, κ also seems to play a role. Safer banks (higher κ) exhibit slower estimated $\hat{\beta}$. This is somewhat puzzling from the model's perspective, since κ is a measure of fundamental fragility rather than network structure, and should not directly affect communication speed. The correlation may reflect omitted factors—for instance, banks with riskier business models might also attract depositors who are better connected through industry networks. Alternatively, it could indicate that information spread depends on depositors' behavior, such as through social learning mechanisms discussed in Section 7.

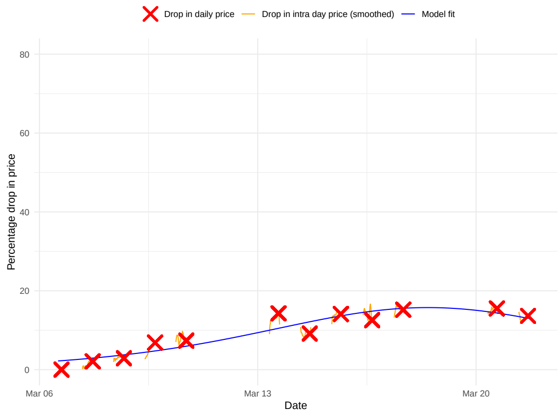
[19] Figure 22 in Appendix B presents the full distribution of R^2 values and additional fit examples.



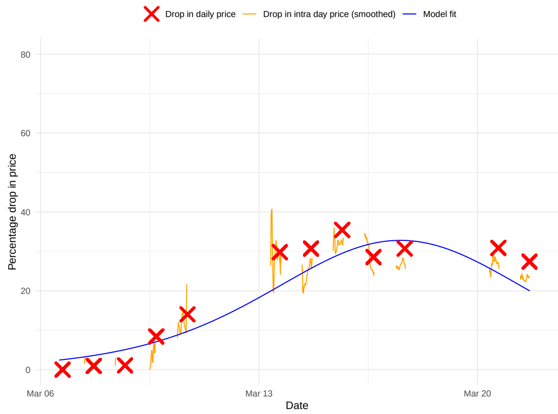
(a) SVB



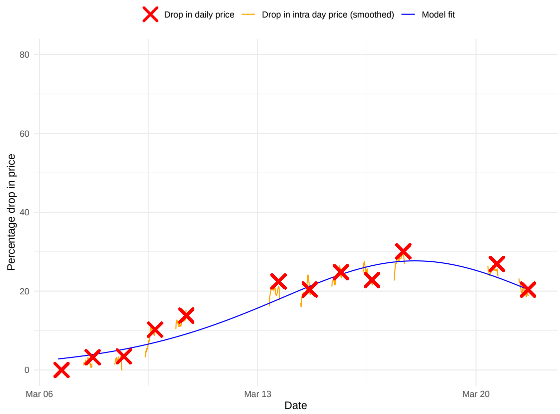
(b) KeyCorp



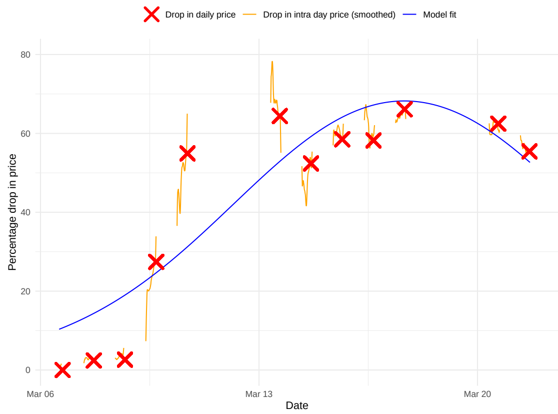
(c) Citi



(d) Bank of Hawaii



(e) US Bank



(f) PacWest

Figure 8: Stock prices and model fit for selected banks.

Dependent Variable:	**Estimated β**			
	(1)	(2)	(3)	(4)
Solvency κ	-0.127***		-0.123***	
	(0.034)		(0.031)	
Depositor HHI		0.475***	0.467***	0.058
		(0.087)	(0.084)	(0.097)
Indicator ($\kappa < 1$)				-0.061**
				(0.024)
$\kappa < 1\times$ Depositor HHI				1.041***
				(0.151)
Uninsured Share	0.411***	0.271***	0.269***	0.214**
	(0.096)	(0.096)	(0.092)	(0.084)
Dom. Deposits (log)	0.006	0.005	0.013	0.007
	(0.010)	(0.009)	(0.009)	(0.008)
Wholesale sh. of dep.	-0.414	-0.578**	-0.426	-0.509**
	(0.302)	(0.287)	(0.278)	(0.254)
Constant	0.153	0.048	0.044	0.056
	(0.143)	(0.139)	(0.133)	(0.125)
Observations	172	172	172	172
R^2	0.191	0.254	0.318	0.433

*Note: *** p<0.01, ** p<0.05, * p<0.1.*

Table 2: Relationship of estimated β with bank characteristics.

The reentry time τ_I presents a different pattern. Table 3 shows that κ is strongly negatively correlated with $\hat{\tau}_I$: safer banks exhibit shorter withdrawal durations. While higher κ mechanically extends run duration in the model (more capacity to absorb), the empirical result is better interpreted through the probability channel: high-solvency banks are perceived as less likely to be fragile (lower p), which by Lemma 4 accelerates belief updating and induces earlier reentry.

The coefficient of HHI on $\hat{\tau}_I$ is weakly positive, reaching marginal significance in some specifications. As discussed in Appendix A (Remark 1), the theoretical effect of connectedness on physical reentry time is ambiguous: while higher β increases reentry time in scaled units, translating to physical time shrinks it back. The weak and less robust HHI coefficient on $\hat{\tau}_I$—in contrast with its strong effect on $\hat{\beta}$—is therefore consistent with these partially offsetting forces.

7 Extensions

This section extends the baseline model in three directions. First, heterogeneity in learning speeds allows some depositors to be better connected than others. Second, deposits may accrue interest at rate $r > 0$. Third, agents learn by observing others' withdrawal behavior rather than through word-of-mouth communication.

Throughout, I focus on run dynamics assuming participation, although the analysis could easily be extended to characterize the no-participation regions. The results obtained in this section are numerical. I refer to Appendix C for a discussion of the numerical implementation.

Dependent Variable:	**Estimated τ**			
	(1)	(2)	(3)	(4)
Solvency κ	-0.970***		-0.955***	
	(0.318)		(0.315)	
Depositor HHI		1.720**	1.658*	1.227
		(0.868)	(0.848)	(1.083)
Indicator ($\kappa < 1$)				0.418
				(0.269)
$\kappa < 1\times$ Depositor HHI				0.919
				(1.688)
Uninsured Share	1.574*	1.087	1.069	0.944
	(0.899)	(0.951)	(0.928)	(0.944)
Dom. Deposits (log)	0.410***	0.376***	0.437***	0.410***
	(0.092)	(0.092)	(0.092)	(0.092)
Wholesale sh. of dep.	4.499	3.275	4.457	4.060
	(2.836)	(2.853)	(2.813)	(2.838)
Constant	-3.518***	-3.873***	-3.908***	-4.580***
	(1.340)	(1.376)	(1.344)	(1.403)
Observations	172	172	172	172
R^2	0.211	0.186	0.229	0.215

*Note: *** p<0.01, ** p<0.05, * p<0.1.*

Table 3: Relationship of estimated τ_I with bank characteristics.

7.1 Heterogeneity in Learning Speed

The baseline model assumes all agents learn at the same rate β. In practice, depositors vary in how connected they are to information networks. Some depositors—such as large institutional investors or those active on social media—learn quickly about rumors, while others learn more slowly. The effects of such heterogeneity are not obvious, as it affects both individual withdrawal strategies and aggregate run dynamics.

I extend the model by dividing agents into K types indexed by k, each learning at rate β_k. Type k comprises a fraction p_k of the population. The learning dynamics for each type are:

$$dG_k(t) = (1 - G_k(t)) \times \omega(t) \times \beta_k \, dt,$$

where $\omega(t) = \sum_{k'} p_{k'} G_{k'}(t)$ is the overall fraction of informed agents. An uninformed agent of type k meets others at rate β_k, with probability $\omega(t)$ of meeting an informed agent. The special case $\beta_k = \beta \times k$ can be explicitly interpreted as a mean-field approximation of rumor spreading on an actual network as we grow the network size to infinity, k being the degree of the agent. As discussed in Newman (2018), the approximation holds when the expected degree of a randomly selected neighbor is independent of one's own degree.[20]

Faster learners become informed earlier, which accelerates information spread to slower learners, coupling the dynamics across types. Equilibria can still be characterized as before: each type has its own exit time $\tau_{O,k}$ and reentry time $\tau_{I,k}$.

Figure 9 illustrates these dynamics in a two-type model with mostly slow agents (90%)

[20] Similar approximations are possible under less restrictive assumptions. In each case, they involve classifying agents in groups based on their structural position in the network — each group would have a different learning dynamic.

and a small fraction of fast agents (10%, learning 100 times faster). Slow agents withdraw immediately upon learning, so aggregate withdrawals primarily track their gradual buildup. Fast agents delay exit and coordinate their departure at the last moment, creating a sharp spike that triggers the crash. Because fast agents can time their exit precisely while slow agents cannot, network position creates inequality: better-connected depositors face less run risk.

The figure also reveals asymmetric recovery dynamics. After the crash, slow agents continue exiting and reach their peak only later, causing aggregate withdrawals to briefly increase again before the long recovery begins—consistent with empirical evidence on protracted recoveries.

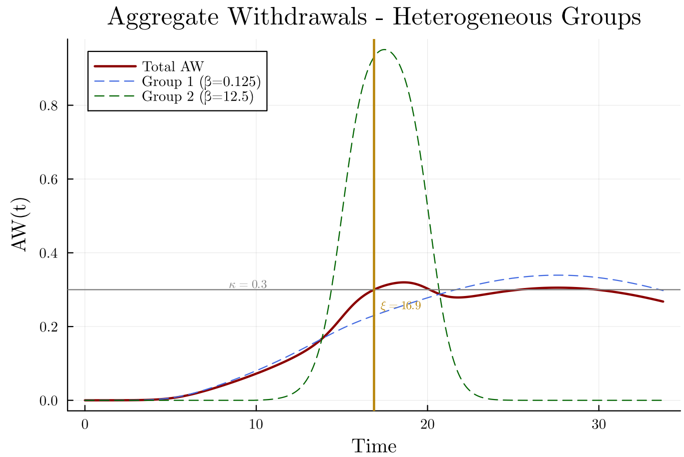


Figure 9: Equilibrium dynamics under heterogeneous learning speeds.

7.2 Interest-Bearing Deposits

Throughout the main analysis, I assumed deposits do not accrue interest, making agents' holding decisions essentially static. I now relax this assumption and show how to solve the problem when deposits earn interest at rate $r > 0$.

With wealth growing at rate r, the agent's continuation value depends on both wealth level W and time since learning τ: $V(W, \tau)$. However, utility's homothetic structure—linear in wealth—implies multiplicative separability of the value function: $V(W, \tau) = W \times V(\tau)$, where $V(\tau)$ is the value of \$1 of wealth.

To prevent V from growing unboundedly when crash risk vanishes, I introduce an exogenous maturity process: deposits mature at Poisson rate $\delta > r$, paying out \$1 upon maturity.[21]

[21] The constraint $\delta > r$ ensures that the value of deposits remains bounded. Economically, it requires that deposits mature faster than they appreciate, preventing the continuation value from diverging. Alternatively, we can think of δ as the rate of an idiosyncratic liquidity shock forcing agents to consume their wealth.

The optimal value $V(\tau)$ solves the Hamilton-Jacobi-Bellman equation:

$$0 = \dot{V}(\tau) + \max_{\alpha(\tau)} \{(u + rV(\tau))(1 - \alpha(\tau)) + h(\tau)(\alpha(\tau) - V(\tau))\} - \delta(V(\tau) - 1)$$
$$\iff \dot{V}(\tau) = (h(\tau) + \delta)(V(\tau) - 1) - \max\{u + rV(\tau) - h(\tau), 0\}.$$

At time ξ when agents are certain the bank is safe, the value equals the present value of a safe deposit: $V(\xi) = (u + \delta)/(r + \delta)$, which serves as the terminal condition for solving the HJB backward.

The tradeoff mirrors Section 4: the marginal cost of cash is now $u + rV(\tau)$, reflecting both utility flow and deposit returns, while the benefit remains $h(\tau)$—the probability of retrieving a nominal dollar upon crash, not the continuation value $V(\tau)$.

As in the main case, linearity in α implies bang-bang solutions: the agent either holds everything in deposits or withdraws everything to cash. However, $V(\tau)$ now depends on future actions and must be solved as part of the equilibrium. The optimal strategy is:

$$\alpha^*(\tau) = \begin{cases} 1 & \text{if } u + rV(\tau) < h(\tau) \\ 0 & \text{if } u + rV(\tau) > h(\tau) \end{cases}$$

The value function $V(\tau)$ inherits properties from $h(\tau)$: it is non-monotonic, reflecting the interplay between capital gains, crash risk, and maturity. Figure 10 shows the hazard rate decomposition in this extended model, in a case where $u = 0$ and deposit attractiveness comes only from returns r. The effect of r parallels that of u: higher interest rates make deposits more attractive, inducing earlier reentry and reducing peak withdrawals.

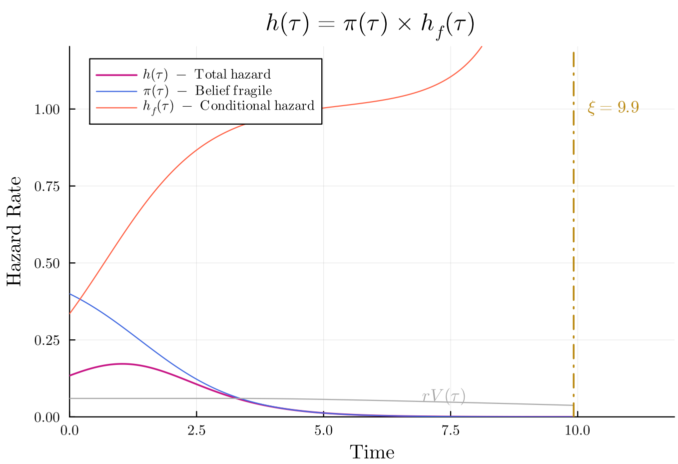


Figure 10: Hazard rate decomposition with interest rates.

7.3 Social Learning

The baseline model assumes agents learn about solvency shocks through word-of-mouth: informed agents directly communicate the rumor to those they meet. A more realistic mechanism is social learning: agents observe others' withdrawal behavior.

Under social learning, agents meet others at rate β as before, but now observe only whether the person they meet is currently withdrawn from the bank. If they observe a withdrawal, they infer a rumor may be circulating. The key difference is that the probability of an informative meeting now equals $AW(t)$, the aggregate withdrawal level, rather than $G(t)$, the fraction of informed agents. Learning dynamics thus depend on withdrawal behavior, not just information diffusion.

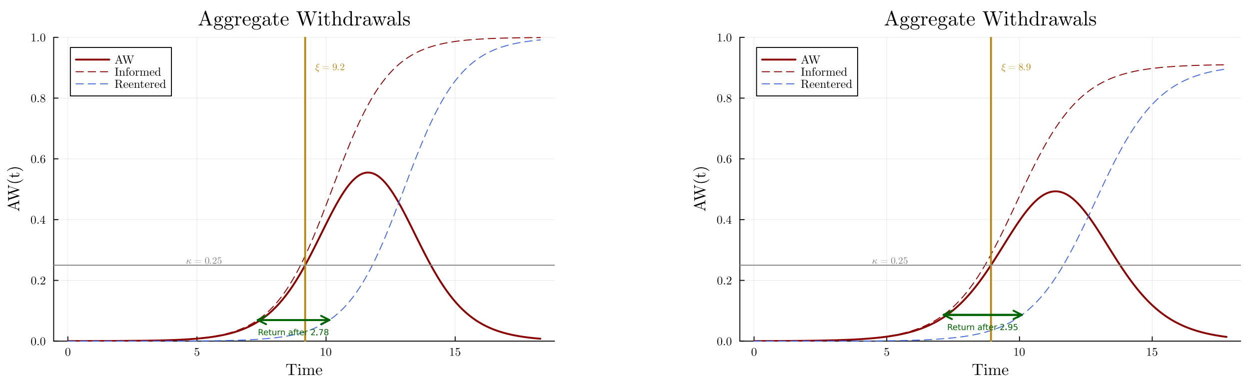
The equilibrium definition must be modified to account for this coupling. A stationary equilibrium now requires that learning and withdrawal dynamics jointly satisfy:

$$\begin{aligned} dG(t) &= (1 - G(t))AW(t)\beta\, dt \\ AW(t) &= G(t) - G(t - \tau_I), \end{aligned}$$

where the second equation uses the fact that agents still withdraw immediately upon learning and reenter at τ_I. The equilibrium is characterized by solving these coupled equations simultaneously. Appendix C explains how to solve this problem numerically.

In the baseline, agents remain informed forever, continuing to spread the rumor even after reentering. Under social learning, agents who reenter no longer contribute to information diffusion. Figure 11 illustrates this mechanism: social learning produces slower, more gradual runs because the "infectious period" is now endogenous, equaling the withdrawal duration τ_I. This creates an amplification effect: increasing u not only induces earlier reentry (reducing the peak directly) but also shortens the infectious period (reducing information spread).

The social learning mechanism offers a potential explanation for the negative correlation between fundamental fragility (κ) and estimated communication speed ($\hat{\beta}$) documented in Section 6. Fundamentally stronger banks induce earlier reentry, shortening the period during which withdrawn agents remain "infectious". This slows information diffusion. The estimated β would thus appear lower for safer banks, even though the underlying network structure (proxied by HHI) remains unchanged. Estimating this variant of the model would be more complex—albeit not infeasible—because the solution relies on numerical methods and we don't obtain the simple reduced-form equation for aggregate withdrawals that I used in the NLS estimation.



(a) Word-of-mouth learning

(b) Learning from others' actions

Figure 11: Comparison of run dynamics under different learning mechanisms.

8 Conclusion

I develop a dynamic model of bank runs where information about potential insolvency spreads gradually through depositor networks following epidemiological diffusion processes. The model shows that coordination frictions—captured through the speed of information transmission—do not merely slow runs but can prevent them entirely: sufficiently high coordination frictions imply peak withdrawals never exceed the bank's solvency capacity, even when fundamentals are genuinely fragile. Beyond the immediate context of banking, this framework applies broadly to many coordination problems in finance (and beyond), where the structure of the asset holder base determines the feasibility of collective attacks (bubbles, currency attacks, sovereign debt crises, etc.). We do observe such phenomena empirically, suggesting a fertile area for future research: Internet-based assets like cryptocurrencies or meme stocks, characterized by concentrated and highly connected holder bases, do face rapid coordinated runs and high volatility. Conversely, "safe" assets like U.S. Treasury debt may source some of their stability from a dispersed base of heterogeneous international investors where information circulates slowly. This general principle suggests that what makes an asset fragile depends not only on its intrinsic value but also on the social structure of its investors.

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A Proofs

Throughout the appendix, I assume that the likelihood intensity $g(t) = \frac{dG(t)}{dt}$ is strictly log-concave, with $g \in C^2[0,\eta]$, $g(0) > 0$, and $g'(0)$ finite. This class includes many standard diffusion models—logistic, normal, gamma densities—and in particular the logistic specification $dG = G(1-G)\beta\, dt$ used in the main text. I maintain the assumption that learning completes within time η. The proof of the results involving comparative statics in β requires making explicit a notion of "speed" in the learning process, and therefore reverts to the logistic case.

A.1 Proof of Lemma 2 (hazard rate)

This subsection presents an extended version of Lemma 2, establishing the closed-form hazard rate and its properties in reversed time $\bar{\tau} = \xi^* - \tau$. Below I define a set of notations that will be used throughout the proof:

$T_0 \sim \text{Exp}(\lambda)$:	solvency shock time (realization t_0)
T:	collapse time (∞ if solvent)
T_i:	learning time for agent i (realization t_i)
$\mathbb{1}_f$:	fragility indicator, $P(\mathbb{1}_f = 1) = p$
$\mathcal{L}_i = T_i - T_0 \in [0,\eta]$:	learning lag (CDF: G)
$h_T(t;\xi^*\|t_i)$:	hazard rate of collapse at t

In a stationary equilibrium, $T = T_0 + \xi^*$ if $\mathbb{1}_f = 1$ and $T = \infty$ otherwise.

Lemma 2 (extended). *Suppose collapse occurs at $t_0 + \xi^*$ for fixed $\xi^* > 0$. Define reversed time $\bar{\tau} = \xi^* - \tau$ where $\tau = t - t_i$.*

1. *The hazard rate depends only on τ, not on t_i. Writing $h(\tau;\xi^*) = h_T(t_i + \tau | t_i)$:*

$$h(\tau;\xi^*) = \frac{p e^{\lambda(\xi^*-\tau)} g(\xi^* - \tau)}{p\int_0^{\xi^*-\tau} e^{\lambda s} g(s)\, ds + (1-p)\int_0^{\eta} e^{\lambda s} g(s) ds}, \quad \xi^* \geq \tau; \tag{1}$$
$$h(\tau;\xi^*) = 0 \quad \textit{otherwise.}$$

2. *Under the change of variable $\bar{\tau} = \xi^* - \tau$, the reversed-time hazard $\bar{h}(\bar{\tau}) := h(\tau;\xi^*)$ depends only on $\bar{\tau}$.*

3. *$\bar{h}(\bar{\tau}) = \pi(\bar{\tau})\bar{h}_f(\bar{\tau})$ where $\pi(\bar{\tau})$ is the posterior fragility belief (non-decreasing) and $\bar{h}_f(\bar{\tau})$ the conditional hazard, with dynamics:* $\begin{cases} d\pi = \pi(1-\pi)\bar{h}_f\, d\bar{\tau}, \\ d\bar{h}_f = -\big(\bar{h}_f - \big(\frac{G''}{G'} + \lambda\big)\big)\bar{h}_f\, d\bar{\tau}. \end{cases}$

4. *$\bar{h}_f(\bar{\tau})$ is non-increasing in $\bar{\tau}$.*

5. *$\bar{h}(\bar{\tau})$ is unimodal in $\bar{\tau}$.*

A.1.1 Proof of point 1.

Step 1: crash time hazard rate. The hazard rate for collapse at time t, conditional on learning at t_i and crash happening at $T_0 + \xi^*$ is

$$h_T(t; \xi^* | T_i = t_i) = f_T(t|t_i)/S_T(t|t_i),$$

where $f_T(t|t_i)$ is the overall posterior density and $S_T(t|t_i)$ is the posterior survival function.

Using the law of total probability, the fact that fragility is independent of learning time, and the fact that $P(T \leq t; t_i, \mathbb{1}_f = 0) = 0$ for finite t, we can express:

$$h_T(t; \xi^* | T_i = t_i) = \frac{p\, f_T(t|t_i, \mathbb{1}_f = 1)}{1 - pP(T \leq t|t_i, \mathbb{1}_f = 1)}. \tag{2}$$

Step 2: shifting to the T_0 random variable. Letting $t_0 = t - \xi^*$, we can relate the required conditional density and CDF of T to those of T_0. Given the independence of the $(T_0, \mathcal{L}_i)$ process from $\mathbb{1}_f$, conditioning on $\mathbb{1}_f = 1$ does not alter the posterior distribution of T_0 given $T_i = t_i$. Therefore:

- $f_T(t_0 + \xi^* | t_i, \mathbb{1}_f = 1) = f_{T_0}(t_0 | t_i, \mathbb{1}_f = 1) = f_{T_0}(t_0|t_i)$
- $P(T \leq t_0 + \xi^* | t_i, \mathbb{1}_f = 1) = P(T_0 \leq t_0 | t_i, \mathbb{1}_f = 1) = P(T_0 \leq t_0|t_i)$

where $f_{T_0}(\cdot|t_i)$ and $P(T_0 \leq \cdot|t_i)$ are the posterior density and CDF of T_0 given $T_i = t_i$. Noting that the event $T = t = t_0 + \xi^*$ is the same as $T_0 = t - \xi^*$, it suffices to study the following quantity[22]:

$$h_{T_0}(t_0 | T_i = t_i) := \frac{p\, f_{T_0}(t_0|t_i)}{1 - pP(T_0 \leq t_0|t_i)}.$$

Step 3: computing posteriors.

The posterior density $f_{T_0}(t_0|t_i)$ can be computed using Bayes' theorem:

$$f_{T_0}(t_0|T_i = t_i) \propto f_{T_0}(t_0) \times \text{Likelihood}(T_i = t_i | T_0 = t_0).$$

Note that the event $\{T_i = t_i \text{ given } T_0 = t_0\}$ is equivalent to the event $\{T_0 + \mathcal{L}_i = t_i \text{ given } T_0 = t_0\}$, which simplifies to $\{\mathcal{L}_i = t_i - t_0\}$ (given the independence of $\mathcal{L}_i$ and T_0). Recall that $g(s)$ characterizes the likelihood intensity associated with the underlying learning process completing at lag s. The above expression can thus be rewritten as:

$$f_{T_0}(t_0|T_i = t_i) \propto f_{T_0}(t_0) g(t_i - t_0) = \lambda e^{-\lambda t_0} g(t_i - t_0).$$

Since we focus on realizations $t_i \geq \eta$, the support for t_0 (given t_i) is $[t_i - \eta, t_i]$. The normalization constant $C(t_i)$ is found by integrating the expression $\lambda e^{-\lambda t_0'} g(t_i - t_0')$ over this support:

$$\begin{aligned} C(t_i) &= \int_{t_i - \eta}^{t_i} \lambda e^{-\lambda t_0'} g(t_i - t_0') dt_0' \\ &= \lambda e^{-\lambda t_i} \int_0^{\eta} e^{\lambda s} g(s) ds, \quad \text{using substitution } s = t_i - t_0'. \end{aligned}$$

[22]The notation of a hazard rate for T_0 is a slight abuse: the event T_0 doesn't depend on bank fragility and the real hazard rate for T_0 should be written with 1 instead of p.

Let $I_\eta := \int_0^\eta e^{\lambda s} g(s) ds$ (note that this constant is independent of t_i). The posterior density of T_0 conditional on $T_i = t_i$ is[23]:

$$f_{T_0}(t_0|T_i = t_i) = \frac{\lambda e^{-\lambda t_0} g(t_i - t_0)}{C(t_i)} = \frac{\lambda e^{-\lambda t_0} g(t_i - t_0)}{\lambda e^{-\lambda t_i} I_\eta} = \frac{e^{\lambda(t_i - t_0)} g(t_i - t_0)}{I_\eta}. \tag{3}$$

This expression holds for $t_0 \in [t_i - \eta, t_i]$. For $t_0 < t_i - \eta$, the posterior density is 0. For $t_0 > t_i$, the posterior density is also 0, as it would imply a negative learning lag.

The posterior cumulative distribution function $P(T_0 \leq t_0|T_i = t_i)$ is obtained by integrating the posterior density (3) over the relevant range. After change of variable we obtain:

$$P(T_0 \leq t_0|T_i = t_i) = \frac{1}{I_\eta} \int_{t_i - t_0}^{\eta} e^{\lambda s} g(s) ds,$$

Step 4: Final formula for hazard rate. Substituting the derived $f_{T_0}(t_0|t_i)$ and $P(T_0 \leq t_0|t_i)$ into the hazard rate formula (2), evaluated at $t = t_0 + \xi^*$, we get:

$$h_{T_0}(t_0|T_i = t_i) = \frac{p\, e^{\lambda(t_i - t_0)} g(t_i - t_0)}{I_\eta - p \int_{t_i - t_0}^{\eta} e^{\lambda s} g(s) ds},$$

which can be re-expressed as:

$$\frac{p\, e^{\lambda(t_i - t_0)} g(t_i - t_0)}{p \int_0^{t_i - t_0} e^{\lambda s} g(s) ds + (1-p) \int_0^\eta e^{\lambda s} g(s) ds}. \tag{4}$$

Step 5: Time since learning.

Let $\tau = t - t_i$ be time since learning. We evaluate the hazard at $t = t_i + \tau$, which corresponds to $t_0 = t - \xi^* = t_i + \tau - \xi^*$.

$$h_T(t_i + \tau, \xi^*|t_i) = h_{T_0}(t_i + \tau - \xi^*|T_i = t_i) = \frac{p\, e^{\lambda(\xi^* - \tau)} g(\xi^* - \tau)}{p \int_0^{\xi^* - \tau} e^{\lambda s} g(s) ds + (1-p) \int_0^\eta e^{\lambda s} g(s) ds}. \tag{5}$$

A.1.2 Proof of point 2.

Immediate from closed form.

A.1.3 Proof of point 3.

Point 3 decomposes the hazard rate into belief and conditional hazard components, and derives their evolution in reversed time. I first establish this decomposition generally, then specialize to our model.

[23]Seemingly innocuous, the final equality below, where $e^{-\lambda t_i}$ is brought to the numerator, is the crucial stepping stone that will allow focusing on stationary equilibrium (as already noted in Abreu and Brunnermeier (2003)). It follows from the memoryless property of the exponential distribution and tells us that we can focus only on the time between t_0 and t_i rather than their calendar values.

Lemma 5. *Let T be a random variable representing the time of failure of a system, with positive support. Suppose the system can be of two types: fragile (with prior probability p) or fail-safe (with prior probability $1-p$). Assume the failure time is $T=\infty$ if the system is fail-safe. Let $\mathbb{1}_f$ be the indicator variable for the fragile state. Define:*

- $h(t) = \lim_{\Delta t \to 0} \frac{P(t \le T < t+\Delta t | T \ge t)}{\Delta t}$ *is the overall hazard rate.*
- $\pi(t) = P(\mathbb{1}_f = 1 \mid T \ge t)$ *is the posterior probability of being fragile given survival up to t.*
- $h_f(t) = \lim_{\Delta t \to 0} \frac{P(t \le T < t+\Delta t | T \ge t, \mathbb{1}_f = 1)}{\Delta t}$ *is the hazard rate conditional on being fragile.*

The following properties hold:

1. $h(t) = \pi(t) h_f(t)$.
2. *Furthermore, if the failure time distribution for the fragile type admits a density then:*

$$\frac{d\pi(t)}{dt} = -\pi(t)(1-\pi(t))h_f(t).$$

Proof. Claim 1 follows from the law of total probability (the fail-safe term vanishes since $T=\infty$ when $\mathbb{1}_f = 0$). For Claim 2, write $\pi(t) = pS_f(t)/[pS_f(t) + (1-p)]$ by Bayes' theorem and differentiate. □

I now turn to the proof of point 3. Applying the lemma to the hazard rate obtained in the proof of point 1, we have:

$$h(t|t_i) = \pi(t|t_i) h_f(t|t_i) \tag{6}$$

and

$$\frac{d\pi(t|t_i)}{dt} = -\pi(t|t_i)(1-\pi(t|t_i))h_f(t|t_i). \tag{7}$$

We need to show that the decomposition holds in the single-variable reversed time formulation, i.e., using $\bar{h}(\bar{\tau})$.

Step 1: independence of learning time. Note that the conditional hazard rate $h_f(t|t_i)$ corresponds to the scenario where fragility is certain ($p=1$). Setting $p=1$ in the derivation for $h(t_i+\tau|t_i)$ (specifically, applying it to the structure derived in (4)), we see that the resulting expression for $h_f(t_i+\tau|t_i)$ depends only on τ. We can write $h_f(\tau) = h_f(t_i+\tau|t_i)$. From (6), since $h(\tau)$ and $h_f(\tau)$ depend only on τ, it follows that $\pi(t_i+\tau|t_i) = h(\tau)/h_f(\tau)$ must also depend only on τ. We write $\pi(\tau) = \pi(t_i+\tau|t_i)$.

Define $\bar{\tau} = \xi^* - \tau$ and $\bar{h}(\bar{\tau}) = h(\tau)$, $\bar{\pi}(\bar{\tau}) = \pi(\tau)$, and $\bar{h}_f(\bar{\tau}) = h_f(\tau)$. The decomposition (6) holds in reversed time:

$$\bar{h}(\bar{\tau}) = \bar{\pi}(\bar{\tau})\bar{h}_f(\bar{\tau}).$$

This establishes the first part of point 3.

Step 2: Dynamics of π. Note that $\frac{d}{dt} = \frac{d\tau}{dt}\frac{d}{d\tau} = \frac{d}{d\tau}$ and $\frac{d}{d\tau} = \frac{d\bar{\tau}}{d\tau}\frac{d}{d\bar{\tau}} = (-1)\frac{d}{d\bar{\tau}}$. Substituting into (7):

$$-\frac{d\pi(\bar{\tau})}{d\bar{\tau}} = -\pi(\bar{\tau})(1-\pi(\bar{\tau}))h_f(\tau) = -\pi(\bar{\tau})(1-\pi(\bar{\tau}))\bar{h}_f(\bar{\tau}).$$

Since $\pi \in (0,1)$ and $\bar{h}_f \geq 0$ (as it's a hazard rate), we have $d\pi/d\bar{\tau} \geq 0$, confirming that $\pi(\bar{\tau})$ is non-decreasing (and increasing if $\bar{h}_f > 0$).

Step 3: Dynamics of h_f.

Set $p = 1$ in (1) to obtain $\bar{h}_f(\bar{\tau}) = e^{\lambda\bar{\tau}}g(\bar{\tau}) / \int_0^{\bar{\tau}} e^{\lambda s}g(s)\, ds$. The ODE is obtained immediately by differentiating the expression.

A.1.4 Proof of point 4.

We want to show that $\bar{h}_f(\bar{\tau})$ is non-increasing in $\bar{\tau}$ for $\bar{\tau} \in [0,\eta]$. From Step 3 (dynamics of h_f) in the proof of point 3, we have the ODE:

$$\frac{d\bar{h}_f}{d\bar{\tau}} = -\left[\bar{h}_f(\bar{\tau}) - k(\bar{\tau})\right]\bar{h}_f(\bar{\tau}),$$

where $k(\bar{\tau}) = \lambda + \frac{G''(\bar{\tau})}{G'(\bar{\tau})}$. Since $\bar{h}_f(\bar{\tau}) \geq 0$, the sign of the derivative is determined by the sign of $-[\bar{h}_f(\bar{\tau}) - k(\bar{\tau})]$. Thus, $\bar{h}_f$ is non-increasing if and only if $\delta(\bar{\tau}) := \bar{h}_f(\bar{\tau}) - k(\bar{\tau}) \geq 0$.

Let's analyze the behavior of $\delta(\bar{\tau})$ near $\bar{\tau} = 0$. Under the stated regularity conditions for g at $\bar{\tau} = 0$:

$$\lim_{\bar{\tau}\to 0^+} \bar{h}_f(\bar{\tau}) = \lim_{\bar{\tau}\to 0^+} \frac{e^{\lambda\bar{\tau}}g(\bar{\tau})}{\int_0^{\bar{\tau}} e^{\lambda s}g(s)ds} = +\infty$$
$$\lim_{\bar{\tau}\to 0^+} k(\bar{\tau}) = \lambda + \frac{g'(0)}{g(0)}, \quad \text{which is finite.}$$

Therefore, $\lim_{\bar{\tau}\to 0^+} \delta(\bar{\tau}) = +\infty > 0$.

Suppose, for the sake of contradiction, that $\delta(\bar{\tau})$ becomes negative for some $\bar{\tau}$ in $(0,\eta]$. Since $\delta(\bar{\tau})$ starts positive and is continuous, there must exist a first point $\bar{\tau}_c \in (0,\eta]$ where $\delta(\bar{\tau}_c) = 0$ and $\delta'(\bar{\tau}_c) \leq 0$. We have at that point:

$$\delta'(\bar{\tau}_c) = -\delta(\bar{\tau}_c)\bar{h}_f(\bar{\tau}_c) - k'(\bar{\tau}_c) = -(0) \times \bar{h}_f(\bar{\tau}_c) - k'(\bar{\tau}_c) = -k'(\bar{\tau}_c).$$

By strict log-concavity, $(\log g)''(\bar{\tau}) = k'(\bar{\tau}) < 0$ for $\bar{\tau} \in (0,\eta]$. Therefore, we must have $\delta'(\bar{\tau}_c) = -k'(\bar{\tau}_c) > 0$, a contradiction.

We can conclude that:

$$\frac{d\bar{h}_f}{d\bar{\tau}} = -\underbrace{\delta(\bar{\tau})}_{>0}\underbrace{\bar{h}_f(\bar{\tau})}_{\geq 0} \leq 0.$$

A.1.5 Proof of point 5.

If $\bar{h}$ has no interior critical point it is monotonic, hence unimodal. Otherwise, suppose $\bar{h}'(\bar{\tau}^*) = 0$ for some $\bar{\tau}^* \in (0, \eta)$. Using the decomposition $\bar{h} = \pi \bar{h}_f$ and the ODEs from points 3–4, the log-derivative simplifies to

$$(\log \bar{h})' = -\bar{h} + \lambda + \frac{g'}{g}.$$

Differentiating once more and evaluating at $\bar{\tau}^*$ where $\bar{h}' = 0$:

$$(\log \bar{h})''|_{\bar{\tau}^*} = (\log g)''(\bar{\tau}^*) < 0,$$

where the inequality is strict log-concavity. Every interior critical point is therefore a local maximum, establishing unimodality.

A.2 Proof of Lemma 3 (Uninformed Hazard Rate)

The derivation parallels Point 1, conditioning on the event $E_{un}(t) := \{\text{uninformed at } t\}$ instead of $T_i = t_i$. The likelihood of remaining uninformed given $T_0 = t_0$ is the survival probability $1 - G(t - t_0)$ rather than the density $g(t_i - t_0)$. By the same Bayesian updating and change of variable, the memoryless property yields a stationary posterior, and the hazard rate becomes:

$$h_{un}(\xi^*) = \frac{p\, e^{\lambda \xi^*}[1 - G(\xi^*)]}{p \int_0^{\xi^*} e^{\lambda s}[1 - G(s)]\, ds + (1-p) \int_0^{\eta} e^{\lambda s}[1 - G(s)]\, ds}.$$

Since h_{un} is constant in calendar time, by Lemma 1 (bang-bang rule), the agent holds deposits if and only if $u > h_{un}(\xi^*)$.

A.3 Properties of the Uninformed Hazard Rate

The following properties of h_{un} are used in the proofs of Proposition 2 (existence of participation thresholds) and Corollary 3 (fragility regimes in β). Since the participation condition requires $u > h_{un}(\xi^*)$, the shape of h_{un} as a function of the conjectured crash time ξ^* determines when participation is sustainable.

Lemma 6 (Properties of h_{un})**.** *Let $k(\xi) := \lambda - g(\xi)/(1 - G(\xi))$. The uninformed hazard rate satisfies:*

1. ***ODE:*** $\dfrac{dh_{un}}{d\xi} = -h_{un}(\xi)\big(h_{un}(\xi) - k(\xi)\big)$.
2. ***Unimodality:*** *The function $h_{un}(\xi)$ is unimodal on $(0, \eta)$, with its peak at the unique ξ satisfying $h_{un}(\xi) = k(\xi)$.*

Proof. ODE. Direct by differentiation.

Unimodality. The argument is identical to Point 5 of Lemma A.1: at any critical point, $(\log h_{un})'' = k'(\xi^*) < 0$ since G's hazard rate $g/(1-G)$ is strictly increasing by log-concavity of g (Bagnoli and Bergstrom (2005)). □

A.4 Proof of Lemma 4

A.4.1 Preliminary Result

I start with a series of useful preliminary lemmas.

Lemma 7 (Unimodality of $AW(\cdot)$)**.** *For any fixed $\tau > 0$, the function $AW(t)$ is single-peaked (unimodal).*

Proof. Step 0. Case $t < \tau$. Suppose $t < \tau$, then $AW(t) = G(t)$ is strictly increasing. Hence it does not achieve a maximum in this region. The rest of the proof assumes $t \geq \tau$.

Step 1. Re-expressing the condition. We analyze the derivative $AW'(t) = g(t) - g(t-\tau)$. A function is unimodal if its derivative crosses zero at most once, from positive to negative. Letting $R(t) = \frac{g(t)}{g(t-\tau)}$, we can write:

$$AW'(t) = g(t-\tau)\left[R(t) - 1\right].$$

Since $g(t-\tau) > 0$, the sign of $AW'(t)$ is determined by the sign of $R(t) - 1$.

Step 2. $R(t)$ is decreasing. Consider the ratio $R(t) = \frac{g(t)}{g(t-\tau)}$ for $t \geq \tau$. Letting $\phi(t) = \log(g(t))$, the derivative of the logarithm of the ratio is:

$$\frac{d}{dt}\log R(t) = \frac{d}{dt}[\phi(t) - \phi(t-\tau)] = \phi'(t) - \phi'(t-\tau).$$

Since $\phi(t)$ is strictly concave, its derivative $\phi'(t)$ is strictly decreasing and we get $\phi'(t) < \phi'(t-\tau)$. Therefore, $\frac{d}{dt}\log R(t) < 0$, $R(t)$ is strictly decreasing in t.

Step 3. Consequence for unimodality. Because $R(t)$ is strictly decreasing, the equation $R(t) = 1$ has at most one solution. Let t^* be such a solution if it exists.

- If t^* exists: For $t < t^*$, $R(t) > 1 \implies AW'(t) > 0$. For $t > t^*$, $R(t) < 1 \implies AW'(t) < 0$.
- If $R(t) > 1$ for all t, then $AW'(t) > 0$ always.
- If $R(t) < 1$ for all t, then $AW'(t) < 0$ always.

In all cases, $AW'(t)$ changes sign at most once, from positive to non-positive. $\square$

Lemma 8 (Relationship between τ_I^* and $\bar{\tau}_I^*$)**.** *Suppose an equilibrium (τ_I^*, ξ^*) exists. Consider $\bar{\tau}_I^* = \xi^* - \tau_I^*$. We have:*

$$\frac{d\tau_I^*}{d\bar{\tau}_I^*} \leq 0.$$

Furthermore, if $\kappa < \max_{t\in[0,\eta]} AW(t)$, then the inequality is strict:

$$\frac{d\tau_I^*}{d\bar{\tau}_I^*} < 0.$$

Proof. Step 1. Weak inequality. Implicitly differentiating the equilibrium condition $G(\xi^*) - G(\bar{\tau}_I^*) = \kappa$ with respect to $\bar{\tau}_I^*$ yields:

$$G'(\xi^*)\frac{d\xi^*}{d\bar{\tau}_I^*} - G'(\bar{\tau}_I^*) \times 1 = 0.$$

Since $G'(\xi^*) > 0$, we have:

$$\frac{d\xi^*}{d\bar{\tau}_I^*} = \frac{G'(\bar{\tau}_I^*)}{G'(\xi^*)}.$$

Finally, differentiating the identity $\tau_I^* = \xi^* - \bar{\tau}_I^*$ gives:

$$\frac{d\tau_I^*}{d\bar{\tau}_I^*} = \frac{d\xi^*}{d\bar{\tau}_I^*} - 1 = \frac{G'(\bar{\tau}_I^*)}{G'(\xi^*)} - 1.$$

The crash time is $\xi^* = \inf\{t : AW(t) = \kappa\}$. Since AW is unimodal (Lemma 7), this infimum corresponds to the first crossing of the threshold κ, which must occur on the increasing branch of AW. Therefore, $AW'(\xi^*) \geq 0$, which gives $G'(\xi^*) - G'(\bar{\tau}_I^*) = AW'(\xi^*) \geq 0$. Hence:

$$\frac{d\tau_I^*}{d\bar{\tau}_I^*} = \frac{G'(\bar{\tau}_I^*)}{G'(\xi^*)} - 1 \leq 0.$$

Step 2. Strict inequality. Equality $\frac{d\tau_I^*}{d\bar{\tau}_I^*} = 0$ holds if and only if $AW'(\xi^*) = G'(\xi^*) - G'(\bar{\tau}_I^*) = 0$. By Lemma 7, $AW(t)$ is unimodal and achieves its maximum at a unique time t^* where $AW'(t^*) = 0$. Thus, equality holds if and only if the crash occurs exactly at the peak of the aggregate withdrawal function, i.e., $\xi^* = t^*$ and $\kappa = AW(t^*)$. Under the assumption $\kappa < AW(t^*)$ stated in the lemma, the crash occurs strictly before the peak ($\xi^* < t^*$). Since AW is strictly increasing on $[\bar{\tau}_I^*, t^*]$, we have $AW'(\xi^*) > 0$, giving $G'(\xi^*) > G'(\bar{\tau}_I^*)$. Hence,

$$\frac{d\tau_I^*}{d\bar{\tau}_I^*} < 0.$$

□

A.4.2 Proof of Comparative Statics for u, p, η

We analyze the local effects of u, p, η on the equilibrium reentry time τ_I^*. By Lemma 8, $\frac{d\tau_I^*}{d\bar{\tau}_I^*} \leq 0$, so it is enough to analyze the effect on $\bar{\tau}_I^*$.

Step 1. Case $\tau_I = \xi^*$. Note that $\tau_I = \xi^* \iff \bar{\tau}_I^* = 0 \iff u < \bar{h}(0)$ (otherwise $\tau_I < \xi^*$). In that case $\bar{\tau}_I$ is constant in u, p and η. Since ξ^* only depends on $\bar{\tau}_I$ and G, it is also constant in those parameters.

Step 2. Case $\tau_I < \xi^*$. In that case $\bar{\tau}_I$ must be interior and $\bar{h}(\bar{\tau}_I^*) = u$. By Point 5 of Lemma A.1, the equilibrium occurs on the increasing branch of the unimodal $\bar{h}$, ensuring

$\partial\bar{h}/\partial\bar{\tau}_I^* > 0$. Applying the Implicit Function Theorem to $\bar{h}(\bar{\tau}_I^*; p, \eta) - u = 0$ gives:

$$\frac{d\bar{\tau}_I^*}{du} = \frac{1}{\partial\bar{h}/\partial\bar{\tau}_I^*} > 0,$$

$$\frac{d\bar{\tau}_I^*}{dp} = -\frac{\partial\bar{h}/\partial p}{\partial\bar{h}/\partial\bar{\tau}_I^*} < 0, \qquad \text{since } \frac{\partial\bar{h}}{\partial p} > 0 \text{ (from Lemma A.1 closed-form)},$$

$$\frac{d\bar{\tau}_I^*}{d\eta} = -\frac{\partial\bar{h}/\partial\eta}{\partial\bar{h}/\partial\bar{\tau}_I^*} > 0, \qquad \text{since } \frac{\partial\bar{h}}{\partial\eta} < 0 \text{ (from Lemma A.1 closed form)}.$$

If $\kappa < \max_t AW(t)$ we must have $\frac{d\tau_I^*}{d\bar{\tau}_I^*} < 0$ (by Lemma 8) and the strictness in comparative statics case 2 is passed on to τ_I.

A.5 Proof of Proposition 2

Proof. **Part 1: Existence and Uniqueness of Run Equilibrium**

The substance of the proof is to show that the set of u that admits an equilibrium is in the form of an interval $[0, \bar{u})$ (as opposed to a collection of intervals for example).

I prove this result constructively. For any u, we attempt to construct a candidate equilibrium pair $(\xi^*(u), \bar{\tau}_I(u))$ satisfying four conditions: (i) reentry optimality; (ii) crash feasibility; (iii) first crossing; and (iv) immediate exit. Each condition yields a threshold beyond which no equilibrium exists. I show these thresholds define an interval, establish uniqueness, and verify $\bar{u} > 0$.

Step 1: Candidate Construction.

Reentry condition ($\bar{u}_1$). From Lemma 2, the reversed hazard rate $\bar{h}(t)$ is continuous and unimodal on $[0, \eta]$ with peak value $\bar{u}_1 = \max_{t\in[0,\eta]} \bar{h}(t)$ achieved at some $t_{peak} \in [0, \eta]$. Define $\bar{\tau}_I(u) = \inf\{s \in [0, \eta] : \bar{h}(s) \geq u\}$ for all $u \geq 0$. For $u \leq \bar{h}(0)$, we have $\bar{\tau}_I(u) = 0$ (corner solution). For $u \in (\bar{h}(0), \bar{u}_1]$, the function $\bar{\tau}_I(u) \in (0, t_{peak}]$ is the unique solution to $\bar{h}(t) = u$ on the increasing branch and is continuous and strictly increasing by the Implicit Function Theorem. For $u > \bar{u}_1$, no solution exists, delimiting the first cutoff $\bar{u}_1$.

Crash feasibility ($\bar{u}_2$). Given $\bar{\tau}_I(u)$, the crash condition $G(\xi^*) - G(\bar{\tau}_I) = \kappa$ uniquely determines $\xi^*(u) = G^{-1}(\kappa + G(\bar{\tau}_I(u)))$, provided $\kappa + G(\bar{\tau}_I(u)) < 1$. Since $\bar{\tau}_I(u)$ is nondecreasing and G is strictly increasing, this constraint becomes binding as u increases. Define $\bar{u}_2 = \sup\{u \in (0, \bar{u}_1) : \kappa + G(\bar{\tau}_I(u)) < 1\}$ (set $\bar{u}_2 = \bar{u}_1$ if the constraint never binds).

First crossing condition ($\bar{u}_3$). The crash must occur on the *increasing branch* of aggregate withdrawals, meaning the solution $\xi^*(u)$ must be the first time AW reaches κ. This requires the slope condition $G'(\xi^*(u)) \geq G'(\bar{\tau}_I(u))$. Define $\Phi(u) := G'(\xi^*(u)) - G'(\bar{\tau}_I(u))$. I prove $\Phi(u)$ is strictly decreasing. Differentiating the crash condition $G(\xi^*) = \kappa + G(\bar{\tau}_I)$ yields:

$$\begin{aligned}
\frac{d\xi^*}{du} &= \frac{G'(\bar{\tau}_I)}{G'(\xi^*)}\frac{d\bar{\tau}_I}{du}, \\
\frac{d\Phi}{du} &= G''(\xi^*)\frac{d\xi^*}{du} - G''(\bar{\tau}_I)\frac{d\bar{\tau}_I}{du} \\
&= G'(\bar{\tau}_I)\left[(\log G')'(\xi^*) - (\log G')'(\bar{\tau}_I)\right]\frac{d\bar{\tau}_I}{du}.
\end{aligned}$$

By strict log-concavity, the function $(\log G')'$ is decreasing. Since $\xi^* > \bar{\tau}_I$ (from the crash condition), we have $(\log G')'(\xi^*) < (\log G')'(\bar{\tau}_I)$. Combined with $G'(\bar{\tau}_I) > 0$ and $d\bar{\tau}_I/du > 0$ (Implicit Function Theorem on the increasing branch), we obtain $d\Phi/du < 0$.

Therefore $\Phi(u)$ is strictly decreasing, and $\{u : \Phi(u) \geq 0\}$ is an interval $[0, \bar{u}_3]$ for some threshold $\bar{u}_3 = \sup\{u \in (0, \bar{u}_2) : \Phi(u) \geq 0\}$. At $u = \bar{u}_3$, the crash occurs exactly at the peak of aggregate withdrawals.

Immediate exit condition ($\bar{u}_4$). Suppose $u < \min(\bar{u}_1, \bar{u}_2, \bar{u}_3)$ so that $\xi^*(u)$ and $\bar{\tau}_I(u)$ are well defined. The candidate equilibrium must also satisfy $\bar{h}(\xi^*(u)) > u$ (so that $\bar{\tau}_O(u)$ is a corner solution). We analyze two cases.

Case (i): $\bar{h}(t)$ is strictly increasing on $[0, \eta]$. Since $\xi^*(u) > \bar{\tau}_I(u)$ (from $G(\xi^*) = \kappa + G(\bar{\tau}_I)$ and G strictly increasing), we immediately have $\bar{h}(\xi^*(u)) > \bar{h}(\bar{\tau}_I(u)) = u$. Set $\bar{u}_4 = \min(\bar{u}_1, \bar{u}_2, \bar{u}_3)$ (no additional binding constraint).

Case (ii): $\bar{h}(t)$ is unimodal or decreasing. Define $\bar{\tau}_O(u) = \sup\{s \in [0, \eta] : \bar{h}(s) \geq u\}$. From the unimodality of $\bar{h}$, the condition $\bar{h}(\xi^*(u)) > u$ is equivalent to $\xi^*(u) < \bar{\tau}_O(u)$.

Consider the gap function $\Delta(u) = \bar{\tau}_O(u) - \xi^*(u)$. If $\bar{\tau}_O(u) = \eta$ (corner: $\bar{h}(\eta) > u$), then $\bar{\tau}_O$ is locally constant; otherwise, the Implicit Function Theorem gives $d\bar{\tau}_O/du < 0$ (since $\bar{h}'(\bar{\tau}_O) < 0$). In both cases $\xi^*(u)$ is strictly increasing, so $\Delta(u)$ is strictly decreasing. If $\Delta(u) < 0$ for all $u \in [0, \min(\bar{u}_1, \bar{u}_2, \bar{u}_3)]$, set $\bar{u}_4 = 0$. If $\Delta(u) > 0$ throughout, set $\bar{u}_4 = \min(\bar{u}_1, \bar{u}_2, \bar{u}_3)$. Otherwise, the strict monotonicity of Δ gives a unique $\bar{u}_4 \in (0, \min(\bar{u}_1, \bar{u}_2, \bar{u}_3))$ where $\Delta(\bar{u}_4) = 0$.

Conclusion. Define $\bar{u} = \min(\bar{u}_1, \bar{u}_2, \bar{u}_3, \bar{u}_4)$. A candidate equilibrium pair exists if and only if $u < \bar{u}$.

Step 2: Uniqueness. For $u \leq \bar{h}(0)$, the reentry time is $\bar{\tau}_I(u) = 0$ (corner solution), then $G(\xi^*) = \kappa$; since G is strictly increasing, both $\bar{\tau}_I^*$ and ξ^* are unique. For $u \in (\bar{h}(0), \bar{u})$, the function $\bar{\tau}_I(u)$ is the unique solution to $\bar{h}(t) = u$ on the strictly increasing branch of the unimodal $\bar{h}$ (by construction in Step 1). Given $\bar{\tau}_I(u)$, the crash time $\xi^*(u) = G^{-1}(\kappa + G(\bar{\tau}_I(u)))$ is uniquely determined by the strict monotonicity of G. Thus, the equilibrium pair $(\xi^*(u), \bar{\tau}_I(u))$ is unique for all $u < \bar{u}$.

Step 3: $\bar{u} > 0$. As $u \to 0$, we have $\bar{\tau}_I(u) \to 0$ (corner solution) and $\xi^*(u) \to G^{-1}(\kappa + G(0))$. Furthermore $\bar{\tau}_O(u) \to \eta$, so $\Delta(u) > 0$ for u near 0. The condition $G(\eta) > \kappa + G(0)$ ensures $\xi^*(0) < \eta = \bar{\tau}_O(0)$, giving $\bar{u} > 0$.

Part 2: Comparative Statics of $\bar{AW}$ with respect to u

For $u \leq \bar{h}(0)$, we have the corner solution $\bar{\tau}_I(u) = 0$. Then $\tau_I = \xi^*$, which is independent of u, and $AW(t) = G(t) - G(t - \xi^*)$ which is also independent of u. So the peak withdrawal $\bar{AW}$ is constant in u.

Suppose now $u > \bar{h}(0)$. Then, $AW(t; \tau_I) = G(t) - G(t - \tau_I)$ and $\tau_I^*(u)$ is interior. Let $t^*(u)$ denote the time that maximizes $AW(t; \tau_I^*(u))$ such that $\bar{AW}(u) = AW(t^*(u), \tau_I^*(u))$. By the Envelope Theorem, the derivative of the maximized value with respect to u is:

$$\frac{d\bar{AW}}{du} = \left.\frac{\partial AW(t, \tau_I)}{\partial \tau_I}\right|_{t=t^*(u), \tau_I = \tau_I^*(u)} \times \frac{d\tau_I^*}{du}.$$

The partial derivative of AW with respect to τ_I is:

$$\frac{\partial AW(t, \tau_I)}{\partial \tau_I} = \frac{\partial}{\partial \tau_I}[G(t) - G(t - \tau_I)] = G'(t - \tau_I).$$

Substituting this into the Envelope Theorem expression:

$$\frac{d\bar{AW}}{du} = G'(t^*(u) - \tau_I^*(u)) \times \frac{d\tau_I^*}{du}.$$

To ensure this derivative is non-zero, we must verify that $t^*(u) > \tau_I^*(u)$. In the run equilibrium, the crash occurs at ξ^* such that $AW(\xi^*) = \kappa$. If $\kappa < AW(t^*(u))$, then $\xi^* < t^*(u)$. Since $\tau_I^*(u) = \xi^* - \bar{\tau}_I^*(u) \leq \xi^*$, we have $\tau_I^*(u) < t^*(u)$. Therefore, $t^*(u) - \tau_I^*(u) > 0$. Since $G'(t^*(u) - \tau_I^*(u)) > 0$ and $d\tau_I^*/du < 0$ by Lemma 4, we conclude that:

$$\frac{d\bar{AW}}{du} < 0.$$

Thus, the peak withdrawal level $\bar{AW}$ is strictly decreasing in u for $u \in (\bar{h}(0), \bar{u})$.

Part 3: Participation Threshold and Non-Participation

3(a) Participation threshold. Define the participation function $V(u) := u - h_{un}(\xi^*(u))$, where $\xi^*(u)$ is the equilibrium crash time from Part 1. Conditional on playing the run equilibrium if the bank forms, agents participate in equilibrium iff $V(u) > 0$.

Lower boundary and corner case. For $u \leq \bar{h}(0)$, we have $\bar{\tau}_I^* = 0$ and $\xi^* = G^{-1}(\kappa)$, both constant in u. Therefore $V(u) = u - h_{un}(G^{-1}(\kappa))$, which gives $V(0) < 0$ (since $h_{un} > 0$) and $V'(u) = 1 > 0$.

Monotonicity (interior case $u > \bar{h}(0)$): We show $V'(u) > 0$ at any zero of V.

For the interior case $u > \bar{h}(0)$: the chain rule gives

$$\frac{d\xi^*}{du} = \frac{g(\bar{\tau}_I^*)}{g(\xi^*)} \cdot \frac{d\bar{\tau}_I^*}{du} = \frac{g(\bar{\tau}_I^*)}{g(\xi^*)} \cdot \frac{1}{\bar{h}'(\bar{\tau}_I^*)},$$

where the first factor comes from differentiating $G(\xi^*) - G(\bar{\tau}_I^*) = \kappa$ and the second from $\bar{h}(\bar{\tau}_I^*) = u$. Therefore:

$$V'(u) = 1 - \underbrace{\frac{h'_{un}(\xi^*)}{\bar{h}'(\bar{\tau}_I^*)}}_{P} \cdot \underbrace{\frac{g(\bar{\tau}_I^*)}{g(\xi^*)}}_{Q}.$$

Bound on P: At any zero of V, both hazard rates equal u: $h_{un}(\xi^*) = \bar{h}(\bar{\tau}_I^*) = u$. Using the ODEs $h'_{un} = h_{un}(k - h_{un})$ and $\bar{h}' = \bar{h}(\gamma - \bar{h})$ with $k(\xi^*) = \lambda - \mathrm{HR}_G(\xi^*)$ and $\gamma(\bar{\tau}_I^*) = (\log g)'(\bar{\tau}_I^*) + \lambda$:

$$\bar{h}'(\bar{\tau}_I^*) - h'_{un}(\xi^*) = u\big[(\log g)'(\bar{\tau}_I^*) + \mathrm{HR}_G(\xi^*)\big].$$

Since $\xi^* > \bar{\tau}_I^*$ and HR_G is increasing (strict IFR):

$$(\log g)'(\bar{\tau}_I^*) + \mathrm{HR}_G(\xi^*) > (\log g)'(\bar{\tau}_I^*) + \mathrm{HR}_G(\bar{\tau}_I^*) = (\log \mathrm{HR}_G)'(\bar{\tau}_I^*) > 0,$$

where the last inequality uses strict IFR. Therefore $\bar{h}'(\bar{\tau}_I^*) > h'_{un}(\xi^*)$. Since $\bar{h}'(\bar{\tau}_I^*) > 0$ (interior equilibrium on the increasing branch of $\bar{h}$), we have $P < 1$.

Bound on Q: By the first crossing condition (Lemma 8), $AW'(\xi^*) = g(\xi^*) - g(\bar{\tau}_I^*) \geq 0$, so $Q = g(\bar{\tau}_I^*)/g(\xi^*) \leq 1$.

Combining: If $h'_{un}(\xi^*) \le 0$, then $PQ \le 0 < 1$. If $h'_{un}(\xi^*) > 0$, then $P \in (0,1)$ and $Q \in (0,1]$, so $PQ < 1$. In both cases $V'(u) = 1 - PQ > 0$.

Existence and uniqueness: Since $V(0) < 0$ and $V'(u) > 0$ at any zero of V, any zero is a crossing from below, so V has at most one zero in $(0, \bar{u})$. Denote this zero $\underline{u}$. If V has no zero—i.e., $V(u) < 0$ for all $u \in (0, \bar{u})$—set $\underline{u} = \bar{u}$ (the middle regime is empty). In either case, $\underline{u} \le \bar{u}$.

3(b) Non-participation for $t \ge \eta$. When $u \le h_{un}(\xi^*)$, the bang-bang rule (Lemma 1) immediately gives $\alpha_{un} = 1$: uninformed agents withdraw all deposits. It remains to verify that non-participation extends to $t < \eta$, where the hazard rate is not yet stationary.

3(c) Unraveling before η. By the bang-bang rule, any symmetric strategy satisfies $\alpha_{un}(t) \in \{0, 1\}$. Suppose a symmetric equilibrium has $\alpha_{un} = 0$ at some $t < \eta$, and let $T := \sup\{t \le \eta : \alpha_{un}(t) = 0\} > 0$. At T, all remaining (uninformed) agents switch to cash, so aggregate withdrawals jump to $1 > \kappa$, triggering insolvency if the bank is fragile (probability $(1 - e^{-\lambda T})p > 0$). A unilateral deviation to cash at $T - \varepsilon$ forgoes $u\varepsilon$ in deposit utility but avoids a discrete expected loss of at least $(1 - e^{-\lambda T})p > 0$ (depositors lose their funds at the crash), which dominates for small ε. Since this rules out any $T > 0$ as a last deposit time, standard unraveling gives $\alpha_{un}(t) = 1$ for all $t \ge 0$: no agent ever deposits, and the bank cannot form.

□

A.6 Information diffusion speed β

A.6.1 Change of Variables

To analyze comparative statics in β, rescale time via $t \mapsto \tilde{t} = \beta t$. The scaled parameters are $\tilde{\beta} = 1$, $\tilde{\lambda} = \lambda/\beta$, $\tilde{u} = u/\beta$, and $\tilde{\eta} = \beta\eta = \bar{\eta}$ (constant); parameters without time dimension (p, κ) are unaffected. Writing $a := 1/\beta$, the scaling jointly varies $\tilde{\lambda} = a\lambda$ and $\tilde{u} = au$ while holding $(\bar{\eta}, p, \kappa)$ constant.

In this section (until the end of the appendix), we specialize to the logistic learning process $dG = G(1 - G)\beta\, dt$, which admits such scaling: writing $G_\beta(t) = G_1(\beta t) = G(\tilde{t})$. A useful identity for the logistic model in scaled coordinates is $\mathrm{HR}_G(\xi) := g(\xi)/(1 - G(\xi)) = G(\xi)$.

A.6.2 Comparative statics

Lemma 9. *Consider the scaled system with $\tilde{\beta} = 1$, and fix base parameters $\lambda_0 > 0$ and $u_0 > 0$. Vary the scaling parameter $a > 0$ such that $\tilde{\lambda}(a) = a\lambda_0$ and $\tilde{u}(a) = au_0$, holding $(\bar{\eta}, p, \kappa)$ constant. Suppose a run equilibrium exists in a neighborhood of the base parameters. If $a\lambda_0\bar{\eta} \le 1$, then:*

$$\frac{d\bar{\tau}_I^*(a)}{da} \ge 0.$$

Remark 1. *Combining Lemmas 9 and 8, the effect on the scaled reentry time satisfies:*

$$\frac{d\tau_I^*}{da} = \frac{d\tau_I^*}{d\bar{\tau}_I^*} \cdot \frac{d\bar{\tau}_I^*}{da} = (\le 0) \times (\ge 0) \le 0.$$

That is, in the scaled system, higher a (equivalently, lower β) reduces reentry time.

To translate this to physical time, note that $\tau_{I,phys} = a \cdot \tau_I^*(a)$ (since physical time = scaled time $/\beta = a\times$ scaled time). Thus:

$$\frac{d\tau_{I,phys}}{da} = \tau_I^*(a) + a \cdot \frac{d\tau_I^*}{da} = \underbrace{\tau_I^*}_{>0} + \underbrace{a \cdot \frac{d\tau_I^*}{da}}_{\leq 0}.$$

The first term (direct scaling) is positive while the second (equilibrium effect) is non-positive. These effects work in opposite directions, so the net effect of β on physical reentry time is ambiguous and depends on parameter values.

Proof. The equilibrium condition requires either $\bar{\tau}_I^* = 0$ (corner) or $\bar{h}(\bar{\tau}_I^*; \tilde{\lambda}(a), \bar{\eta}, p) - \tilde{u}(a) = 0$ (interior). In the former case $\bar{\tau}_I^*(a)$ is constant in a, so the derivative is 0. In the latter, we apply the Implicit Function Theorem:

$$\frac{d\bar{\tau}_I^*}{da} = -\frac{(\partial\bar{h}/\partial\tilde{\lambda})(\partial\tilde{\lambda}/\partial a) - \partial\tilde{u}/\partial a}{\partial\bar{h}/\partial\bar{\tau}_I^*} = \frac{u_0 - \lambda_0(\partial\bar{h}/\partial\tilde{\lambda})}{(\partial\bar{h}/\partial\bar{\tau}_I^*)}$$

The equilibrium $\bar{\tau}_I^*$ occurs on the increasing branch of $\bar{h}$, so $\partial\bar{h}/\partial\bar{\tau}_I^* > 0$. To prove the lemma's claim that $\frac{d\bar{\tau}_I^*}{da} \geq 0$, we need to show $u_0 - \lambda_0(\partial\bar{h}/\partial\tilde{\lambda}) \geq 0$.

We compute the derivative of $\bar{h}(\bar{\tau}; \tilde{\lambda})$ with respect to $\tilde{\lambda}$. Applying the quotient rule to the closed form from Lemma 2 yields:

$$\frac{\partial\bar{h}}{\partial\tilde{\lambda}} = \bar{h}(\bar{\tau}; \tilde{\lambda})\left(\bar{\tau} - E_{\tilde{\lambda}}[s|\bar{\tau}]\right),$$

where $E_{\tilde{\lambda}}[s|\bar{\tau}]$ denotes the posterior mean learning lag, given by:

$$E_{\tilde{\lambda}}[s|\bar{\tau}] = \frac{p\int_0^{\bar{\tau}} se^{\tilde{\lambda}s}g(s)ds + (1-p)\int_0^{\bar{\eta}} se^{\tilde{\lambda}s}g(s)ds}{p\int_0^{\bar{\tau}} e^{\tilde{\lambda}s}g(s)ds + (1-p)\int_0^{\bar{\eta}} e^{\tilde{\lambda}s}g(s)ds}.$$

At equilibrium $\bar{h}(\bar{\tau}_I^*) = au_0$, the condition $u_0 - \lambda_0(\partial\bar{h}/\partial\tilde{\lambda}) \geq 0$ reduces to $a\lambda_0(\bar{\tau}_I^* - E_{\tilde{\lambda}}[s|\bar{\tau}_I^*]) \leq 1$. Since $E_{\tilde{\lambda}}[s|\bar{\tau}_I^*] \geq 0$ and $\bar{\tau}_I^* \leq \bar{\eta}$, this follows from the condition $a\lambda_0\bar{\eta} \leq 1$. □

A.6.3 Proof of Corollary 3

Proof. **Part 1: Existence Threshold $\underline{\beta}$.**

The proof is constructive and follows a similar structure as the proof of Proposition 2. I switch to scaled time coordinates. A run equilibrium is characterized by a pair $(\tilde{\xi}^*, \bar{\tau}_I^*)$ satisfying four conditions analogous to Proposition 2: (i) reentry $\bar{\tau}_I^*(a) = \inf\{t \in [0, \bar{\eta}] : \bar{h}(t; a\lambda) \geq au\}$ is attained; (ii) crash feasibility $G(\tilde{\xi}^*) - G(\bar{\tau}_I^*) = \kappa$; (iii) first crossing $G'(\tilde{\xi}^*) \geq G'(\bar{\tau}_I^*)$; and (iv) immediate exit $\bar{h}(\tilde{\xi}^*; a\lambda) > au$. I construct the set of a values for which such an equilibrium exists, then translate back to β.

Step 1: candidate construction.

Reentry condition a_1. From Lemma 2, $\bar{h}(t; a\lambda)$ is continuous and unimodal in t on $[0, \bar{\eta}]$, with $\bar{h}_0(a) := \bar{h}(0; a\lambda) > 0$ and peak value $\bar{h}_{\max}(a) = \max_{t\in[0,\bar{\eta}]} \bar{h}(t; a\lambda)$ achieved at some

$t_{peak}(a) \in [0, \bar{\eta}]$. Define $\bar{\tau}_I^*(a) = \inf\{s \in [0, \bar{\eta}] : \bar{h}(s; a\lambda) \geq au\}$ for all $a > 0$. Following the same construction as in the proof of Proposition 2 let $A_1 = \{a > 0 : au \leq \bar{h}_{\max}(a)\}$.

We need to establish that A_1 is an interval of the form $(0, a_1]$ for some $a_1 > 0$ (possibly $a_1 = \infty$). It holds if $\bar{h}_{max}(a)/a$ is decreasing. From the derivative formula for $\bar{h}(\bar{\tau}; \tilde{\lambda})$ with respect to $\tilde{\lambda}$ (derived in Lemma 9, applied with $\lambda_0 = \lambda$ and $u_0 = u$), evaluated at $\bar{\tau} = t_{peak}(a)$:

$$\frac{d}{da}\left(\frac{\bar{h}_{max}(a)}{a}\right) = \frac{1}{a^2}\left(a\lambda\bar{h}_{max}(a)(t_{peak}(a) - E_{a\lambda}[s|t_{peak}(a)]) - \bar{h}_{max}(a)\right) \leq \frac{\bar{h}_{max}(a)}{a^2}(a\lambda t_{peak}(a) - 1),$$

where the second inequality follows from $E_{a\lambda}[s|t_{peak}(a)] \geq 0$. Under the maintained assumption $a\lambda\bar{\eta} \leq 1$ and since $t_{peak}(a) \leq \bar{\eta}$, we have $a\lambda t_{peak}(a) \leq 1$, implying $\frac{d}{da}(\bar{h}_{max}(a)/a) \leq 0$. Thus $\bar{h}_{max}(a)/a$ is non-increasing in a.

This establishes that if $a \in A_1$, then $(0, a] \subset A_1$. Taking $a_1 = \sup A_1$, we obtain $A_1 = (0, a_1]$.

Crash feasibility (a_2) and first crossing (a_3). These steps follow the same argument as the proof of Proposition 2, with a replacing u and $d\bar{\tau}_I^*/da \geq 0$ (Lemma 9) replacing $d\bar{\tau}_I^*/du > 0$. The crash condition defines $\tilde{\xi}^*(a) = G^{-1}(\kappa + G(\bar{\tau}_I^*(a)))$, yielding threshold a_2 when $\kappa + G(\bar{\tau}_I^*) = 1$. For the first crossing, $\Phi(a) := G'(\tilde{\xi}^*(a)) - G'(\bar{\tau}_I^*(a))$ is non-increasing in a by strict log-concavity (the same $d\Phi$ computation as in the Proposition, with da replacing du), yielding threshold a_3.

Immediate exit condition (a_4). As in the proof of Proposition 2, a_4 is chosen to ensure $\bar{h}(\tilde{\xi}^*(a); a\lambda) > au$. Again, there are two cases. The case where $\bar{h}(t; a\lambda)$ is strictly increasing resolves as in Proposition 2. The case where $\bar{h}(t; a\lambda)$ is unimodal also follows the proof but applies the IFT to $\bar{h}(\bar{\tau}_O; a\lambda) = au$ and uses $a\lambda\bar{\eta} \leq 1$ (as in Lemma 9) to get $d\bar{\tau}_O/da \leq 0$.

Conclusion. Define $\underline{a} = \min(a_1, a_2, a_3, a_4)$. A run equilibrium exists if and only if $a \in (0, \underline{a})$.

Step 4: Uniqueness. Follows the same argument as Proposition 2.

Step 5: Finiteness of $\underline{a}$. We show that $\underline{a} < \infty$ when $\kappa + G(\eta) > 1$. If $a_1 < \infty$ the conclusion is immediate, so suppose $a_1 = \infty$. Then $\bar{\tau}_I^*(a)$ is well defined for all $a > 0$. Rewrite the hazard rate as:

$$\bar{h}(\bar{\tau}; a\lambda) = \frac{pg(\bar{\tau})}{\int_0^{\bar{\tau}} e^{a\lambda(s-\bar{\tau})}g(s)ds + (1-p)\int_{\bar{\tau}}^{\eta} e^{a\lambda(s-\bar{\tau})}g(s)ds}.$$

For any fixed $\bar{\tau} < \eta$, as $a \to \infty$ the first integral vanishes (as $s - \bar{\tau} < 0$) while the second grows unboundedly (as $s - \bar{\tau} > 0$), so $\bar{h}(\bar{\tau}; a\lambda)/a \to 0$. Since the equilibrium condition $\bar{h}(\bar{\tau}_I^*; a\lambda) \geq au$ requires $\bar{h}(\bar{\tau}_I^*)/a \geq u > 0$, it follows that $\bar{\tau}_I^*(a) \to \eta$.

Equilibrium condition (ii) requires $G(\tilde{\xi}^*(a)) = \kappa + G(\bar{\tau}_I^*(a)) < 1$. But $\kappa + G(\bar{\tau}_I^*(a)) \to \kappa + G(\eta) > 1$, so condition (ii) cannot hold for sufficiently large a. Therefore $a_2 < \infty$, which implies $\underline{a} \leq a_2 < \infty$.

Step 6: Conclusion.

Translating back to the original parameter $\beta = 1/a$, the condition $a < \underline{a}$ becomes $\beta > 1/\underline{a}$. Defining $\underline{\beta} = 1/\underline{a}$ (with the convention that $\underline{\beta} = 0$ if $\underline{a} = \infty$), we obtain the desired result: a unique run equilibrium exists if and only if $\beta > \underline{\beta}$, and no equilibrium exists if $\beta \leq \underline{\beta}$. When

$\kappa + G(\eta) > 1$, Step 5 ensures $\underline{a} < \infty$, hence $\underline{\beta} > 0$.

Part 2: Comparative Statics of $\bar{AW}$

We first establish the effect in the scaled system. By the Envelope Theorem applied to $AW(t;\tau_I) = G(t) - G(t-\tau_I)$, we have $\frac{d\bar{AW}}{d\tau_I^*} = G'(t^* - \tau_I^*) > 0$, where t^* is the time at which the peak is attained (strict positivity holds when $\kappa < AW(t^*)$). Combined with $\frac{d\tau_I^*}{da} \leq 0$ (from Remark 1), we have:

$$\frac{d\bar{AW}}{da} = \frac{d\bar{AW}}{d\tau_I^*} \cdot \frac{d\tau_I^*}{da} = (>0) \times (\leq 0) \leq 0.$$

Finally note:

$$\frac{d\bar{AW}}{d\beta} = \frac{d\bar{AW}}{da} \cdot \frac{da}{d\beta} = (\leq 0) \times \left(-\frac{1}{\beta^2}\right) \geq 0.$$

Thus, $\bar{AW}$ is non-decreasing in β for $\beta > \underline{\beta}$. The increase is strict if $\kappa < AW(t^*)$ and the inequality in Lemma 9 is strict (e.g., if $\eta < 1/\lambda$).

Part 3: Participation Threshold $\bar{\beta}$.

I follow again Proposition 2. Define the participation function in scaled coordinates $V(a) := au - h_{un}(\xi^*(a); a\lambda)$.

Lower boundary. At $a = 0^+$: $au \to 0$ while $h_{un}(\xi^*) > 0$ (bounded below), so $V(0^+) < 0$.

Monotonicity. Differentiating $V(a)$:

$$V'(a) = u - h'_{un}(\xi^*)\frac{d\xi^*}{da} - \frac{\partial h_{un}}{\partial(a\lambda)}\lambda.$$

I show $V'(a) > 0$ at any zero of V, which suffices for uniqueness. The argument proceeds in two steps.

Step (i): Computing $\partial h_{un}/\partial(a\lambda)$. The same quotient-rule computation as in Lemma 9 (applied to h_{un} instead of $\bar{h}$, with $(1-G)$ replacing g) yields: $\partial h_{un}/\partial(a\lambda) = h_{un}(\xi^* - E^{un}[s])$, where $E^{un}[s] > 0$ is the posterior mean shock age under the uninformed agent's measure.

Step (ii): Evaluating $V'(a)$ at a zero. At any zero of V (where $au = h_{un}(\xi^*)$), grouping terms: $V'(a) = \underbrace{u - \frac{\partial h_{un}}{\partial(a\lambda)}\lambda}_{\text{Terms 1+3}} + \underbrace{|h'_{un}(\xi^*)| \cdot \frac{d\xi^*}{da}}_{\text{Term } 2 \geq 0}$. For Term 2: the uninformed ODE gives $h'_{un}(\xi^*) = au(k(\xi^*) - au)$ with $k = a\lambda - \mathrm{HR}_G(\xi^*)$. This is ≤ 0 iff $a(\lambda - u) \leq \mathrm{HR}_G(\xi^*)$. Using $a\lambda\bar{\eta} \leq 1$: $a(\lambda - u) = a\lambda(1 - u/\lambda) \leq (1 - u/\lambda)/\bar{\eta} \leq \kappa \leq G(\xi^*) = \mathrm{HR}_G(\xi^*)$, where the second inequality uses $\kappa\bar{\eta} + u/\lambda \geq 1$ and the third uses the crash condition. Combined with $d\xi^*/da \geq 0$ (Lemma 9), Term 2 ≥ 0. For Terms 1+3: substituting $h_{un} = au$ gives $u[1 - a\lambda(\xi^* - E^{un}[s])]$. Since $E^{un}[s] > 0$ and $a\lambda\xi^* \leq a\lambda\bar{\eta} \leq 1$, the bracket is positive, so Terms 1+3 > 0.

Combining: $V'(a) > 0$ at any zero of V.

Conclusion. Since $V(0^+) < 0$ and $V'(a) > 0$ at any zero, V crosses zero at most once. If such a zero $\bar{a} \in (0, \underline{a})$ exists, define $\bar{\beta} := 1/\bar{a}$; otherwise set $\bar{a} = \underline{a}$ (i.e., $\bar{\beta} = \underline{\beta}$). In either case $\underline{\beta} \leq \bar{\beta}$, and the three regimes follow:

- $\beta \leq \underline{\beta}$ (i.e., $a \geq \underline{a}$): No run equilibrium exists, $V(a) > 0$, agents participate.

- $\beta \in (\underline{\beta}, \bar{\beta})$ (i.e., $a \in (\bar{a}, \underline{a})$): Run equilibrium exists, $V(a) > 0$, agents participate.
- $\beta \geq \bar{\beta}$ (i.e., $a \leq \bar{a}$): $V(a) \leq 0$, agents do not participate.

When $V(a) \leq 0$, the non-participation conclusion for $t \geq \eta$ and the unraveling to $t = 0$ for $t < \eta$ follow from the same argument as Part 3(b)–(c) of Proposition 2. $\square$

B Data and Empirical Results

This appendix provides detailed documentation of the data sources, methodology, and robustness checks underlying the empirical analysis. I also include a short review of the existing empirical literature on the 2023 banking crisis, placing my contribution within that field.

B.1 Perspectives on the 2023 Regional Bank Crisis

The regional banking crisis of March 2023 emerged following Federal Reserve interest rate hikes during 2022–2023 that eroded the value of fixed-rate securities held by many regional banks. Silicon Valley Bank (SVB) announced a $1.8 billion loss on securities sales on March 8, 2023, triggering a rapid deposit run; two days later, SVB failed after $42 billion in deposit outflows. Signature Bank followed on March 12, and First Republic collapsed on May 1, while several other regional banks faced severe stress but survived. Regulatory interventions announced on March 12, including the Federal Reserve's Bank Term Funding Program and FDIC guarantees of uninsured deposits, sought to prevent contagion. The crisis prompted commentary attributing failures to structural balance sheets, digital banking technologies, and social media—with one puzzle: why did some banks with similar fundamentals survive while others failed? In empirical finance, two complementary strands of literature have emerged to answer this question.[24]

The first emphasizes fundamental risk stemming from a shift in the composition of bank value. A series of papers[25] document that monetary tightening had strong adverse effects on bank balance sheets. Banks like SVB that invested heavily in long duration assets experienced significant mark-to-market depreciation after the rate hikes. Loss of asset value, however, does not suffice to explain the sudden, *coordinated*, nature of depositor withdrawals. Drechsler et al. (2025) highlights a complementary effect of monetary tightening: while asset values declined, deposit franchise values actually increased because higher market rates allowed banks to capture larger spreads on deposits. Increased reliance on deposit franchises subjected banks to solvency runs, particularly when most depositors are uninsured (as was the case for SVB). Indeed, those franchises are fragile: if depositors run, the franchise is destroyed, destroying the bank's value overall and thereby justifying the run in the first place.

The second strand focuses on depositor characteristics. Some papers emphasize the *individual* behavior of depositors, documenting that certain clienteles—such as uninsured depositors or those from sectors under pressure (crypto, venture capital)—were particularly sensitive to fragility concerns.[26] Other papers, motivated by the historical speed of the run, emphasize the *collective* behavior of depositors, examining how relationships among depositors facilitated coordination. Cookson et al. (2023) demonstrate that banks with high

[24] I refer to Kelly and Rose 2025 and Cipriani et al. 2024 for a very rich account of the crisis and the subsequent literature.

[25] See notably Jiang et al. (2024), Choi et al. (2023) and Koont et al. (2024).

[26] Chang et al. (2023) emphasize the fundamental role of uninsured depositors in propagating the crisis, documenting that uninsured depositors represent valuable clients whose loan demand creates core lending relationships, rather than merely unstable funding sources. Kelly and Rose (2025) show that banks with business models oriented toward sectors under pressure (crypto, venture capital) faced greater stress, independent of their fundamental balance sheet exposures.

pre-existing Twitter exposure lost 4.3 percentage points more stock value during the SVB run, showing that social media relationships acted as a catalyst for coordinating depositor withdrawals. Benmelech et al. (2024) find that bank branch density—a proxy for physical coordination frictions—significantly affected run vulnerability. My paper proposes a model formalizing insights from that strand of the literature.

B.2 Data

My empirical analysis combines multiple datasets to construct measures of bank connectedness and vulnerability during the 2023 regional banking crisis.

Data sources.

- **Summary of Deposits (SOD) Data.** Branch-level deposit information for 2022 was obtained from the FDIC's Summary of Deposits (SOD) database. Data were accessed via the FDIC public API endpoint (`https://banks.data.fdic.gov/api/sod`). SOD data report branch conditions annually as of June 30.

- **Call Report Data.** Balance sheet quantities and deposit franchise estimates are obtained following Drechsler et al. (2025). Their dataset is constructed from quarterly regulatory filings (FFIEC Call Reports, forms 031/041) accessed via the Wharton Research Data Services (WRDS) platform, following standard procedures for variable definitions and time-series harmonization.

- **Stock Price Data.** Daily stock price and return data are obtained from the Center for Research in Security Prices (CRSP) database. Intraday trade data are obtained from the Trade and Quote (TAQ) database. Both CRSP and TAQ data are accessed via Wharton Research Data Services (WRDS).

- **Matching.** Data from the FDIC (SOD) and FFIEC (Call Reports) are matched at the Bank Holding Company (BHC) level using the holding company's RSSD ID. These regulatory data are then linked to CRSP daily stock data by mapping BHC RSSD IDs to CRSP PERMCOs using standard WRDS linking tables. For intraday TAQ data, historical stock identifiers (tickers) associated with the BHC's CRSP PERMCO are retrieved from CRSP's name history.

Sample. My analysis focuses on medium-to-large commercial banks with significant domestic deposit bases. The sample includes Bank Holding Companies (BHCs) with at least \$3 billion in total assets (as of Q4 2021) and substantial domestic deposit operations (deposit-to-asset ratio above 65%, excluding broker-dealers, credit card banks, custodians, and foreign-owned banks).[27] The focus on medium-to-large institutions ensures the sample includes

[27] Sample criteria are inherited from Drechsler et al. (2025), whose deposit franchise estimates I employ, see below. Their estimates cover commercial banks (not BHCs) with at least \$1 billion in total assets, though I further restrict the analysis to BHCs above \$3 billion.

banks comparable in scale to those most affected during the 2023 crisis and maintains sufficient variability in depositor dispersion.[28] I exclude banks with deposits entirely concentrated in a single branch. All results are reported at the BHC level, with branch-level and bank-level data aggregated accordingly. The sample comprises 278 BHCs, of which 176 are publicly traded.[29]

Data Access Timing. All data retrieved from the FDIC API and WRDS were accessed on April 2, 2026.

B.3 Variable Construction and Descriptive Statistics

This section documents the construction of the key empirical measures used in Section 2: the solvency measure κ, depositor concentration (HHI), and cumulative stock returns during the crisis period. I then present descriptive statistics characterizing the distributions and properties of these measures across the sample of Bank Holding Companies.

B.3.1 Methodology

Deposit Franchise Value Construction. I obtain deposit franchise values following Drechsler et al. (2025)'s methodology. The total economic value of a bank is decomposed into its equity value and its deposit franchise value:

$$\underbrace{A - D}_{\text{MTM Equity Value}} + \underbrace{DF_I + DF_U}_{\text{Deposit Franchise Value}},$$

where A is the marked-to-market value of assets, D is the book value of total deposits, DF_I is the franchise value of insured deposits, and DF_U is the franchise value of uninsured deposits. I briefly review their construction here; see Drechsler et al. (2025) for complete details.

The marked-to-market equity value, $A-D$, is computed by adjusting reported equity (Q4 2022 Call Reports) for unrealized gains or losses on securities and loans through February 2023, using Bloomberg price indices following Drechsler et al. (2025).

The deposit franchise value for each deposit type (insured/uninsured) equals the net present value of expected future profits from these deposits. Drechsler et al. (2025) provides a methodology to estimate deposit betas for uninsured and insured deposits separately, combining these with estimated operating costs to determine deposit franchise value for each group. I use their deposit franchise values as of February 2023.

I obtain deposit franchise and equity values separately for each bank within a BHC from Drechsler et al. (2025), then aggregate to the holding company level using asset-weighted means. All components ($A - D$, DF_I, DF_U) are expressed as ratios to total book assets for comparability across institutions.

[28] Smaller banks are generally local and concentrated, making it hard to separate effects stemming from concentration and from size. The $3 billion threshold is chosen to match regulatory guidance: BHCs under $3 billion are not required to file the consolidated FR Y-9C (they file instead FR Y-9SP).

[29] A few publicly traded banks are not represented in the TAQ dataset, further reducing the sample for high-frequency analyses.

I introduce the solvency measure κ. It is defined as the critical fraction of total deposit franchise value $(DF_I + DF_U)$ that must be lost for the bank's total value to fall to a distress threshold, $\underline{v}$, set at 3% of book assets in the implementation.[30] That is, κ solves:

$$A - D + (1-\kappa)(DF_I + DF_U) = \underline{v},$$

where $A - D$, DF_I and DF_U are all expressed as ratios to total book assets. In principle $\kappa \in [0, 1]$, where $\kappa = 1$ implies robust solvency (economic equity remains above $\underline{v}$ even if the entire deposit franchise is lost). However, I do not cap κ at 1 in my empirical analysis. It allows me to better gauge the distance from this robust solvency point.[31]

HHI Calculation. I measure depositor concentration using the Herfindahl-Hirschman Index (HHI) of deposits for each BHC. The HHI is defined as:

$$\text{HHI} = \sum_{k \in \text{Branches}} \left(\frac{\text{Deposits in branch } k}{\text{Total Deposits in BHC}} \right)^2,$$

where the summation is over all branches k belonging to the BHC. This measure captures the probability that two randomly selected dollars of deposits within the BHC come from the same branch.

Cumulative Stock Returns. I compute cumulative stock returns for each BHC over the crisis period from March 6, 2023 (close of trading) to March 13, 2023 (close of trading), using daily total returns (including dividends) from the CRSP database. The cumulative return for BHC i over this period is calculated by compounding the daily returns:

$$\text{Cumulative Return}_{i,[06/Mar,13/Mar]} = \left(\prod_{t=\text{Mar } 7}^{\text{Mar } 13} (1 + \text{ret}_{i,t}) \right) - 1,$$

where $\text{ret}_{i,t}$ is the daily total return for BHC i on trading day t. The product is taken over the trading days from March 7, 2023, to March 13, 2023, inclusive.

B.3.2 Descriptive Statistics

Before discussing the robustness of the main findings of Section 2, I provide further descriptive analysis of my primary measures of risk: solvency κ and depositor HHI. This analysis doesn't require stock data and is performed on the full sample of BHCs, including those that are not publicly traded.

[30] 3% is chosen to match the Basel III capitalization ratios. Banks under 3% are considered severely undercapitalized.

[31] A κ of 2 for example would mean that we would need to destroy double the deposit franchise to make the bank insolvent.

Distribution of Solvency Measure κ Figure 12 presents the distribution of the solvency measure κ across BHCs in the sample. In blue are publicly traded BHCs used in regressions (176 entities), while the grey bars show the full sample (278 BHCs). The distribution shows heterogeneity: while many banks cluster around $\kappa = 1$, a significant fraction falls below this threshold, indicating vulnerability to deposit franchise losses. Banks with $\kappa > 1$ tend to stay near unity but do include a notable right tail of highly robust institutions.

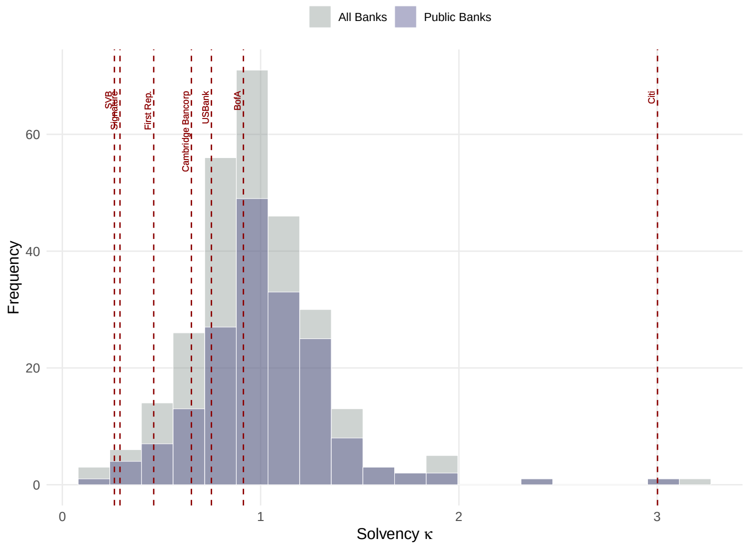
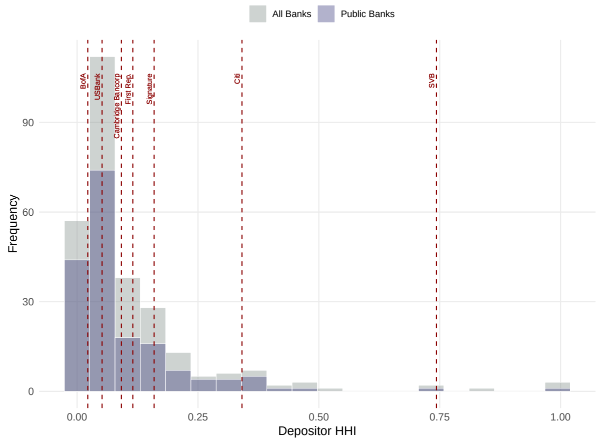


Figure 12: Distribution of κ across BHCs. This figure shows a histogram of the solvency measure κ across BHCs in the sample. Most banks cluster near $\kappa = 1$ (solvent), with a notable left tail of vulnerable banks ($\kappa < 1$) including crisis-affected institutions.

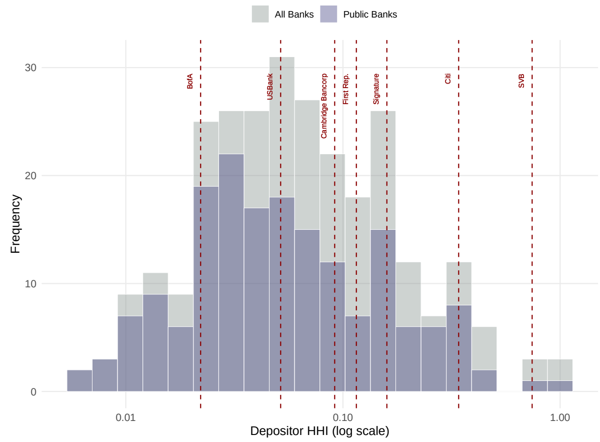
Distribution of HHI. Figure 13a shows the HHI distribution across BHCs, ranging from near-zero to close to one. Figure 13b represents the same distribution, but the x-axis is in log scale. Most banks exhibit relatively low HHI values (below 0.1), indicating diversified deposit bases, though a substantial minority shows high concentration. Traditional retail banks like Bank of America have low HHI, while SVB and Citigroup appear as high-concentration outliers. All banks experiencing difficulties during the 2023 crisis had high HHI values, with SVB representing the most extreme case.

Evolution of HHI. Figure 14 shows the evolution of average deposit HHI for BHCs from 1994 to 2022.[32] I also plot a selected sample of individual banks to illustrate their evolution. Note the y-axis is in log scale. The data reveal a secular increase in average HHI since 1994, somewhat slowed by the financial crisis and COVID-19, reaching historically high levels in the 2020s. Average HHI nearly doubled: from around 0.03 in 1994 to about 0.06 in 2022. While some larger banks such as Citi and US Bank substantially increased their HHI since 2005, the banks most impacted by the 2023 crisis displayed high HHI well before the event.

[32] The national average includes all commercial banks with over \$3 billion in assets (in 2021-equivalent dollars, adjusted for inflation using CPI) and over 0.65 domestic deposits to asset ratio. The average is weighted by BHC asset size.



(a) Distribution of HHI across BHCs (linear scale).



(b) Distribution of HHI across BHCs (log scale).

Figure 13: Distribution of HHI across BHCs. This figure shows histograms of depositor HHI in linear (left) and log (right) scales. Most banks exhibit low HHI (diversified deposits), but a substantial minority shows high concentration. Crisis-affected banks (SVB, First Republic, Citi) appear as high-HHI outliers.

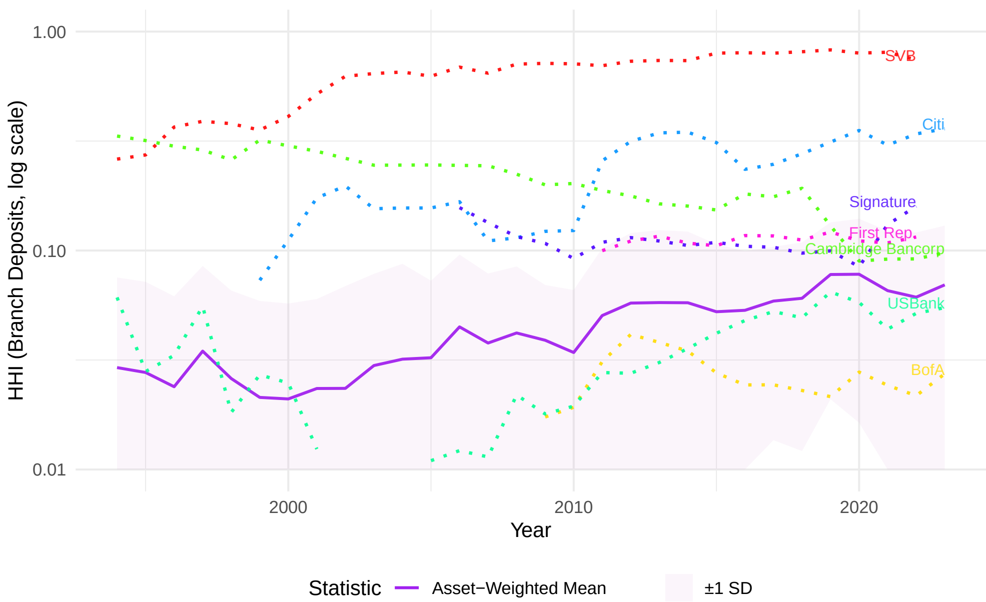


Figure 14: Evolution of Average Deposit HHI for medium to large BHCs (1994–2022). This figure plots the asset-weighted average HHI (purple line) with ±1 SD band (shaded) and selected individual banks (dotted), using a log scale on the y-axis. Average HHI shows a secular increase from ∼0.03 (1994) to ∼0.06 (2022), with crisis-affected banks already exhibiting high HHI pre-2023.

Correlation with Deposit Rates. Table 4 and Figure 15 explore the relationship between BHC-level HHI and the average deposit rates paid by these institutions.[33] I find a strong positive correlation, suggesting that banks with a more concentrated depositor base (higher HHI) tend to offer higher deposit rates. In the context of my model, this correlation could be explained by an equilibrium effect: banks with a more concentrated deposit base compensate elevated risk by offering higher rates. More generally, such correlation might be explained by a "deposit clientele effect" as proposed by Benmelech et al. (2024).[34]

Dependent Variable:	**Deposit Rate**				
	(1)	(2)	(3)	(4)	(5)
Depositor HHI	0.022***	0.023***	0.022***	0.023***	0.022***
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Dom. Deposits (log)		0.000		0.000	0.001***
		(0.000)		(0.000)	(0.000)
Uninsured Share			0.001	-0.000	-0.004**
			(0.002)	(0.002)	(0.002)
Wholesale sh. of dep.					0.044***
					(0.006)
Constant	0.006***	0.001	0.005***	0.000	-0.005*
	(0.000)	(0.003)	(0.001)	(0.003)	(0.003)
Observations	278	278	278	278	278
R^2	0.400	0.405	0.400	0.405	0.501

*Note: *** p<0.01, ** p<0.05, * p<0.1.*

Table 4: Relationship between Depositor HHI and Deposit Rates (2022Q4).
Note: OLS regressions showing positive correlation between HHI and deposit rates. Column (4) with full controls shows an HHI coefficient of 0.022, suggesting concentrated banks offer higher rates.

One may worry that the effect of HHI on run severity could be confounded by deposit rates. Indeed, higher deposit rates can reflect higher deposit betas (implying lower deposit franchise values) and may influence depositor behavior by making deposits more attractive to retain (as argued in my model). Notice, however, that such factors would imply *smaller* runs, not larger ones, as I document.

Depositor concentration in the literature. I discuss here two related papers that also study depositor concentration, albeit they propose different measures. Already mentioned is the study by Benmelech et al. (2024). They study "bank density", defined as the number of branches over the deposit volume per BHC. They find that banks with lower density (fewer branches per dollar of deposits) were more vulnerable to runs during the 2023 crisis, consistent with my HHI findings. While related, bank density does not capture the possibility

[33] Deposit rate is computed at the BHC level as the ratio between total interest paid on deposits (net) and total deposits. Wholesale share of deposit is the share of time deposits above $\$250k$ on total domestic deposits.

[34] See the discussion of that paper below.

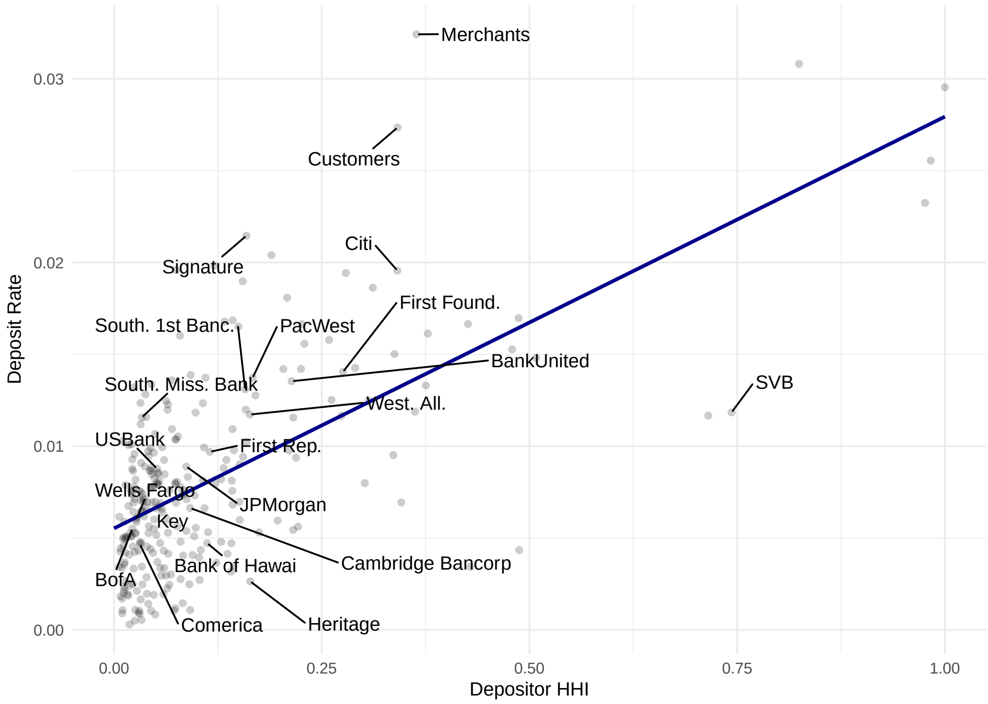


Figure 15: Scatter plot of HHI vs. Average Deposit Rate (2022). This figure shows a scatter plot with fitted regression line illustrating the positive relationship between depositor concentration (HHI) and deposit rates. Selected banks are labeled, including crisis-affected institutions at the high-HHI end.

that banks might have concentrated deposits in a few branches while having a large number of branches overall—a nuance better captured by HHI. Benmelech et al. (2024) interpret their results as evidence of a "deposit clientele effect", where concentrated banks cater to urban, educated, financially sophisticated depositors. My interpretation focuses more directly on HHI's role in facilitating depositor communication. Overall my view is that both frameworks are in agreement: they emphasize that information spreads faster within concentrated deposit bases.

Kundu et al. (2025) provide yet another alternative measure of depositor concentration: the share of deposits accumulated within the single largest branch (or county) of a BHC. They find that, on average, 30% of a bank's deposits are concentrated in a single county. Their measure is, in a sense, at the opposite end of the spectrum from Benmelech et al. (2024) bank density: the latter abstracts completely from the within-bank repartition of deposits, while this measure focuses on the largest single unit (branch or county) and ignores smaller ones. My HHI measure lies somewhere in the middle, capturing the overall distribution. The study by Kundu et al. (2025) is not focused on the 2023 runs but rather on the macroeconomic effects of the geographical concentration of deposits. Methodologically, Kundu et al. (2025) provide strong evidence supporting the validity of using SOD data for computing depositor concentration, particularly documenting that reporting errors should not be a major concern.[35]

B.4 Robustness and Graphical Evidence

This section validates the main empirical findings from Section 2. I first present graphical evidence illustrating the interaction between depositor concentration and fundamental solvency documented in the regression analysis. I then assess robustness of the main results across alternative risk measures, control specifications, and sample restrictions.

Specification. As discussed in the main text, I focus on two baseline models:

$$\text{Stock Return}_i = \beta_1 \text{HHI}_i + \beta_2 \kappa_i + \text{Controls}_i + \epsilon_i, \tag{8}$$

$$\text{Stock Return}_i = \beta_1 \text{HHI}_i + \beta_2 \mathbb{1}_{\kappa_i<1} + \beta_3 \text{HHI}_i \times \mathbb{1}_{\kappa_i<1} + \text{Controls}_i + \epsilon_i, \tag{9}$$

where the dependent variable is the total stock return from March 6 to March 13, 2023. Specifications of controls vary across different robustness exercises, as discussed below.

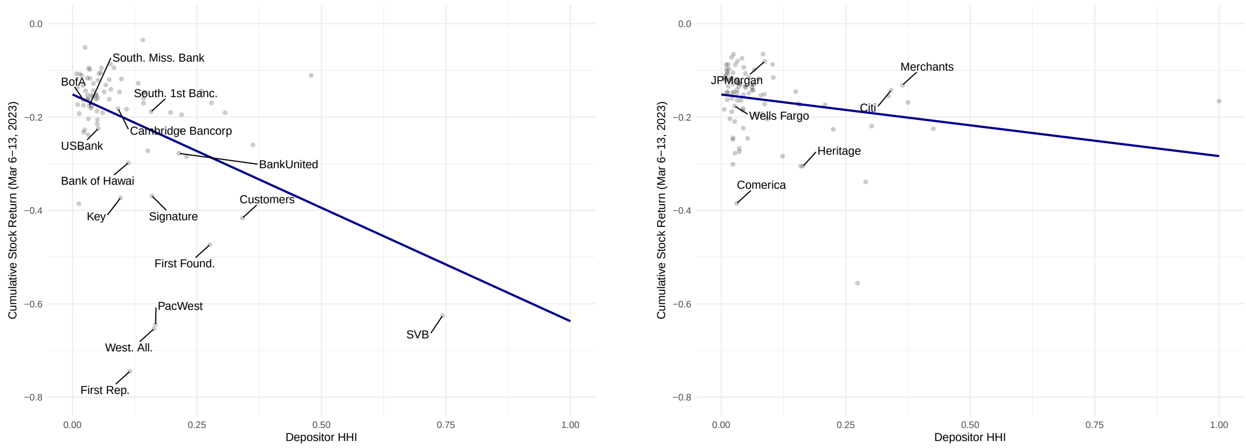
B.4.1 Graphical Evidence

Figure 16 provides visual evidence for the interaction effect documented in the main text (Table 1). The figure plots the relationship between depositor concentration (HHI) and cumulative stock returns during the crisis period (March 6–13, 2023), separately for vulnerable banks ($\kappa < 1$, left panel) and robust banks ($\kappa \geq 1$, right panel).

Among vulnerable banks ($\kappa < 1$), the negative relationship between HHI and stock returns is pronounced. Banks with high depositor concentration experienced substantially

[35] Indeed, a potential issue of the SOD data is that banks can choose where to report deposits. Interpreting SOD branch reporting requires careful consideration of potential biases in such choices.

larger price declines, with the linear fit capturing a steep downward slope. In contrast, among robust banks ($\kappa \geq 1$), the relationship is notably weaker. Most observations cluster in a region of moderate price drop regardless of HHI level, and the fitted line is much flatter. This visual evidence directly confirms the regression-based interaction effect: depositor concentration's impact on crisis outcomes appears clearly when fundamental vulnerabilities are present, but plays little role when banks have strong solvency buffers.



(a) $\kappa < 1$ (Vulnerable Banks) (b) $\kappa \geq 1$ (Robust Banks)

Figure 16: Scatter plots of Cumulative Stock Returns vs. HHI by Solvency Subsample. This figure shows the split-sample analysis: among vulnerable banks ($\kappa < 1$, left panel), there is a steep negative relationship—banks with high HHI experienced the largest declines. Among robust banks ($\kappa \geq 1$, right panel), the relationship is flat, with stock returns largely unaffected by HHI.

B.4.2 Robustness

Alternative risk measures and controls. To verify that the coordination risk captured by HHI is distinct from fundamental vulnerabilities, I assess the stability of the HHI coefficient across alternative specifications. I substitute κ with: (i) marked-to-market losses (as percentage of 2021 Q4 book value); and (ii) uninsured solvency κ_u, which is defined similarly to κ but only uninsured deposits can run ($\underline{v} = A - D + DF_I + (1 - \kappa_u)DF_U$). Control variables include log of domestic deposits, uninsured deposit share, and wholesale deposit share.[36] Figure 17 presents coefficient stability plots showing the point estimate of the HHI coefficient and confidence intervals across specifications, with results for both standard and winsorized HHI using conventional and robust standard errors.

The point estimate remains consistently negative, ranging between -0.2 and -0.6 across specifications. Controlling for the uninsured share of deposits seems to reduce the coefficient magnitude. Winsorizing HHI strongly increases the coefficient magnitude across specifications (i.e., more negative), an effect likely due to the skewness of the HHI distribution. The main specification (non-winsorized) is therefore conservative. Finally, using robust standard

[36]Defined as the share of time deposits above \$250k on total domestic deposits.

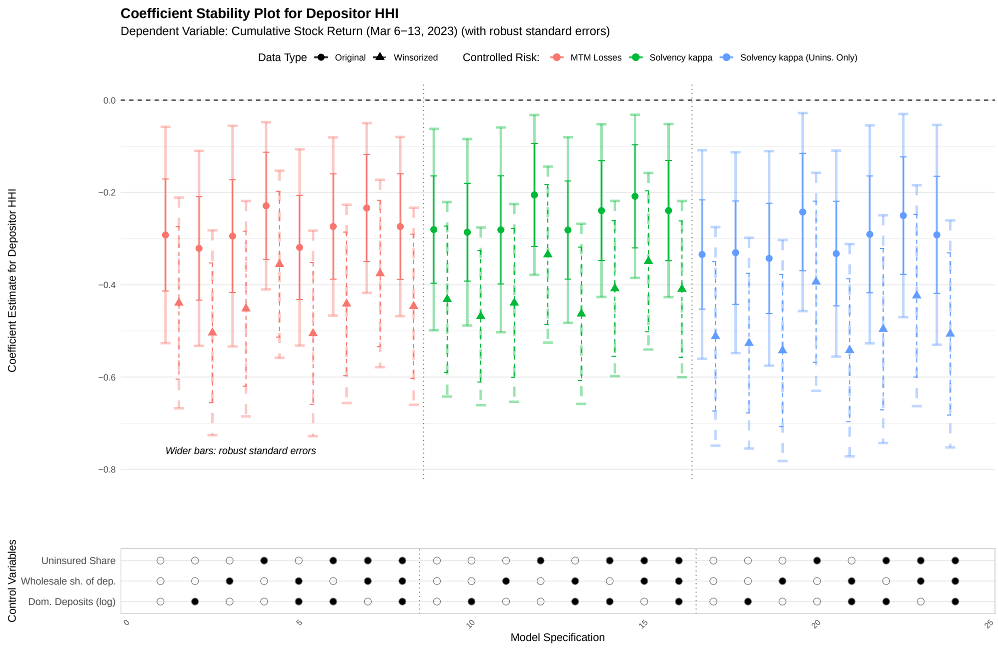


Figure 17: Coefficient stability plot for HHI. This figure shows the estimated HHI coefficient ($\hat{\beta}_1$) across specifications varying fundamental risk measures (colors) and controls. The coefficient is consistently negative (-0.2 to -0.6) across all specifications. Winsorized HHI yields larger magnitudes. Robust standard errors (wider intervals) suggest heteroskedasticity, but estimates remain significant.

errors increases confidence bounds noticeably, suggesting some heteroskedasticity in the data (again, likely due to the skewness of the HHI distribution). Estimates remain nonetheless significant.

To better visualize the role of controls, Table 5 presents the results of the main regression specifications using κ as the primary risk measure and with a full set of controls. Throughout, bank size and the share of uninsured deposits are significant, with larger banks and those with a higher share of uninsured deposits experiencing more negative stock returns (consistent with findings throughout the literature). Coefficients on HHI are robust to introducing these controls, including when considering the interaction term.

Dependent Variable:	**Cumulative Stock Return (Mar 6-13, 2023)**			
	(1)	(2)	(3)	(4)
Solvency κ	0.091***		0.089***	
	(0.021)		(0.020)	
Depositor HHI		-0.244***	-0.239***	-0.122*
		(0.058)	(0.055)	(0.071)
Indicator ($\kappa < 1$)				-0.014
				(0.017)
$\kappa < 1 \times$ Depositor HHI				-0.286**
				(0.110)
Uninsured Share	-0.231***	-0.172***	-0.162***	-0.147**
	(0.059)	(0.061)	(0.058)	(0.059)
Dom. Deposits (log)	-0.019***	-0.018***	-0.023***	-0.020***
	(0.006)	(0.006)	(0.006)	(0.006)
Wholesale sh. of dep.	-0.032	0.082	-0.025	0.019
	(0.193)	(0.192)	(0.183)	(0.185)
Constant	0.138	0.199**	0.193**	0.239***
	(0.090)	(0.091)	(0.087)	(0.090)
Observations	176	176	176	176
R^2	0.265	0.261	0.338	0.325

*Note: *** p<0.01, ** p<0.05, * p<0.1.*

Table 5: Main Regression Results: Depositor HHI and Stock Returns.

Excluding Failed Banks. To ensure that the main findings are not disproportionately influenced by institutions that failed during the March 2023 crisis period (namely, Silicon Valley Bank, Signature Bank, and First Republic Bank[37]), I re-estimate the set of regression specifications analyzed in the paragraph above, but this time excluding these failed banks from the sample. Figure 18 presents the results in the form of a coefficient stability plot.

The estimated HHI coefficient remains consistently negative and mostly statistically significant across the various specifications, albeit slightly smaller in magnitude. Confidence intervals cross zero (with robust standard errors) in some very specific cases: when using κ_u as the measure of fundamental risk and controlling for uninsured deposits simultaneously. If HHI values are winsorized, the estimates are robust.

[37]Silvergate Bank, having failed prior to the March 6–13, 2023 estimation window, is already excluded from the primary sample.

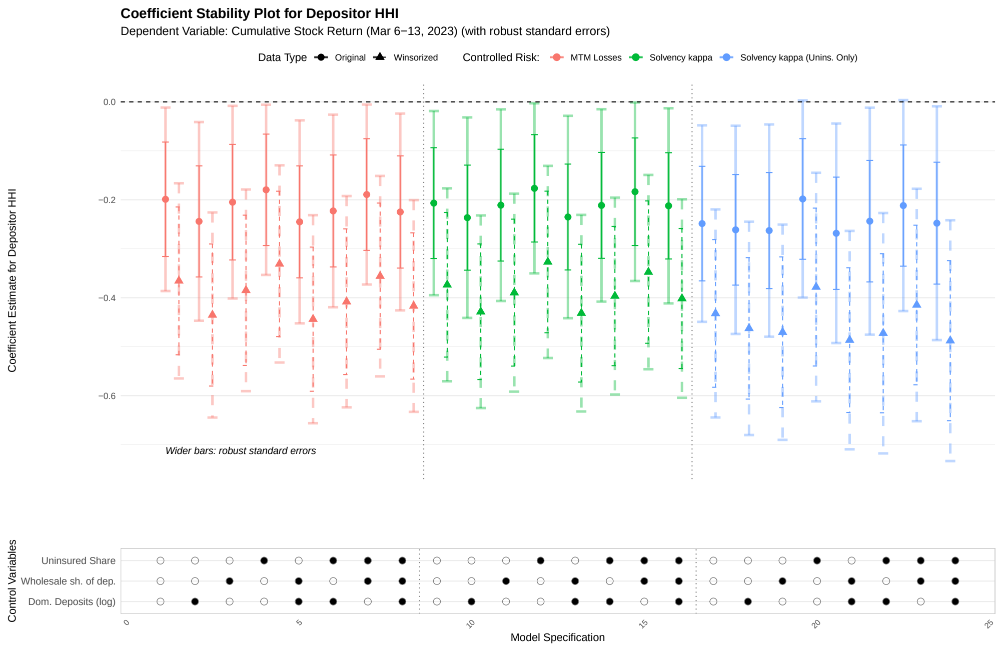


Figure 18: Coefficient stability plot for HHI (excluding failed banks). This figure repeats the stability analysis excluding SVB, Signature, and First Republic. The HHI coefficient remains negative and mostly significant, though slightly smaller in magnitude. Some specifications with κ_u and uninsured controls cross zero with robust standard errors, but winsorized estimates remain robust.

B.5 Model Fitting

This section provides details on the method and results used in Section 6 when fitting the model to high-frequency stock return data.

B.5.1 Methodology

I use high-frequency intraday stock price data sourced from the Trade and Quote (TAQ) database (available on WRDS). TAQ provides comprehensive trade and quote information for U.S. equities, identified by ticker. I map the RSSD IDs of banks in my sample to their CRSP PERMCOs, then to TAQ via tickers using CRSP name history. TAQ data is queried from March 6, 2023 (9:00 AM) to March 21, 2023 (4:50 PM)[38], covering trading days during and immediately following the acute crisis period. Individual trade prices are aggregated to 10-minute intervals, then smoothed using a one-hour centered rolling window. Returns $R_{obs}(t)$ are derived as $R_{obs}(t) = (1 - P_s(t)/P_s(0))$, where $P_s(t)$ is the smoothed price at time t and $P_s(0)$ is the initial smoothed price (March 6, 2023, 9:00 AM). SVB's time series is truncated at the close of March 9, 2023, the last trading day before closure.[39]

B.5.2 Additional Results

Model Fit Examples. The Nonlinear Least Squares (NLS) estimation procedure yields a satisfactory fit to observed stock price dynamics for most banks in the sample. Figures 19 to 21 provide illustrative examples beyond those in the main text, comparing smoothed observed stock returns (cumulative percentage decline) against values predicted by the fitted model. Examples are organized into three categories based on severity: (1) large shocks, (2) medium shocks, and (3) small or no shocks. Each group uses its own scale for clarity.

For banks experiencing large shocks, the estimated curve $R_{model}(t)$ effectively captures the overall decline dynamics, including the initial rapid drop whose speed reflects the estimated $\hat{\beta}$ (particularly evident for SVB). Among banks that survived, the model captures the beginning of stabilization as peak withdrawals are reached (e.g., Customers Bancorp, Western Alliance), with correspondingly high estimated peaks. For medium-shock banks, estimated peaks are substantially lower, and the fitted curves exhibit the beginning of price recovery. Finally, for banks with small or no shocks, model fit is generally weaker, though the logistic curve still captures some price movements. These banks exhibit low estimated peaks and relatively flat fitted curves.

In all groups, instances of poorer fit or non-convergence can arise when a bank's stock price stabilizes for an extended period after the initial decline without a prompt reversal, or if the recovery is very slow. Since the specified $R_{model}(t)$ implies a symmetric profile around the peak withdrawal intensity (proxied as maximum price decline rate), it is less adept at capturing such slow, asymmetric recoveries. This limitation is visible for banks like PacWest Bancorp, where an initially rapid price decline is not followed by a correspondingly swift

[38]While I computed cumulative returns only through March 13 for the results in Section 2, I continue through March 21 to include some of the recovery phase in model fitting. Each trading day extends from 9:00 AM to 4:50 PM to capture regular trading hours.

[39]Signature Bank, seized on March 12, is excluded from the NLS estimation: its stock declined only 40% before the halt, leaving insufficient curvature for the NLS procedure to converge.

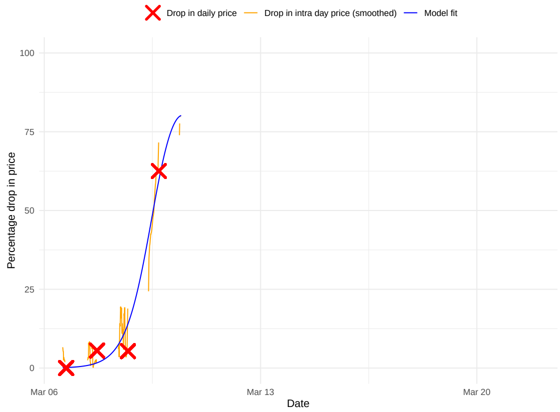
reversal within the estimation window. In such cases, the NLS procedure may estimate a lower $\hat{\beta}$ than warranted by the initial drop, effectively attempting to "postpone" the model-implied reversal beyond the observed data to achieve a better overall fit.

Richer models might better capture these asymmetric recovery patterns (see the extension section). Given the illustrative purpose of this model fitting exercise in the broader context of the paper, I retain the current specification which has the advantage of obtaining a simple closed form for aggregate withdrawals.

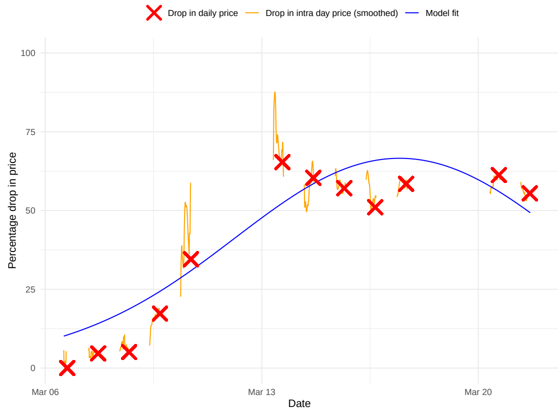
Goodness of Fit. To complement the individual examples, I assess overall model fit across all banks using the R^2 from the NLS estimation, which quantifies the proportion of variance in observed stock returns explained by the fitted model $R_{model}(t)$. Figure 22 presents the distribution of values.

The median R^2 is approximately 0.82, indicating that the model explains a substantial portion of variance in stock returns for at least half of the banks. The distribution is somewhat left-skewed, with a tail of lower R^2 values corresponding to banks where model fit was less precise.

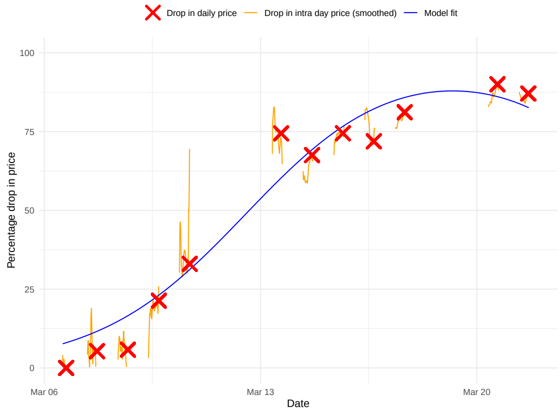
These R^2 values generally indicate strong explanatory power, though goodness-of-fit measures for logistic-based models warrant cautious interpretation. Logistic functions are inherently flexible and can adapt to various S-shaped patterns, potentially yielding high R^2 values even when the underlying data generating process deviates from a true logistic form. Nevertheless, combined with visual evidence from the fit examples, the overall high R^2 values support the model's utility for extracting dynamic parameters $(\hat{t}_{mid}, \hat{\beta}, \hat{\tau})$ that characterize observed stock price movements during the crisis.



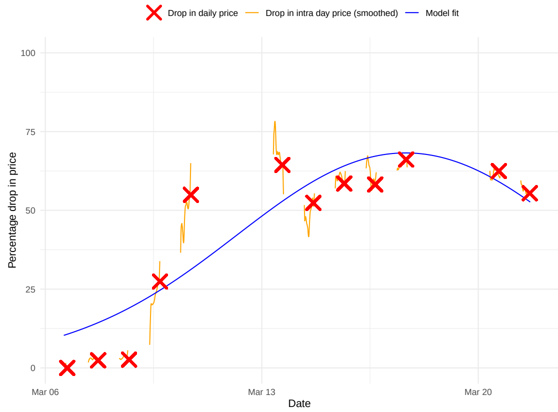
(a) SVB



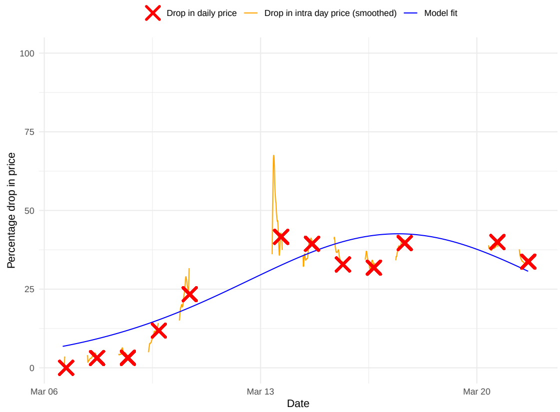
(b) Western All.



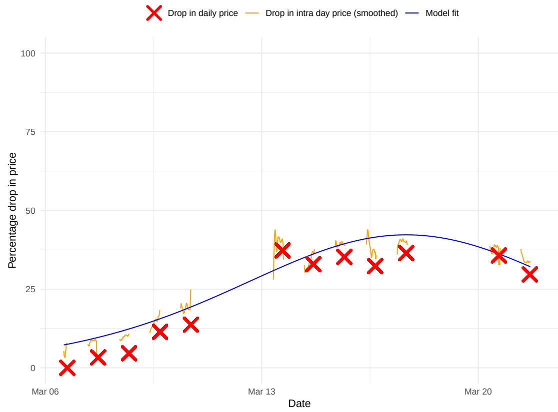
(c) First Rep.



(d) PacWest

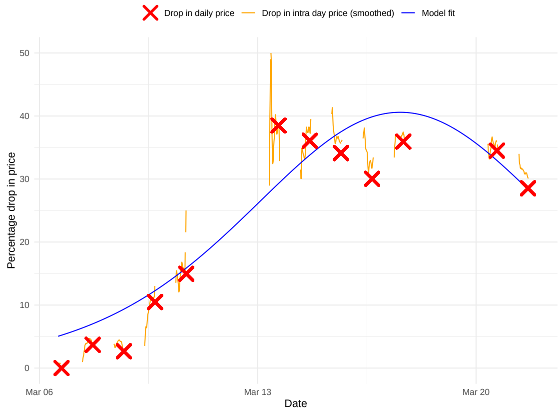


(e) Customers

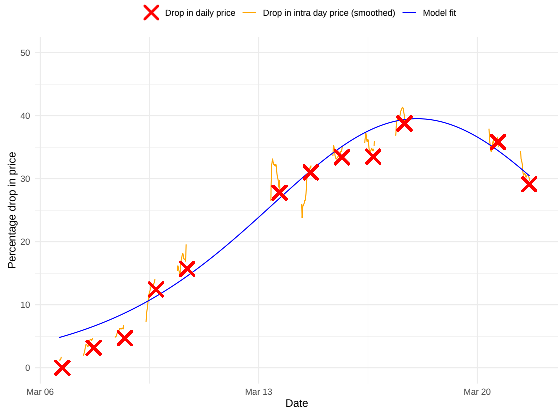


(f) KeyCorp

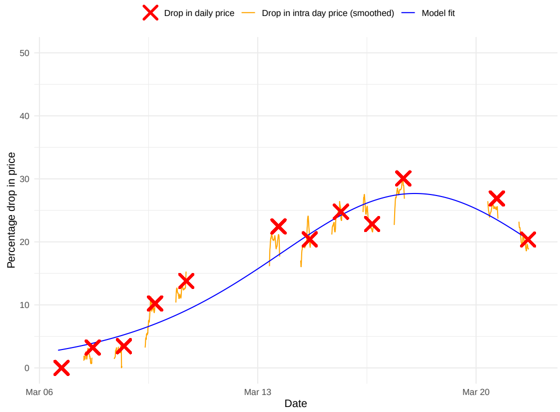
Figure 19: Stock prices and model fit for selected banks (Group: Large Shock). This figure shows TAQ intraday prices (orange), CRSP daily returns (red), and fitted logistic curves (blue) for banks experiencing large shocks. The model captures the rapid initial decline (high $\hat{\beta}$) and stabilization onset. SVB shows a truncated series at closure.



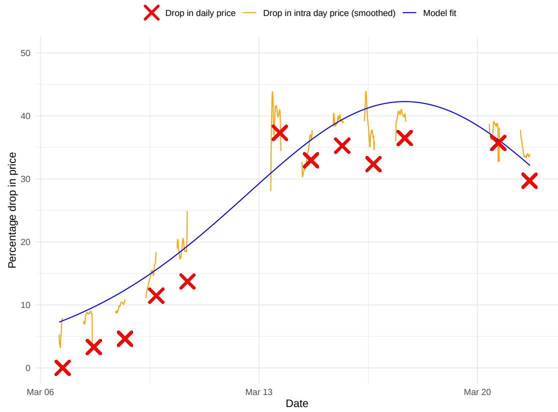
(a) Comerica



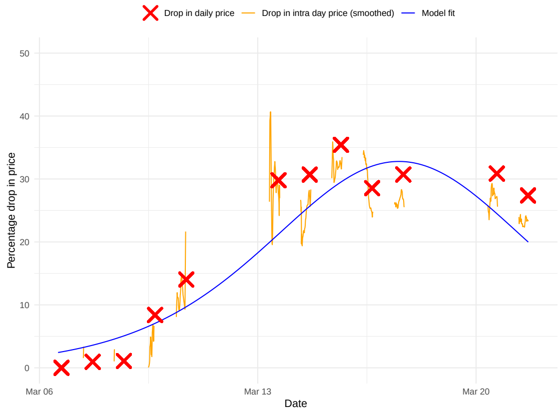
(b) BankUnited



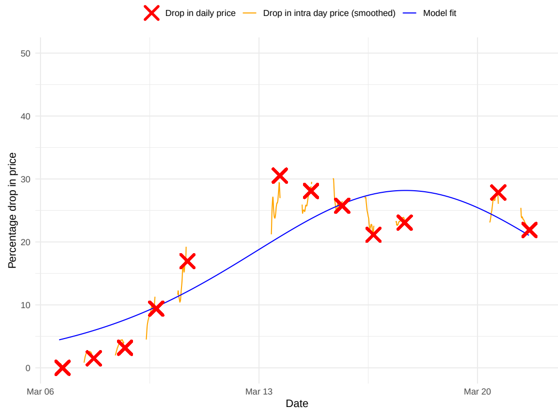
(c) USBank



(d) Key

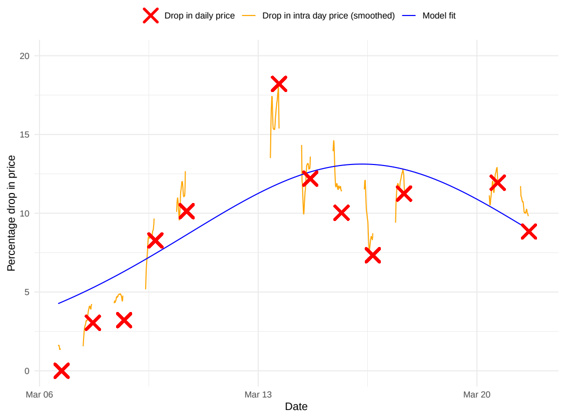


(e) Bank of Hawaii

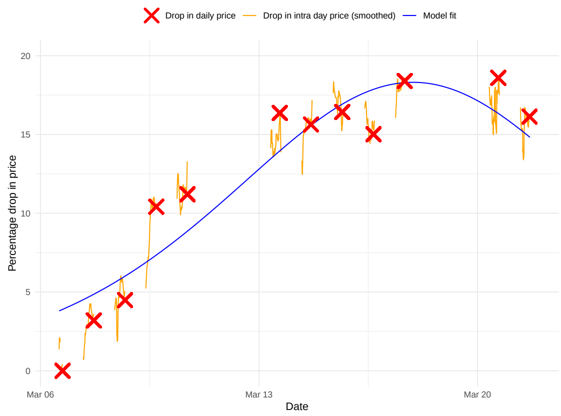


(f) Heritage

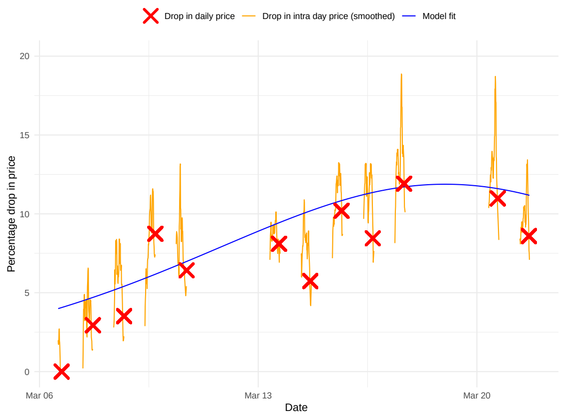
Figure 20: Stock prices and model fit for selected banks (Group: Medium Shock). This figure shows model fits for banks with moderate shocks. Lower estimated peaks ($\hat{\tau}_I$) compared to the large-shock group. Fitted curves capture the decline and beginning of recovery. Mid-range $\hat{\beta}$ values produce intermediate-speed dynamics.



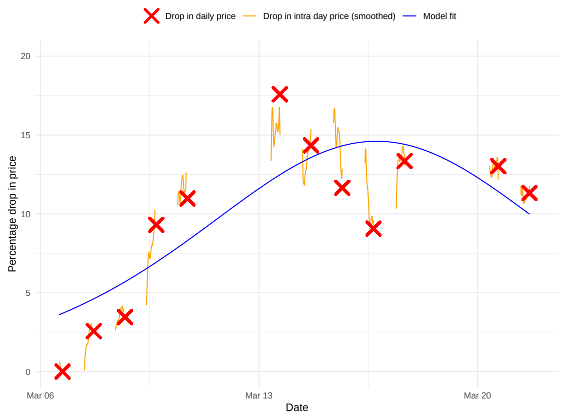
(a) Cambridge Bancorp



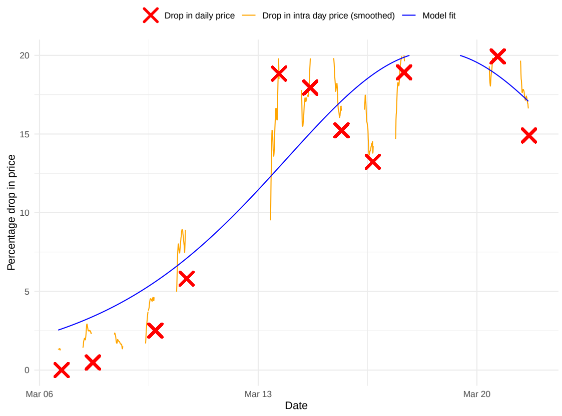
(b) BofA



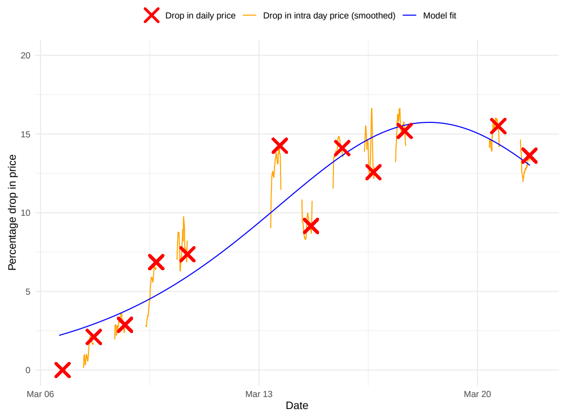
(c) JPMorgan



(d) South. Miss. Bank



(e) South. 1st Banc.



(f) Citi

Figure 21: Stock prices and model fit for selected banks (Group: Small Shock). This figure shows results for banks with minimal or no shocks. The model fit is generally weaker but captures some price movements. Low estimated peaks and relatively flat fitted curves characterize this group. Large banks (JPMorgan, Bank of America) show minimal price response despite high liquidity.

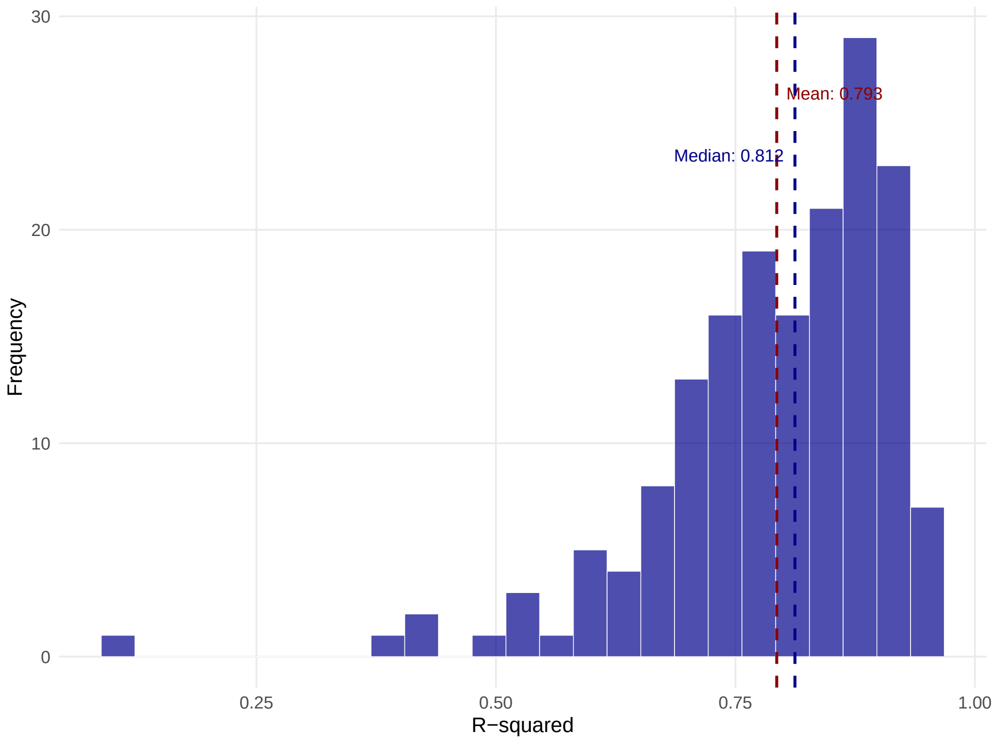


Figure 22: Histogram of R^2 of estimated NLS model on individual banks. This figure shows the distribution of R^2 from NLS estimation across all banks. Median ≈ 0.82 indicates the model explains substantial variance. The left-skewed distribution has a tail of lower values for banks with poorer fit (asymmetric recoveries, stabilization periods).

C Computational Methods

This appendix provides detailed documentation of the numerical methods used to solve the model's equilibrium. While analytical characterization is possible for certain aspects, I solve equilibria numerically because (1) it provides more insight into how the model actually works, and (2) it extends naturally to alternative models from Section 7.

The complete Julia implementation of all computational methods described in this appendix is publicly available at:

`https://github.com/Robin-Lenoir/replication-social-bank-runs`.

C.1 Mathematical Framework

C.1.1 Problem Statement and Computational Challenge

The goal is to solve the equilibrium system formally defined in Proposition 1. The primary computational challenge arises from a fixed-point problem: the optimal withdrawal times for agents depend on the perceived hazard rate $h(\tau)$, which in turn depends on the equilibrium bank collapse time, ξ^*. Simultaneously, the collapse time ξ^* is determined by the aggregate withdrawals, which are a function of those same optimal withdrawal times. This interdependency makes a direct solver approach complex, as it would require iterating on the entire hazard rate function.

C.1.2 The Reversed Time Approach

My solution hinges on a change of variables to "reversed time", a concept formalized in Lemma A.1 of the proof appendix. Reversed time ($\bar{\tau} = \xi^* - \tau$) represents the time remaining until a potential collapse. The key observation from the lemma is that the hazard rate in reversed time, $\bar{h}(\bar{\tau})$, is completely independent of the crash time ξ^*. This allows one to solve for unconstrained optimal agent behavior *before* determining the equilibrium crash time, effectively decoupling the multi-variate fixed-point into a sequence of root-finding problems.

C.1.3 The Two Separable Equations

In reversed time coordinates, one can rewrite the equilibrium system in the following way:

Unconstrained Optimal Withdrawal Times The unconstrained optimal withdrawal policies solve:

$$\bar{\tau}_{OUT}^{UNC} = \sup\{\bar{\tau} : \bar{h}(\bar{\tau}) > u\} \tag{10}$$

$$\bar{\tau}_{IN}^{UNC} = \inf\{\bar{\tau} : \bar{h}(\bar{\tau}) > u\} \tag{11}$$

In equilibrium, the domain of $\bar{\tau}_I$ is $[0, \xi^*]$, forcing potential corner solutions. It is useful, however, to first compute the unconstrained values, because they are independent of ξ^*.

Constraint Application and Equilibrium From the unconstrained withdrawal times, we can write the aggregate withdrawal function at the moment of collapse, ξ^*, as:

$$\mathcal{A}(\xi^*) =: AW(\xi^*; \xi^*) = G(\min(\xi^*, \bar{\tau}_{OUT}^{UNC})) - G(\min(\xi^*, \bar{\tau}_{IN}^{UNC})), \tag{12}$$

where the first input in AW indicates that we evaluate aggregate withdrawal at $t = \xi^*$ and the second input makes explicit the crash time in the equilibrium under consideration. The equilibrium ξ^* must satisfy the condition $\mathcal{A}(\xi^*) = \kappa$: at crash time, equilibrium withdrawals attain the bank capacity. As discussed below, one must also verify it is the first time the implied $AW(\cdot, \xi^*)$ crosses κ.

While the final equation for ξ^* is still non-linear, the system is now decoupled. One can first solve for the unconstrained $\bar{\tau}$ values and then take them as given when solving the final equation for ξ^*. Each step involves solving for one unknown in one equation, for which standard root-finding algorithms are well-suited.

C.1.4 Staged Computational Architecture

The solution method employs a three-tier staged approach that exploits this decoupled structure:

1. **Stage 1**: Solve learning dynamics $G(t)$.
2. **Stage 2**: Solve for unconstrained optimal withdrawal times $\bar{\tau}_{IN}^{UNC}$ and $\bar{\tau}_{OUT}^{UNC}$.
3. **Stage 3**: Solve the equilibrium condition for the crash time ξ^* using $\mathcal{A}(\xi)$.

C.2 Stage 1: Learning Dynamics Solver

C.2.1 Logistic Model Formulation

The learning process follows a logistic ODE describing information diffusion through the depositor population:

$$\frac{dG}{dt} = \beta G(t)(1 - G(t)) \tag{13}$$

where $G(t)$ represents the cumulative fraction of agents who have learned about potential bank fragility by time t, and $\beta > 0$ is the communication speed parameter.

C.2.2 Problem Setup

The ODE is solved over a time domain $[0, T]$ with initial condition $G(0) = G_0$, where G_0 is a small positive number (typically 10^{-4}) to initialize the process.

C.2.3 Technical Implementation

The implementation uses standard adaptive ODE solvers. I use a 5th-order Runge-Kutta with automatic step size control. The logistic equation's multi-scale nature—rapid initial growth followed by slower convergence—is handled efficiently by the solver's automatic step-size adjustments. I set tolerances to machine epsilon ($\approx 1 \times 10^{-16}$) to ensure maximum precision for subsequent equilibrium calculations. The adaptive grid generated by the solver is inherited by all subsequent computational stages to maintain numerical consistency. The discrete solution points $(t_i, G(t_i))$ are converted to a continuous function $G(t)$ via linear interpolation.

C.2.4 PDF Computation

The learning probability density function is computed analytically from the ODE's right-hand side:

$$g(t) = \frac{dG}{dt} = \beta G(t)(1 - G(t)) \tag{14}$$

where $G(t)$ is the numerical solution from the ODE solver. I evaluate $g(t_i)$ at each grid point $\{t_i\}$ and create a corresponding interpolating function $g(t)$.

C.3 Stage 2: Computing Unconstrained Optimal Withdrawal Times

C.3.1 Problem Statement and Approach

The goal of this stage is to find the unconstrained optimal withdrawal times that solve:

$$\bar{h}(\bar{\tau}_{IN}^{UNC}) = u \quad \text{and} \quad \bar{h}(\bar{\tau}_{OUT}^{UNC}) = u \tag{15}$$

I solve this by first computing $\bar{h}(\bar{\tau})$ over its full domain and then extracting the roots from the list of values given by the precomputed function, leveraging its known unimodal structure.

C.3.2 Hazard Rate Function Computation

Grid Construction and Inheritance Strategy The computation operates on a reversed time grid derived from the forward time grid $\{t_i\}$ from Stage 1, preserving the adaptive density.

Efficient One-Shot Integration The hazard rate uses the closed-form expression from Lemma A.1, which contains an integral term that is computationally expensive to evaluate repeatedly:

$$\bar{h}(\bar{\tau}) = \frac{pe^{\lambda\bar{\tau}} g(\bar{\tau})}{p \int_0^{\bar{\tau}} e^{\lambda s} g(s) ds + (1-p) \int_0^{\eta} e^{\lambda s} g(s) ds} \tag{16}$$

I resolve this bottleneck by computing the integral for all grid points in a single pass using cumulative summation with the trapezoidal rule. This operation avoids the high cost of repeated numerical integration calls inside an iterative root-finder. Once the integral term is computed for all grid points, $\bar{h}(\bar{\tau})$ can be evaluated efficiently on all grid points using vectorized operations.

C.3.3 Root Extraction via Grid Search and Interpolation

With $\bar{h}(\bar{\tau})$ evaluated at all grid points, root extraction proceeds as follows:

1. **Grid Scan**: Create an indicator array $\mathbb{I}[\bar{h}(\bar{\tau}_j) > u]$ to identify the grid points that bracket the roots. $\bar{\tau}_{IN}^{UNC}$ is bracketed by the first transition from 0 to 1, and $\bar{\tau}_{OUT}^{UNC}$ by the last transition from 1 to 0.
2. **Interpolation**: Once the bracketing grid points are found, a precise root is obtained via linear interpolation. This is computationally inexpensive while being more accurate than simply snapping to the nearest grid point.
3. **Boundary Cases**: If no grid points satisfy $\bar{h}(\bar{\tau}_j) > u$, both withdrawal times are set to the domain boundary η.

C.4 Stage 3: Equilibrium Solver via Adaptive Bisection

C.4.1 Problem Statement and Mathematical Framework

The equilibrium crash time ξ^* satisfies $\xi^* = \inf\{t : AW(t; \xi^*) \geq \kappa\}$, where $AW(t; \xi)$ represents aggregate withdrawals at time t for a given reentry time $\tau_I = \xi - \bar{\tau}_{IN}^{UNC}$. I solve this by finding the root of $\mathcal{A}(\xi) = \kappa$, where:

$$\mathcal{A}(\xi) := AW(\xi; \xi) = G(\min(\xi, \bar{\tau}_{OUT}^{UNC})) - G(\min(\xi, \bar{\tau}_{IN}^{UNC})) \tag{17}$$

This formulation automatically handles constraint binding: as ξ varies, the effective exit and reentry times adjust via the min operators. By construction, $\bar{\tau}_{OUT}^{UNC} > \bar{\tau}_{IN}^{UNC}$ (agents re-enter after exiting). This implies $\mathcal{A}(\xi)$ is weakly increasing in ξ with three cases: (1) If $\xi < \bar{\tau}_{IN}^{UNC}$: $\mathcal{A}(\xi) = 0$ (no withdrawals yet). (2) If $\bar{\tau}_{IN}^{UNC} < \xi < \bar{\tau}_{OUT}^{UNC}$: $\mathcal{A}(\xi) = G(\xi) - G(\bar{\tau}_{IN}^{UNC})$ (strictly increasing, main case). (3) If $\xi > \bar{\tau}_{OUT}^{UNC}$: $\mathcal{A}(\xi) = G(\bar{\tau}_{OUT}^{UNC}) - G(\bar{\tau}_{IN}^{UNC})$ (constant; if this is less than κ, no run equilibrium exists).

Because $\mathcal{A}(\xi)$ is monotonic, the bisection will find at most one root ξ^* satisfying $\mathcal{A}(\xi^*) = \kappa$. However, this mathematical root is necessary but not sufficient for a valid equilibrium. The implied time-series $AW(t; \xi^*)$ may have crossed κ at some earlier time $\bar{t} < \xi^*$. In such cases, ξ^* is a "false equilibrium": the bank would have crashed at $\bar{t}$, not ξ^*. When a false equilibrium is detected, there is no valid run equilibrium—monotonicity of $\mathcal{A}(\xi)$ ensures no other candidate exists.

Remark 2. *It is essential to distinguish between $AW(t; \xi^*)$, the time-series of withdrawals for a fixed collapse time parameter ξ^*, and the bisection object $\mathcal{A}(\xi) := AW(\xi; \xi)$. The latter conflates evaluation time with collapse time parameter, giving withdrawal volume at the candidate crash time.*

C.4.2 Bisection Algorithm

I solve $\mathcal{A}(\xi) = \kappa$ using a bisection algorithm complemented by a finite-difference check to ensure the root found is indeed an equilibrium. In particular, the algorithm validates equilibrium candidates using a finite-difference slope check. For each candidate ξ, I compute:

$$\Delta AW = AW(\xi + \epsilon; \xi) - AW(\xi; \xi) \tag{18}$$

where ϵ is the local grid spacing of the learning CDF interpolation. If $\Delta AW < 0$ at convergence, ξ lies on the decreasing branch of $AW(t; \xi)$, indicating a false equilibrium.

Algorithm Setup

- **Initialization**: I set search bounds, e.g., $\xi_{\min} = 0$ and $\xi_{\max} = T$.
- **Function evaluation**: $\mathcal{A}(\xi_{guess}) = G(\min(\xi_{guess}, \bar{\tau}_{OUT}^{UNC})) - G(\min(\xi_{guess}, \bar{\tau}_{IN}^{UNC}))$.
- **Error assessment**: $error = \mathcal{A}(\xi_{guess}) - \kappa$.

Iterations

1. **Case 1: Overshoot** ($\mathcal{A} > \kappa + \text{tol}$). The search has gone too far. Reduce upper bound: $\xi_{\max} \leftarrow \xi_{guess}$.
2. **Case 2: Undershoot, Valid Direction** ($\mathcal{A} < \kappa - \text{tol}$). The search is approaching the first root from below. Increase lower bound: $\xi_{\min} \leftarrow \xi_{guess}$.
3. **Case 3a: Converged, Valid Equilibrium** ($|\mathcal{A} - \kappa| \leq \text{tol}$ and $\Delta AW \geq 0$). Root found on increasing branch. Valid equilibrium: ξ^* is the first crossing of κ. Return ξ^*.
4. **Case 3b: Converged, False Equilibrium** ($|\mathcal{A} - \kappa| \leq \text{tol}$ and $\Delta AW < 0$). Root found on decreasing branch (indicating false equilibrium). The peak of $AW(t; \xi^*)$ occurred before ξ^* and exceeded κ, meaning the bank would have crashed earlier. No valid run equilibrium exists. Return failure.

If the search interval $|\xi_{\max} - \xi_{\min}|$ collapses to machine precision without convergence, no run equilibrium exists.

The No-Run Case If a run equilibrium does not exist for a given set of parameters, the algorithm identifies this through: (1) the search interval $|\xi_{\max} - \xi_{\min}|$ collapsing below tolerance without finding a root, (2) detection of a false equilibrium (Case 3b), indicating that while a mathematical root exists, the time-series dynamics would have caused an earlier crash.

C.5 Participation Constraint Verification

Figures 6 and 7 overlay regions where the participation constraint $u > h_{un}(\xi^*)$ from Lemma 3 fails. The uninformed hazard rate $h_{un}(\xi)$ has the same structure as the informed hazard rate from Stage 2, with the density g replaced by the survival function $(1 - G)$:

$$h_{un}(\xi) = \frac{p\, e^{\lambda \xi}[1 - G(\xi)]}{p \int_0^{\xi} e^{\lambda s}[1 - G(s)]\, ds + (1 - p) \int_0^{\eta} e^{\lambda s}[1 - G(s)]\, ds} \tag{19}$$

The integral is computed via the same cumulative trapezoidal rule on the inherited adaptive grid from Stage 1. Unlike the informed case, no root-finding is needed: h_{un} is evaluated at the equilibrium ξ^* from Stage 3 and compared directly to u. In the parameter sweeps of Figures 6 and 7, this check is performed for each parameter pair after the equilibrium is solved.

C.6 Extensions to the Baseline Model

This section documents the computational methods for the three extensions: heterogeneous learning speeds, interest rates on deposits and endogenous social learning from observed withdrawals. Each extension preserves the staged approach but requires specific modifications to handle the increased complexity.

C.6.1 Heterogeneous Learning Speeds

Modified Learning Dynamics The key computational change is the coupled system of ODEs governing information diffusion:

$$\frac{dG_k}{dt} = \beta_k(1 - G_k(t)) \sum_{j=1}^{K} p_j G_j(t) \tag{20}$$

This represents agents in group k learning from the overall pool of informed agents $\omega(t) = \sum_j p_j G_j(t)$. Closed-form solutions exist for certain distributions[40] but involve complex expressions and do not extend to the social learning setting below. I solve this system numerically using the same adaptive ODE solver as the baseline model.

Aggregate Withdrawal Computation The aggregate withdrawal function $\mathcal{A}(\xi) =: AW(\xi;\xi)$ to use in the bisection becomes a weighted sum across groups:

$$AW(\xi) = \sum_{k=1}^{K} p_k \left[G_k(\min(\xi, \bar{\tau}_{OUT,k}^{UNC})) - G_k(\min(\xi, \bar{\tau}_{IN,k}^{UNC}))\right] \tag{21}$$

Each group k has its own learning CDF $G_k(t)$ and hence, its own hazard rate. The unconstrained optimal withdrawal times $\bar{\tau}_{IN,k}^{UNC}$ and $\bar{\tau}_{OUT,k}^{UNC}$ are computed using the same root-finding approach as the baseline model, applied to each group. One key advantage of the approach described above is that one can remain agnostic on which group will have binding constraint and which will not (in particular it is possible for a group to have both $0 < \bar{\tau}_{IN,k}^{UNC}, \bar{\tau}_{OUT,k}^{UNC} < \xi$).

Computational Workflow

1. Solve the coupled ODE system for $\{G_k(t)\}_{k=1}^{K}$ on the adaptive grid.

2. Extract optimal withdrawal times for each group using the baseline algorithm.

3. Solve for equilibrium ξ^* using the weighted aggregate withdrawal function.

The equilibrium solver (Stage 3) remains unchanged except for the modified $AW(\xi)$ calculation.

[40] The closed-form solutions involve the moment-generating functions of the distribution, and are available whenever MGFs admit closed forms.

Validation for Heterogeneous Groups Because the implied time-series $AW(t;\xi^*)$ may be multimodal (multiple humps from different group dynamics), the slope check I perform in the main case is insufficient. Rather, I validate that the root is a correct equilibrium by checking if $AW(t;\xi^*)$ crosses back below κ at any $t < \xi^*$, which would indicate ξ^* is a second crossing rather than the first. This requires computing $AW(t;\xi^*)$ over the grid for $t < \xi^*$, which is $O(n)$ but necessary only after convergence.

C.6.2 Interest Rates and Value Functions

Problem Formulation With positive interest rate $r > 0$, agents maximize expected discounted utility. This introduces a value function $V(\bar{\tau})$ representing the continuation value of holding deposits in reversed time, where $\bar{\tau} = \xi^* - \tau$ is the time remaining until potential collapse (with τ being time since learning). The optimal holding decision becomes:

$$\text{Hold deposits if and only if } \bar{h}(\bar{\tau}) < u + rV(\bar{\tau}) \tag{22}$$

This modifies the threshold condition from $\bar{h}(\bar{\tau}) < u$ to $\bar{h}(\bar{\tau}) - rV(\bar{\tau}) < u$.

Value Function Computation The value function satisfies an ordinary differential equation derived from the Hamilton-Jacobi-Bellman equation. In reversed time, the HJB becomes:

$$V'(\bar{\tau}) = (\bar{h}(\bar{\tau}) + \delta)(1 - V(\bar{\tau})) + \max\{u + rV(\bar{\tau}) - \bar{h}(\bar{\tau}), 0\} \tag{23}$$

where $\delta > r$ is the deposit maturity rate. The boundary condition is $V(0) = (u+\delta)/(r+\delta)$, representing the present value of holding \$1 in deposit: convenience yield u, interest r, and maturity payoff 1 at rate δ, evaluated at the crash time $\bar{\tau} = 0$ (which corresponds to $\tau = \xi^*$ in forward time).

I solve this ODE from $\bar{\tau} = 0$ to $\bar{\tau} = \eta$ using the adaptive grid inherited from the learning dynamics solver. This ensures consistency across all computational stages and avoids interpolation errors.

Modified Optimal Buffer Computation Although the value function implicitly contains the optimal holding strategy, I explicitly recompute the withdrawal times for clarity and numerical stability:

1. Define the modified threshold function: $\tilde{h}(\bar{\tau}) = \bar{h}(\bar{\tau}) - rV(\bar{\tau})$

2. Apply the baseline root-finding algorithm to find:

$$\bar{\tau}_{IN}^{UNC} = \inf\{\bar{\tau} : \tilde{h}(\bar{\tau}) > u\} \tag{24}$$

$$\bar{\tau}_{OUT}^{UNC} = \sup\{\bar{\tau} : \tilde{h}(\bar{\tau}) > u\} \tag{25}$$

This approach reuses the efficient grid-based root extraction from the baseline model while incorporating the value function effects. The subsequent equilibrium computation (Stage 3) proceeds unchanged with these modified withdrawal times.

C.6.3 Social Learning from Withdrawals

The Endogenous Learning Challenge Social learning fundamentally alters the model's structure by making learning dynamics depend on equilibrium behavior. Agents learn about bank fragility by observing others' withdrawals, creating a fixed-point problem: aggregate withdrawals determine learning, learning determines optimal behavior, and optimal behavior determines aggregate withdrawals.

Iterative Solution via Damped Fixed-Point Iteration I solve this fixed-point problem through a damped fixed-point iteration scheme on the aggregate withdrawal function in its entirety (rather than simply at collapse time—$AW(\xi;\xi)$):

1. **Initialization**: Obtain initial guess $AW^{(0)}(t)$ by solving the baseline model (word-of-mouth learning) without social learning effects.
2. **Iteration**: At each iteration $n \geq 1$:
   - Solve learning ODE with forcing term $AW^{(n-1)}(t)$: $\frac{dG}{dt} = \beta(1-G(t)) \times AW^{(n-1)}(t)$
   - Given $G(t)$, compute hazard rates and optimal withdrawal times $\bar{\tau}_{IN}^{UNC}$, $\bar{\tau}_{OUT}^{UNC}$
   - Compute implied equilibrium collapse time ξ^* and withdrawal function: $\tilde{AW}^{(n)}(t)$ (I reuse the bisection algorithm on ξ from the baseline but replace the withdrawal times and learning functions by what is obtained in step 1 and step 2).
   - Apply damping with fixed parameter $\alpha = 0.5$:

   $$AW^{(n)}(t) = (1-\alpha)AW^{(n-1)}(t) + \alpha \tilde{AW}^{(n)}(t) \tag{26}$$

     Alternative damping schemes are possible but practice shows a uniform rule works well.
   - Check convergence: $\|AW^{(n)} - AW^{(n-1)}\|_\infty < \epsilon$
3. **Convergence**: Stop when iteration error falls below tolerance, or maximum iterations is reached.

I assume the unimodal structure of AW persists under social learning dynamics, which is verified numerically in all computed equilibria. Throughout the iteration, η is fixed to its baseline value; it is not recomputed from the endogenous G curve.

Inner Loop Failure For a given $AW^{(n-1)}(t)$, the equilibrium solver (Stage 3 from the baseline model) may find that there is no "inner" equilibrium. That is, fixing the learning curve from $dG^{(n)}/dt = \beta(1 - G^{(n)})AW^{(n-1)}(t)$, we cannot find crash time ξ^* satisfying the equilibrium condition. Without a crash time, the next outer iteration's aggregate withdrawal function cannot be constructed. I address this by setting $\xi^{(n)} = \xi^{(n-1)} + \eta/500$, and compute $AW(t) = G(t) - G(t - \xi^{(n)} + \bar{\tau}_I)$, allowing the iteration to continue. While this lacks theoretical foundation, it permits the algorithm to continue exploring for a potential solution.

Handling No-Run Cases The algorithm identifies no-run cases by failure to converge. In contrast to the baseline case, the identification is not fully grounded in theory and simply a pragmatic solution. The theoretical shortcoming is that there is a possibility that one may fail to capture a run equilibrium (and label it as non-run), in particular because the search path depends on the initial guess and the update rule in case of non-run in the inner loop. Empirically, non-convergence seems to reliably indicate that no social learning run equilibrium exists for the given parameters: varying the initial guess, damping, or update rule in case of inner failure do not "save" equilibria. Non-convergent sequences typically oscillate from the first few iterations or stay at $AW = 0$.

Computational Cost Convergence typically requires approximately 40 iterations to achieve accuracy of order 10^{-5}. Each iteration involves solving an ODE with the time-varying forcing term $AW^{(n-1)}(t)$, which is computationally more demanding than the baseline logistic ODE. Total computation time is approximately 20 seconds, compared to 0.5 seconds for the baseline model.

C.7 Implementation Details and Figure Parameters

This section documents the exact parameter choices used to generate the figures in the main text.

C.7.1 Baseline Model Parameters

The baseline model parameters, used as the foundation for all figures unless otherwise specified, are in Table 6.

Parameter	Symbol	Value
Communication speed	β	1.0
Raw awareness window	$\bar{\eta}$	15.0
Awareness window	η	15.0 ($= \bar{\eta}/\beta$)
Utility flow from deposits	u	0.1
Prior fragility probability	p	0.5
Solvency threshold	κ	0.6
Exponential rate for t_0	λ	0.01
Initial learning condition	$G(0)$	0.0001
Time span	—	$(0, 2\eta) = (0, 30)$

Table 6: Baseline model parameters used throughout the paper.

C.7.2 Figure-Specific Parameters

Table 7 summarizes remaining parameter values used for each figure in the main text. When a cell shows a range (e.g., $[0.001, 0.2]$), this indicates comparative statics over that parameter range. All other parameters are held at baseline values unless otherwise specified.

Figure	β	u	κ	r	Description
Fig. 1: Learning	$\{0.5, 1.0, 2.0\}$	0.1	0.6	—	Multiple β comparison
Fig. 2: Hazard Rate	1.0	0.1	0.6	—	Main scenario
Fig. 3a: Main Eq.	1.0	0.1	0.6	—	Main scenario
Fig. 3b: Fast Comm.	3.0	0.1	0.6	—	High β case
Fig. 3c: Low Utility	1.0	0.01	0.6	—	Low u case
Fig. 4: u Effects	1.0	$[0.001, 0.2]$	0.6	—	Comp. statics in u
Fig. 5: Cross-Effects	$[1, 10^4]$	$[0.001, 1]$	0.6	—	2D parameter sweep
Fig. 9: Heterogeneity	$\{0.125, 12.5\}$	0.1	0.3	—	Two-type model (90%/10%)
Fig. 10: Interest-rate	1.0	0.0	0.6	0.06	Interest-bearing deposits
Fig. 11: Social Learning	0.9	0.5	0.25	—	Social learning dynamics

Table 7: Parameter values for main text figures. All use $p = 0.5$, $\lambda = 0.01$, $\eta = 15/\beta$, time span $(0, 2\eta)$ unless noted. Ranges indicate comparative statics. Figure 4 uses 5000 grid points, Figure 5 uses 500×500 grid with $\beta = 1/\text{meeting time}$ where meeting time $\in [0.0001, 1]$. Figure 9 uses heterogeneous β values with population shares $(0.9, 0.1)$, $p = 0.9$, $\lambda = 0.1$, $\bar{\eta} = 30.0$. Figure 11 uses $p = 0.99$, $\lambda = 0.25$, $\bar{\eta} = 30.0$ with social learning dynamics.